

MODEL THEORY AND GROUPS

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I certify that this project report has been written by me, is a record of work carried out by me, and is essentially different from work undertaken for any other purpose or assessment

ABSTRACT. This project aims to bring a reader interested in group theory up to scratch with classical model theory. We give an introduction to model theory and prove some elementary equivalence results using the compactness theorem and quantifier elimination. We investigate homogeneity, proving Fraïssé’s Theorem and some structural results for homogeneous groups, before considering Hall’s universal group. We then prove the fundamental Ryll-Nardzewski, Engeler, Svenonius Theorem for ω -categoricity (by way of the Omitting Types Theorem) and survey the theory of countable ω -categorical groups. Finally, we see how ω -categoricity, quantifier elimination, and homogeneity are all neatly related.

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1. INTRODUCTION

Some ideas throughout history radically changed how we perceive mathematics. For example, a 10th century European mathematician would be alarmed at our free-wheeling use of the number zero, or at our comfort with infinity. Model theory is built on one of these perspective-changing ideas: We can treat mathematical formulae as genuine mathematical objects and consider those structures in which they are realised, often along combinatorial, algebraic, and topological lines of inquiry.

Model theory slowly distinguished itself from the rest of mathematical logic during the early 20th century. The first proper mention of model theory comes from Alred Tarski in 1954, though perhaps the first model theoretic results were the Löwenheim-Skolem Theorems, originally proved by Löwenheim in 1915, and generalised by Skolem in 1922 and 1929. We would be nowhere without the Compactness Theorem for first-order logic, proved first by Gödel in 1930. Even earlier, notions similar to categoricity (which we study here in detail) were first considered by Veblen in 1904.

Tarski's influence is all over model theory. Quantifier elimination grew from his Warsaw seminar in 1927 and he first defined elementary equivalence, a relation between structures we investigate thoroughly in this project. Hodges [5] also notes Abraham Robinson as an important developer of the tools in use. Indeed, Robinson introduced diagrams and elementary diagrams, which are used in the proof of Theorem 4.4.1.

This project is mainly a story of two seemingly different perspectives on a similar problem, and how they are in fact very similar. We regularly take scenic diversions into group theory. The intended audience is a reader familiar with group theory (at the level of MT5864 and MT5869), algebra, and some first-order logic. Some topology is also needed for some of Chapter 6. A pure mathematics student at the 5000 level at St Andrews should find the material challenging but accessible. All of our technical preliminaries will be laid out in detail below before we begin doing any mathematics.

The first perspective (the 'structural' one) comes from Roland Fraïssé in 1953 [16], who applied back-and-forth arguments to build countable structures which are rigid under homogeneity and the prescription of finitely generated substructures. In Chapter 5, we investigate homogeneity, present a proof of Fraïssé's Theorem and look at some structures we can build with Fraïssé's method. Though the original language of this work is model theoretic (and French), it is often developed without much logic. We see in Chapter 7 that logic still lurks beneath the surface in Fraïssé's theory.

The best-known examples of Fraïssé limits are relational structures, but in 1959 Hall (seemingly without any knowledge of Fraïssé's work) published work on a homogeneous locally finite group universal for the class of finite groups. Hall's group is nice in many ways, including simplicity, homogeneity, order conjugacy, and verbal completeness, all of which we show here. The rich structure of Hall's group is in contrast to the huge control these properties exert over finite groups. We consider this in Appendix A, presenting the joint work of Cherlin, and Felgner on homogeneity [9, 10] and the work of Fitzpatrick on order conjugacy [15].

The second perspective (the 'logical' one) comes from 1959, when Erwin Engeler, Czesław Ryll-Nardzewski, and Lars Svenonius each independently proved a vital theorem linking ω -categoricity of a structure (which one could consider as rigidity with respect to its first-order theory among countable structures) to its automorphism group and the size of the type space of its theory. We see a proof of this in Chapter 6 via ω -saturated structures and the Omitting Types Theorem. With this fundamental theorem, group theorists were able to begin work on ω -categorical groups. Rosenstein laid some of these

foundations in 1973 [34]. Wilson [44] and his student Apps [1] classified the characteristically simple finite ω -categorical groups and left us with some conjectures which remain unresolved.

We mainly present results for countable structures, as the landscape is very different in uncountable cardinalities. Morley [28] showed that categoricity in any two uncountable cardinalities are equivalent, and Jónsson [23] generalised some of Fraïssé's theory to higher cardinalities. Such results often require some heavy set theory (in Jónsson's case, the generalised continuum hypothesis). Along the way, we give some examples showing how the theory for countable structures fails if we try to apply it to uncountable structures.

Also, the results we present could be described as classical model theory. Shelah and his classification theory was successful in determining which theories admit reasonable classifications of their models, and everything which grew from this is best described as modern model theory.

Used throughout were textbooks by Hodges [20], Tent and Zeigler [40], and Marker [27]. Unless we say otherwise, every solution to an exercise in one of these books is original. Hodges [5] and Vaught [41] were our main sources on the history of model theory.

2. NOTATION

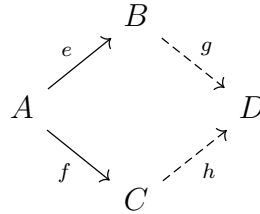
Notation for Logic and Set Theory

- We sometimes take unions of functions (e.g. $f_1 \cup f_2$), or consider one function as a subset of another (e.g. $f_1 \subseteq f_2$). This is using the formal definition of a function $f : A \rightarrow B$ as a subset of $A \times B$ with certain properties.
- To add clarity, we sometimes add a hook to an arrow (e.g. $\hookrightarrow$) to represent embeddings and add another head to the arrow (e.g. $\twoheadrightarrow$) to represent a surjection. We may combine these to denote an isomorphism.
- When doing general model theory we use left actions (i.e. $f(x) = y$).
- We sometimes write disjoint unions as $\sqcup$.
- We are very free with our use of tuples. We often write a tuple as a letter with an overline, e.g. $\bar{x} = (x_1, \dots, x_n)$ for an n -tuple over some set X . When it would be wordy to write ' $\bar{x}$ is an n -tuple over X ', we may write $\bar{x} \in X$. For $f : X \rightarrow Y$, we may write $f(\bar{x})$ for $(f(x_1), \dots, f(x_n))$.
- We define formulae with a colon and sometimes vary the size of our brackets for visual clarity. For example, we will see the following formula later on:

$$\text{size}_n : (\exists x_1) \dots (\exists x_n) \left(\left(\bigwedge_{1 \leq i < j \leq n} x_i \neq x_j \right) \wedge (\forall y) \left(\bigvee_{1 \leq k \leq n} x_k = y \right) \right).$$

The colon is necessary to work around the fact that our formulae often contain a symbol for equality.

- In a diagram, a dashed line refers to a map we are building using some property, while a filled in line refers to some map which already exists. For example, when we discuss the amalgamation property we use the following diagram:



Which says that given appropriate A, B, C, e , and f , there exists some D, g , and h making the diagram commute.

- We readily switch between ω and $\mathbb{N}$ for our countable indexing.

Notation for Group Theory

- We write $G * H$ for the free product of the groups G and H .
- We write $G \rtimes H$ for the semidirect product of the group G by the group H .
- We write $G \oplus H$ for the direct sum of the groups G and H . Note the infinite direct sum $\bigoplus_{i < \omega} G_i$ is the subgroup of the infinite direct product where only finitely many of the components contribute a non-identity element.
- We write $G^{(\omega)}$ for the direct sum of countably many copies of G , i.e.

$$G^{(\omega)} := \bigoplus_{i < \omega} G.$$

- We write $[x, y]$ for the commutator $x^{-1}y^{-1}xy$.
- We write G' for the derived subgroup of the group G (often also written $[G, G]$).
- When doing group theory we use left actions (i.e. $(x)f = y$).

3. PRELIMINARIES

Preliminaries from Logic and Set Theory.

Without the Compactness Theorem (4.3.2) this project would never get anywhere. Compactness relies on the Boolean Prime Ideal Theorem, which is nearly equivalent to Zorn's Lemma. If one is happy to accept the Boolean Prime Ideal Theorem, one might as well go all the way.

Lemma 3.0.1. (*Zorn's Lemma*) *If a every chain in a partially ordered set $(X, <)$ has an upper bound in X , then X has some maximal element with respect to $<$.*

Other than Zorn's Lemma and a familiarity with first-order logic, we assume little from logic and set theory.

Preliminaries from Group Theory.

We assume a knowledge of group theory including some parts of MT5864 and MT5869. In particular, the reader should have some familiarity with:

- Soluble groups
- Nilpotent groups
- Characteristically simple groups
- Free groups and free products
- Semidirect products

The other preliminaries we need from group theory are some rather disjoint results. In the proof of Lemma 5.4.6 we will apply the following, which is Exercise 69.1 in Munkres [29].

Theorem 3.0.2. *Let G and H be groups, then*

$$G * H / (G * H)' \cong (G/G') \oplus (H/H').$$

In the proof of Theorem 4.4.1 and Proposition 5.3.1 we will use the following classification of finitely generated abelian groups, which is a restatement of Theorem 15.5 in Fuchs [17].

Theorem 3.0.3. *Let A be a finitely generated abelian group. Then*

$$A \cong \mathbb{Z}^n \oplus \bigoplus_{i=1}^k C_{p_i}^{m_i}$$

for some $n, k, p_1, m_1, \dots, p_k, m_k \in \mathbb{N}_0$ with $p_1, \dots, p_k$ prime. In particular, if A is torsion free then $A \cong \mathbb{Z}^n$.

The following is Theorem 17.2 in Fuchs [17], and is used in our proof of Lemma 6.4.3

Theorem 3.0.4. *An abelian group of bounded exponent is a direct sum of finite cyclic groups.*

The following definition is used in Chapter 5 and the appendix, and is not found in MT5864 or MT5867.

Definition 3.0.5. We say a group is *homocyclic* if it is a direct sum of subgroups of the same order.

With all this out of the way, we can finally do some mathematics!

4. INTRODUCTION TO MODEL THEORY

We begin by inviting the reader to model theory via Jules Verne [42]:

‘Descend into the crater of Yocul of Sneffels, which the shade of the Scartaris caresses, before the kalends of July, audacious traveller, and you will reach the centre of the earth. I did it.’

This chapter is our dive into the crater, where we build most of the tools we will need to tell our main story in the following chapters. Along the way we will prove some striking results on elementary equivalence, one using the Compactness Theorem (4.3.2), the other using quantifier elimination.

4.1. First Definitions. Model theory talks about structures very broadly, making statements about graphs, groups, fields, vector spaces, all at once! For this, we require some flexible notation.

Definition 4.1.1. A language L is a set of constants, function symbols of arity at least 1, and relation symbols of arity at least 1. Given a language L , an L -structure A is a tuple $(D, (Z^A)_{Z \in L})$ where:

- D is an underlying non-empty set called the *domain* of A . We often use the notation $D = \text{dom}(A)$.
- $Z^A \in D$, if Z is a constant
- $Z^A \subseteq D^n$, if Z is an n -ary relation symbol
- $Z : A^n \rightarrow A$, if Z is an n -ary function symbol

We can also say L is the *signature* of the structure A . In practice, a structure is often identified with its underlying set.

Remark 4.1.2. Our definition does not allow an empty structure, which is entirely a matter of taste.

Example 4.1.3. A group $(G, \cdot)$ is a $\{1, \cdot, {}^{-1}\}$ -structure with domain G , where 1 is a constant, $\cdot$ is a binary function symbol, and ${}^{-1}$ is a unary function symbol. Equally, $(G, \cdot)$ is a $\{1, \cdot\}$ -structure. We shall see why the first convention is more appropriate than the second when we consider substructures. We will also use the more natural language $\{0, +, -\}$ to discuss abelian groups.

Example 4.1.4. A graph (V, E) is a $\{\sim\}$ structure with domain V , where $\sim$ is a binary relation symbol corresponding to the edge relations in graphs.

A structure like above with no functions or constants is a *relational* structure. While we want to build to questions about groups, relational structures will serve as nice visual examples throughout the project.

As in all of mathematics, when we have a structure the natural considerations are then homomorphisms and substructures.

Definition 4.1.5. Let L be some signature of structures A and B . A function $f : \text{dom}(A) \rightarrow \text{dom}(B)$ is a *homomorphism* if it respects the signature L as follows:

- (1) For all constants c , $f(c^A) = c^B$.
- (2) For all n -tuples $\bar{a}$ over A , and n -ary relations R , if $\bar{a} \in R^A$, then $f(\bar{a}) \in R^B$.
- (3) For all n -tuples $\bar{a}$ over A , and n -ary functions F , we have $f(F(\bar{a})) = F(f(\bar{a}))$.

We then define an *embedding* to be an injective homomorphism where (2) is strengthened to an if and only if statement. An *isomorphism* is a surjective embedding.

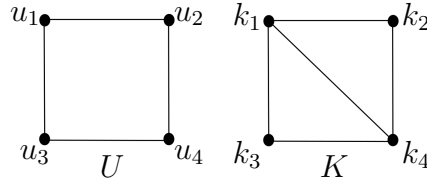


FIGURE 1

Example 4.1.6. Figure 1 shows that we really do need the if and only if condition to sensibly define an isomorphism. Consider the two graphs U and K ; the map $u_i \rightarrow k_i$ is a bijective homomorphism but not an isomorphism.

We observe that, as groups are structures in relation-free languages, a bijective group homomorphism is an isomorphism. A proof of the following lemma would be better written than read.

Lemma 4.1.7. *L -structure homomorphisms, embeddings, and isomorphisms are preserved under composition.*

In particular we can consider the automorphism group of any L -structure A , write $\text{Aut}(A)$. The 'size' of an automorphism group plays a huge role in Chapters 5 and 6.

Definition 4.1.8. For A and B two L -structures with $\text{dom}(B) \subseteq \text{dom}(A)$, we say B is a *substructure* of A (write $B \leq A$) if the inclusion map from $\text{dom}(B)$ to $\text{dom}(A)$ is an embedding.

Applying Lemma 4.1.7 gives that substructures are inherited, as one would hope.

Example 4.1.9. Consider the additive group $(\mathbb{Z}, +)$ and the languages

$$L_1 = \{0, +, -\} \text{ and } L_2 = \{0, +\}.$$

The naturals $\mathbb{N}$ are an L_2 -substructure of $\mathbb{Z}$, but not an L_1 -substructure, as we cannot define $-$ in a way which would respect an embedding into $\mathbb{Z}$. More generally, the L_2 -substructures of a group are subsemigroups, while L_1 substructures are subgroups. If we want to do group theory, the signature L_1 is more appropriate. In light of this, we will call L_1 the natural language of abelian groups and call $\{1, \cdot, {}^{-1}\}$ the natural language of groups.

We want to be able to talk about the subgroup generated by some subset of a group. As in group theory, we first need a lemma before we can make such a definition.

Lemma 4.1.10. *Let A be some L -structure, and let $B \subseteq \text{dom}(A)$. Let $(A_i)_{i \in I}$ be the set of substructures of A whose domains contain B . Then $\bigcap_{i \in I} \text{dom}(A)_i$ is the domain of a substructure of A .*

Proof. Write $X := \bigcap_{i \in I} \text{dom}(A)_i$. For all constants c and for all $i \in I$, as A_i is a substructure we have $c^A \in \text{dom}(A)_i$. Thus $c^A \in X$. Similarly for all $n \in \mathbb{N}$, for all n -ary function symbols f in the language L , and for all $i \in I$, as A_i is a substructure we have that $\text{dom}(A_i)^n$ is closed under the application of f^A . Thus X^n is closed under the application of f^A . We have shown we can well-define the structure $\langle B \rangle$ with domain X such that:

- $c^{(B)} = c^A$ for all constants c .
- $R^{(B)} = R^A \cap X^n$ for all n -ary relation symbols R .

- $f^{(B)} = f|_{X^n}$ for all n -ary function symbols f .

That the inclusion map is an embedding is immediate. $\square$

In particular, if some subset of the domain containing the constants is closed under the application of the functions, we know it is the domain of a substructure.

Definition 4.1.11. For $B \subseteq \text{dom}(A)$, the structure $\langle B \rangle$ defined in Lemma 4.1.10 is the *substructure generated by B* . If a substructure can be generated by some finite set, we say it is *finitely generated*.

The class of finitely generated substructures of a structure is an important consideration in Chapter 5. Note that the substructure generated by a set is more interesting when the structure has functions.

Lemma 4.1.12. *If L is a countable/finite language, then a finitely generated L -structure is at most countable.*

Example 4.1.13. We may want to talk about the same mathematical object in many different ways. For example, the set of real numbers $\mathbb{R}$ can be viewed as:

- An abelian group in the language $(0, +, -)$
- A field in the language $(0, 1, +, -, \cdot, ^{-1})$
- A vector space over $\mathbb{Q}$ in the language $(0, +, -, \hat{q}_1, \hat{q}_2, \dots)$ where $\mathbb{Q} = \{q_1, q_2, \dots\}$ and each $q_i \in \mathbb{Q}$ has some associate unary function symbol $\hat{q}_i$ corresponding to scalar multiplication by q_i .
- A total order in the language $(<)$

We give some notation to help in situations like this. Given a language L , we can add a set C of constants, relations and functions to get a new language $L(C)$. Any $L(C)$ -structure A can be viewed as an L -structure simply by forgetting the constants, relations, and functions in C . We write $A|_L$ for this structure and call $A|_L$ the *L -reduct* of A .

Changing the language to fit our requirements and taking a reduct is one of the key ideas in model theory, and will be used throughout the project.

We now want to make formal statements about L -structures in the language L and analyse these statements as mathematical objects.

Definition 4.1.14. Given a language L we can define terms as follows:

- Every variable is a term
- Every constant symbol is a term
- Given $t_1, \dots, t_n$ terms and f an n -ary function symbol, $f(t_1, \dots, t_n)$ is a term

Terms are placeholders for elements of our structures about which we make statements. From terms, relations, and certain logical symbols we can recursively build formulae. Let $t_1, \dots, t_n$ be terms, then:

- $t_1 = t_2$ is a formula
- $R(t_1, \dots, t_n)$ is a formula

Formulae of the kind above are called *atomic*. Let ψ, γ be some formulae and $\bar{x}$ be a tuple of variables, then :

- $\psi(\bar{x}) \wedge \gamma(\bar{x})$ is a formula
- $\neg\psi(\bar{x})$ is a formula
- $(\exists \bar{x})\psi(\bar{x})$ is a formula

Formulae correspond to the statements about our structures.

Some authors adopt an order of operation for logical symbols to limit the number of brackets in formulae. Here, we will simply use a lot of brackets to avoid adding more

rules. We also define the following shorthand for certain important formulae. Working this way shortens any induction proofs on formulae.

- $\psi(\bar{x}) \vee \gamma(\bar{x})$ will mean $\neg((\neg\psi(\bar{x})) \wedge (\neg\gamma(\bar{x})))$
- $(\forall \bar{x})\psi(\bar{x})$ will mean $\neg((\exists \bar{x})(\neg\psi(\bar{x}))$
- $\psi(\bar{x}) \rightarrow \gamma(\bar{x})$ will mean $\neg(\psi(\bar{x}) \wedge \neg\gamma(\bar{x}))$
- $\psi(\bar{x}) \leftrightarrow \gamma(\bar{x})$ will mean $(\psi(\bar{x}) \rightarrow \gamma(\bar{x})) \wedge (\gamma(\bar{x}) \rightarrow \psi(\bar{x}))$

A formula where all the variables are within the scope of some quantifier $\forall$ or $\exists$ is a *sentence*.

Example 4.1.15. In $\{1, \cdot, ^{-1}\}$, the language of Example 4.1.3, consider the following sentences, which should be true of any group:

$$\begin{aligned} &(\forall x)(\forall y)(\forall z)(x \cdot (y \cdot z) = (x \cdot y) \cdot z) \\ &(\forall x)((x \cdot 1 = x) \wedge (1 \cdot x = x)) \\ &(\forall x)(x \cdot x^{-1} = 1) \end{aligned}$$

We also use standard group theory abbreviations in our first-order sentences, for example writing x^4 for $x \cdot x \cdot x \cdot x$, or writing x^y for $y^{-1} \cdot x \cdot y$.

Now given some L -structure A , some formulae, and some elements of A corresponding to the variables in the formula A , how can we say whether or not our formula is true in A when we substitute in our elements to the variables?

Tarski in 1933 and 1949 (with Vaught) laid the groundwork for us here, showing that our definition of truth below does exactly what we would like it to do. Hodges [5] discusses the history here in more detail.

Definition 4.1.16. Let A be an L -structure. If $t(\bar{x})$ is some L -term and $\bar{a}$ a tuple in A of the same length as $\bar{x}$, we see that from the definition of a term that we can define $t(\bar{a}) \in A$.

For some L -formula $\varphi(\bar{x})$, we can assign elements the elements in $\bar{a}$ to the formula φ , write $\varphi(\bar{a})$. If this formula is true in A , we write $A \models \varphi(\bar{a})$, and say $\bar{a}$ *satisfies* φ .

Truth values are assigned to atomic formulae according to equality in the structure A , and we then recursively assign truth values to all formulae as follows. Throughout let $\phi(\bar{x}), \sigma(\bar{x})$ be formulae:

- $A \models \phi(\bar{a}) \wedge \sigma(\bar{a})$ if $A \models \phi(\bar{a})$ and $A \models \sigma(\bar{a})$.
- $A \models \neg\phi(\bar{a})$ if $A \models \phi(\bar{a})$ does not hold.
- If $\phi(\bar{x}) = (\exists y)(\sigma(y, \bar{x}))$, then $A \models \phi(\bar{a})$ if there is some $b \in A$, such that $A \models \sigma(b, \bar{a})$.

The following lemma is immediate from the definitions.

Lemma 4.1.17. Let A be some L -structure and $\phi(\bar{x}), \sigma(\bar{x})$ two L -formulae.

- $A \models \phi(\bar{a}) \vee \sigma(\bar{a})$ if and only if $A \models \phi(\bar{a})$ or $A \models \sigma(\bar{a})$.
- If $\phi(\bar{x}) = (\forall y)(\sigma(y, \bar{x}))$, then $A \models \phi(\bar{a})$ if and only if for all $b \in A$, we have $A \models \sigma(b, \bar{a})$.
- $A \models \phi(\bar{a}) \rightarrow \sigma(\bar{a})$ if and only if it is not true that both $A \models \phi(\bar{a})$ and $A \not\models \sigma(\bar{a})$.
- $A \models \phi(\bar{a}) \leftrightarrow \sigma(\bar{a})$ if and only if both or neither of $A \models \phi(\bar{a})$ and $A \models \sigma(\bar{a})$ hold.
- $A \not\models \phi(\bar{a}) \wedge (\neg\phi(\bar{a}))$.

Remark 4.1.18. To save ink, we will allow ourselves to be very free with lengths of tuples. For example we may write the atomic formula $R(t_1(\bar{x}_1), \dots, t_n(\bar{x}_n))$ as $R(t_1(\bar{x}), \dots, t_n(\bar{x}))$ where $\bar{x}$ contains all the variables in tuples $\bar{x}_1, \dots, \bar{x}_n$, allowing terms $t_1, \dots, t_n$ to not use every variable.

The following lemma says that embeddings behave well under assignments.

Lemma 4.1.19. *Let A, B be L -structures and $f : A \rightarrow B$ an embedding.*

- (1) *For every term $t(\bar{x})$ and A -tuple $\bar{a}$, we have $f(t(\bar{a})) = t(f(\bar{a}))$.*
- (2) *For every atomic formula $\varphi(\bar{x})$ and A -tuple $\bar{a}$, we have*

$$A \models \varphi(\bar{a}) \text{ if and only if } B \models \varphi(f(\bar{a})).$$

- (3) *If f is an isomorphism, then for every formula $\varphi(\bar{x})$ and A -tuple $\bar{a}$, we have*

$$A \models \varphi(\bar{a}) \text{ if and only if } B \models \varphi(f(\bar{a})).$$

Proof. (1). We use the recursive definition of terms and regularly apply Definition 4.1.5. If $t(x)$ is a variable, then for all $a \in A$, $t(a) = a$, so

$$f(t(a)) = f(a) = t(f(a)).$$

If t is some constant c , then

$$f(t(\bar{a})) = f(c^A) = c^B = t(f(\bar{a})).$$

Now for the recursive step, if $t(\bar{x}) = F(t_1(\bar{x}), \dots, t_n(\bar{x}))$ for some function symbol F where $t_1, \dots, t_n$ are terms for which the claim holds, then:

$$\begin{aligned} f(t(\bar{x})) &= f(F(t_1(\bar{x}), \dots, t_n(\bar{x}))) = F(f(t_1(\bar{x})), \dots, f(t_n(\bar{x}))) \\ &= F(t_1(f(\bar{x})), \dots, t_n(f(\bar{x}))) = t(f(\bar{x})). \end{aligned}$$

- (2). Let $s(\bar{x}), t(\bar{x})$ be terms, let $\varphi(\bar{x})$ be an atomic formula, and let $\bar{a}$ be some tuple over A . Suppose first that

$$\phi(\bar{x}) : s(\bar{x}) = t(\bar{x}).$$

Then $A \models \phi(\bar{a})$ if and only if $s(\bar{a}) = t(\bar{a})$, giving $f(s(\bar{a})) = f(t(\bar{a}))$. Applying (1), we see this happens if and only if $s(f(\bar{a})) = t(f(\bar{a}))$, so $B \models \varphi(f(\bar{a}))$. If instead

$$\phi(\bar{x}) : R(t_1(\bar{x}), \dots, t_n(\bar{x})),$$

we know from Definition 4.1.5

$$A \models R(t_1(\bar{a}), \dots, t_n(\bar{a})) \iff B \models R(f(t_1(\bar{a})), \dots, f(t_n(\bar{a}))).$$

Now the left hand satisfaction is exactly $A \models \phi(\bar{a})$. Applying (1), the right hand satisfaction is equivalent to $B \models \phi(f(\bar{a}))$, as required.

- (3). This follows from the above ideas using the recursive definition of formulae. □

Definition 4.1.20. We call an atomic formula or a negation of an atomic sentence a *closed literal*.

One can then easily note that condition (2) in Lemma 4.1.19 above can be extended to closed literals.

Definition 4.1.21. A set of L -sentences T is an L -theory. If there is some L -structure A which satisfies every sentence in T , we write $A \models T$ and say A is a *model* of T . We say a theory is *consistent* if it has a model.

We now have some natural questions: Which theories are consistent? How many models can a theory have? How different can models of a theory look?

Example 4.1.22. We show there is a sentence encoding the size of a finite set. Let $0 < n < \omega$ be given and define:

$$\text{size}_n : (\exists x_1) \dots (\exists x_n) \left(\left(\bigwedge_{1 \leq i < j \leq n} x_i \neq x_j \right) \wedge (\forall y) \left(\bigvee_{1 \leq k \leq n} x_k = y \right) \right)$$

So $A \models \text{size}_n$ forces $|\text{dom}(A)| = n$.

Example 4.1.23. We give an example to show how theories can force embeddings from other structures. Given an L -structure A , we can add a constant a to the language for each $a \in A$, creating the language $L(A)$. Now A is naturally an $L(A)$ structure, so we can define the following theory:

$$T_A = \{\phi \mid \phi \text{ is a closed literal in } L(A) \text{ such that } A \models \phi\}.$$

Now consider any model M of the above theory. The substructure generated by

$$\{a^M \mid a \in A\}$$

is isomorphic to A under the map $a \mapsto a^M$, so A embeds into M . This idea will be very useful in the proof of Theorem 4.4.1.

We now define the related notion of consequence. We will go the entire project without mentioning deduction or proofs (as model theory is often done in languages which are not first-order, and so do not have a natural deduction system), though the reader with some experience in first-order logic is welcome to think of consequence in this way.

Definition 4.1.24. Let T be an L -theory and ϕ be some L -sentence. Then ϕ is a *consequence* of T , write $T \vdash \phi$, if for all L -structures A , $A \models T$ implies $A \models \phi$. We say two L -formulae $\gamma(\bar{x}), \sigma(\bar{x})$ are *equivalent modulo T* , write $\gamma(\bar{x}) \equiv_T \sigma(\bar{x})$, if

$$T \vdash (\forall \bar{x})(\gamma(\bar{x}) \leftrightarrow \sigma(\bar{x})).$$

Lemma 4.1.25. *Let T be some L -theory.*

- (1) *Two formulae $\gamma(\bar{x}), \sigma(\bar{x})$ are equivalent modulo T if and only if $A \models \gamma(\bar{a})$ and $A \models \sigma(\bar{a})$ are equivalent for all tuples $\bar{a}$ over A and all $A \models T$*
- (2) *Equivalence of formulae modulo T is an equivalence relation.*

Proof. (1). We prove the forward direction, and the proof can be reversed to give the backward direction. Suppose $\gamma(\bar{x}) \equiv_T \sigma(\bar{x})$, then

$$T \vdash (\forall \bar{x})(\gamma(\bar{x}) \leftrightarrow \sigma(\bar{x})).$$

Let $A \models T$ and $\bar{a} \in A$ be given, then Definition 4.1.16 gives $A \models \gamma(\bar{a}) \leftrightarrow \sigma(\bar{a})$, which means $A \models \gamma(\bar{a})$ and $A \models \sigma(\bar{a})$ are equivalent statements.

(2). Now we have (1) this is a very quick check. □

We shall see in Chapter 6 that the number of equivalence classes of relations modulo some theory tells us a lot about models of the theory!

4.2. Theories and Elementary Equivalence.

Definition 4.2.1. We say a consistent theory T is *complete* if for all sentences ϕ , either $T \vdash \phi$ or $T \vdash \neg\phi$. Given an L -structure A , we define the *complete L -theory* of A :

$$\text{Th}_L(A) = \{\phi \mid \phi \text{ is an } L\text{-sentence with } A \models \phi\}$$

It is a quick check to see that the complete theory of a structure really is complete.

Lemma 4.2.2. *Every consistent theory is contained in some maximal complete consistent theory.*

Proof. Let T be some consistent theory. Then we may find some model $A \models T$, giving that T is contained in the complete theory $\text{Th}(A)$. From the definition of $\text{Th}(A)$, we see it contains ϕ or $\neg\phi$ for all L -sentences ϕ , so $\text{Th}(A)$ is maximal. □

Definition 4.2.3. We say two L -structures A and B are *elementarily equivalent* with respect to L , write $A \equiv_L B$, if $\text{Th}_L(A) = \text{Th}_L(B)$.

When the language is clear we will drop the prefixes and subscripts. For example when we say two groups are elementarily equivalent, we generally mean they are elementarily equivalent with respect to $\{1, \cdot, ^{-1}\}$.

One should think of elementarily equivalent structures as those which the language (in first-order logic) cannot tell apart.

We devote much of this project to the study of elementary equivalence, often specialising to groups. This chapter builds to Theorem 4.4.1 and Theorem 4.5.5, which are two striking elementary equivalence results for abelian groups. Most of the results show elementary equivalence is weak for groups, but Chapter 6 is spent studying those cases where elementary equivalence is strong for countable groups.

Lemma 4.2.4. *Let T be a consistent L -theory. The following are equivalent:*

- (1) T is complete.
- (2) Any two models of T are elementarily equivalent.

The following proof is just unpacking definitions.

Proof. (1) $\implies$ (2). Suppose T is complete. For any two models $A, B \models T$, for all L -sentences ϕ we have:

$$A \models \phi \iff T \vdash \phi \iff B \models \phi,$$

giving $\text{Th}(A) = \text{Th}(B)$ and hence $A \equiv B$.

(2) $\implies$ (1). Suppose any two models of T are elementarily equivalent. Now let ϕ be some L -sentence. The theory T is consistent so has some model $A \models T$. If $A \models \phi$, then for all models $B \models T$ we also have $B \models \phi$, implying $T \vdash \phi$. Otherwise if $A \models \neg\phi$ we similarly get $T \vdash \neg\phi$. We have shown either $T \vdash \phi$ or $T \vdash \neg\phi$ for the arbitrary L -sentence ϕ , so T is complete. $\square$

Lemma 4.2.5. *Let A and B be two L -structures. If $\text{Th}(A) \subseteq \text{Th}(B)$ then $\text{Th}(A) = \text{Th}(B)$, and so $A \equiv B$.*

Proof. Suppose, looking for a contradiction, that $\text{Th}(B) \not\subseteq \text{Th}(A)$. Then we may find some $\phi \in \text{Th}(B) \setminus \text{Th}(A)$, so $A \not\models \phi$. From Definition 4.1.16, $A \models \neg\phi$. Thus

$$A \models \phi, A \models \neg\phi \implies A \models \phi \wedge (\neg\phi)$$

which is a contradiction by Lemma 4.1.17. $\square$

As in other areas of mathematics, isomorphism is the strongest notion of two structures being equal. It would be alarming if two isomorphic groups did not share a theory.

Lemma 4.2.6. *Let A and B be L -structures. If $A \cong B$, then $A \equiv B$.*

Proof. Follows from Lemma 4.1.19. $\square$

The converse of Lemma 4.2.6 does not hold, and model theory would not be very interesting if it did! The Löwenheim-Skolem Theorems (4.3.8 and 4.3.9) give the most striking counter-examples, and we shall see some group-theoretic counter examples below. First we show that the converse does hold in the finite case.

Proposition 4.2.7. ([40], Exercise 1.3.3)

Let A and B be finite L -structures. If $A \equiv B$, then $A \cong B$.

Proof. Our argument is in two parts. First assume L has finitely many constants, relation symbols and function symbols. The structure A has some finite enumeration $A = \{a_1, \dots, a_k\}$. The idea is to associate variables x_i to each a_i and encode all the information we have about A into a single sentence with these variables.

Write C for the set of constants in L . For every $c \in L$ we have $c^A = a_i$ for some i , so define the formula $\psi_c : c = x_i$. Let P_n be the set n -tuples over $\{1, \dots, k\}$, and write $\mathfrak{R}_n$ for the set of n -ary relation symbols in L , $\mathfrak{F}_n$ for the set of n -ary function symbols in L . For every $R \in \mathfrak{R}_n$, and every $(i_1, \dots, i_n) \in P_n$, set

$$\phi_{R,(i_1,\dots,i_n)}(x_{i_1}, \dots, x_{i_n}) = \begin{cases} R(x_{i_1}, \dots, x_{i_n}) & \text{if } A \models R(a_{i_1}, \dots, a_{i_n}) \\ \neg R(x_{i_1}, \dots, x_{i_n}) & \text{if } A \models \neg R(a_{i_1}, \dots, a_{i_n}) \end{cases}$$

For every $f \in \mathfrak{F}_n$, write $\iota_f(i_1, \dots, i_n)$ for the index of $f^A(a_{i_1}, \dots, a_{i_n})$ in our enumeration of A above.

Define the sentence:

$$\lambda : (\exists x_1) \dots (\exists x_n) \left(\left(\bigwedge_{1 \leq i < j \leq n} x_i \neq x_j \right) \wedge (\forall y) \left(\bigvee_{1 \leq k \leq n} x_k = y \right) \right) \wedge \left(\bigwedge_{c \in C} \psi_c \right) \wedge \\ \left(\bigwedge_{1 \leq n \leq k} \bigwedge_{(i_1, \dots, i_n) \in P_n} \left(\left(\bigwedge_{R \in \mathfrak{R}_n} \phi_{R,(i_1, \dots, i_n)}(x_{i_1}, \dots, x_{i_n}) \right) \wedge \left(\bigwedge_{f \in \mathfrak{F}_n} f(x_{i_1}, \dots, x_{i_n}) = x_{\iota_f(i_1, \dots, i_n)} \right) \right) \right)$$

The first big bracketed part forces any model of λ to have exactly n -elements. The second big bracketed part ensures the assignment of constants is preserved under isomorphism. The third big bracketed part encodes every n -ary relation and non-relation of n -tuples and writes out every application of every n -ary function to every n -tuple, like writing out the Cayley table of a group.

By construction the assignment $x_i \mapsto a_i$ witnesses $A \models \lambda$. By assumption $A \equiv B$, so $B \models \lambda$, requiring some assignment $x_i \mapsto b_i$. Now define $\Phi : A \rightarrow B$ by $\Phi(a_i) = b_i$. By our discussion above, this is an isomorphism, so $A \cong B$.

Now for the second part of the argument, we can suppose L is some arbitrary language. We still have $A \models \text{size}_k$, so $B \models \text{size}_k$, so A and B are in bijection. In particular there are finitely many bijections $f_1, \dots, f_m$ from A to B . Suppose, looking for a contradiction, that none of these bijections are isomorphisms. Then for each $f_i : A \rightarrow B$ a bijection, f_i must fail one of the conditions in Definition 4.1.5, i.e. there is some constant, function or relation $Z_i \in L$ which witnesses that f_i is not an isomorphism. Consider $L' = \{Z_1, \dots, Z_m\}$. By construction, none of the f_i 's are L' -structure isomorphisms either, so there is no isomorphism between A and B as L' -structures. But $A \equiv_L B$ and $L' \subseteq L$, so certainly $A \equiv_{L'} B$. The language L' is finite, so $A \cong_{L'} B$ by our work above. We have arrived at the desired contradiction, so one of the bijections must in fact be an isomorphism, and $A \cong_L B$. $\square$

The proposition below shows that showing two clearly different groups are not elementarily equivalent can require some work.

Proposition 4.2.8. ([40], Exercise 1.3.2)

For $m, n \in \mathbb{N}$ with $m \neq n$, we have $\mathbb{Z}^m \not\equiv \mathbb{Z}^n$.

Proof. The best way to prove two structures are not elementarily equivalent is by giving a sentence satisfied by exactly one of the two structures!

Suppose $m < n$ and consider:

$$\phi_m : (\exists x_1, \dots, x_m) (\forall y) (\exists z_1, \dots, z_m) ((y = 2z_1 + x_1 + \dots + 2z_m + x_m) \vee \dots \vee (y = 2z_1 + \dots + 2z_m))$$

where each term in the disjunction varies whether each x_i is in the sum. To motivate why this is used consider

$$\phi_1 : (\exists x) (\forall y) (\exists z) ((y = 2z + x) \vee (y = 2z)).$$

In $\mathbb{Z}$, assigning the element 1 to x witnesses $\mathbb{Z} \models \phi_1$, and so ϕ_1 says there is a well-defined notion of parity. The sentence ϕ_m says there is a well-defined ' m -dimensional' notion of parity.

We see immediately that the standard generators for $\mathbb{Z}^m$:

$$((1, 0, \dots, 0), (0, 1, 0, \dots, 0), \dots, (0, \dots, 0, 1))$$

witness that $\mathbb{Z}^m \models \phi_m$. We now show $\mathbb{Z}^n \not\models \phi_m$.

Define

$$f : \mathbb{Z}^n \rightarrow \{0, 1\}^n \text{ where } f(y_1, \dots, y_n) = (y_1 \pmod{2}, \dots, y_n \pmod{2}).$$

Write $\bar{x} = (x_1, \dots, x_m)$, $\bar{z} = (z_1, \dots, z_m)$. Note that there are 2^m terms $y = t(\bar{x}, \bar{z})$ in the disjunction in ϕ_m , as for each $1 \leq j \leq m$ we decide whether or not x_j appears in $t(\bar{x}, \bar{z})$. Define the set:

$$U = \{t(\bar{x}, \bar{z}) \mid y = t(\bar{x}, \bar{z}) \text{ is a term in a disjunction in } \phi_m\}.$$

For each term $t \in U$, upon an assignment of the the variables $\bar{x}, \bar{z}$ to $\mathbb{Z}^n$, the choices of $\bar{z}$ will not affect $f(t(\bar{x}, \bar{z}))$.

Now fix some m -tuple $\bar{a} = (a_1, \dots, a_m)$ over $\mathbb{Z}^n$ and consider the assignment $\bar{x} \mapsto \bar{a}$. Define

$$g_{\bar{a}} : U \rightarrow \{0, 1\}^n \text{ where } g_{\bar{a}}(t) = f(t(\bar{a}, \bar{0})).$$

Then as

$$|g_{\bar{a}}(U)| \leq |U| \leq 2^m < 2^n = |\{0, 1\}^n|,$$

we may pick some

$$b \in \{0, 1\}^n \setminus g_{\bar{a}}(U),$$

so $b \in \mathbb{Z}^n$. Now for all choices of assignments $\bar{z} \mapsto \bar{c}$ into $\mathbb{Z}^n$, for all $t(\bar{x}, \bar{z}) \in U$, we have

$$b \neq t(\bar{a}, \bar{c})$$

as the parities do not match in each coordinate by construction of $b \in \mathbb{Z}^n$. In particular

$$\begin{aligned} & \mathbb{Z}^n \not\models (\exists z_1, \dots, z_m)((b = 2z_1 + a_1 + \dots + 2z_m + a_m) \vee \dots \vee (b = 2z_1 + \dots + 2z_m)) \\ \implies & \mathbb{Z}^n \not\models (\forall y)(\exists z_1, \dots, z_m)((y = 2z_1 + a_1 + \dots + 2z_m + a_m) \vee \dots \vee (y = 2z_1 + \dots + 2z_m)). \end{aligned}$$

As the assignment $\bar{x} \mapsto \bar{a}$ was arbitrary, $\mathbb{Z}^n \not\models \phi_m$, as required. $\square$

We will see many more elementary equivalence results as we go along. For now, we look at special substructures behaving well with respect to satisfaction, which are a necessity for Chapter 6 and useful for Theorem 4.4.1.

Definition 4.2.9. Let A be some L -structure. We define a very important $L(A)$ -theory, called the *elementary diagram* of A :

$$\text{eldiag}(A) := \text{Th}_{L(A)}(A)$$

where A is viewed as an $L(A)$ -structure under the assignment of constants $a^A = a$ for all $a \in A$. If some $L(A)$ -structure M is a model of $\text{eldiag}(A)$, we call the L -reduct of M an *elementary extension* of A .

In particular with the same setup as above, as $\text{Th}_L(A) \subseteq \text{eldiag}(A)$, we see any elementary extension M is a model for $\text{Th}_L(A)$, so by Lemma 4.2.5 we have $M \equiv_L A$. The next lemma says that the elementary diagram encodes whether a structure admits a particular type of embedding from A .

Lemma 4.2.10. (*Elementary Diagram Lemma*) Let A and M be L -structures. The following are equivalent

- (1) M is an elementary extension of A .

(2) *There is some embedding*

$$F : A \hookrightarrow M$$

such that, for all formula $\phi(\bar{x})$ and $\bar{a} \in A$, the following holds:

$$A \models \phi(\bar{a}) \iff M \models \phi(F(\bar{a})).$$

Proof. (1) $\implies$ (2).

Suppose condition (1) holds. Then we may find some assignment of the constant a^M for all $a \in A$, such that M is an $L(A)$ -structure with

$$M \models \text{eldiag}(A).$$

Now define the function

$$F : A \hookrightarrow M \text{ where } F(a) = a^M.$$

One can easily check that $\text{eldiag}(A)$ forces F to be an embedding. Now we have

$$(\dagger) \quad A \models \phi(\bar{a}) \iff \phi(\bar{a}) \in \text{eldiag}(A) \iff M \models \phi(\bar{a}^M)$$

so we are done as $F(a) = a^M$ for all $a \in A$.

(2) $\implies$ (1).

Suppose condition (2) holds. Then we define assign the constants $a \in A$ so that $a^M = F(a)$ and view M as an $L(A)$ -structure under this assignment. Then the equivalences in $(\dagger)$ hold again, giving that $M \models \text{eldiag}(A)$. $\square$

4.3. Compactness and Corollaries. So far we have studied the theories of structures which already exist. What if we want to go the other way? The Compactness Theorem (4.3.2), along with its descendants we will meet later, lets us build structures satisfying certain theories. In Chapter 6 with the Omitting Types Theorem (6.2.3), we shall see that it also gives us a lot of control over the structures we build.

Definition 4.3.1. We call a theory T *finitely consistent* if every finite subset of T is consistent.

Theorem 4.3.2. (*Compactness Theorem*)

A theory is consistent if and only if it is finitely consistent.

The forward direction is immediate, so we focus on the backward direction. This proof was sketched in MT5867, but a key lemma is also used when we prove the Omitting Types Theorem (6.2.3), so we will give the full proof here. First-order compactness was first proved by Gödel, but we follow Henkin's strategy here. There are more beautiful ways to prove the compactness theorem, but Henkin's proof is the most useful.

Definition 4.3.3. Let T be a theory in a language L . We will say T is *finitely consistent* if every finite subset of T is consistent. Given some countable set of constants C not in L , say a $L(C)$ -theory T^* is a *Henkin theory* if for every $L(C)$ formula $\phi(x)$, there is some $c \in C$ such that

$$((\exists x)\phi(x)) \rightarrow \phi(c) \in T^*$$

Some authors call the constants C in the above definition *witnesses*, but we will call them *Henkin constants*. They make sure whenever $(\exists x)\phi(x)$ is true, we have some constant witnessing that it is true.

Lemma 4.3.4. *Let $(T_i)_{i \in I}$ be some collection of finitely consistent theories totally ordered with respect to inclusion. Then $T = \bigcup_{i \in I} T_i$ is finitely consistent.*

Proof. Let $\Sigma = \{\gamma_1, \dots, \gamma_n\}$ be some finite subset of T ; we must show Σ is consistent. For each $1 \leq j \leq n$, $\gamma_j \in T_{i_j}$ for some $i_j \in I$. As we have a total order, there is some $k \in I$ such that $\Sigma \subseteq \bigcup_{j=1}^n T_{i_j} \subseteq T_k$. The theory T_k is finitely consistent by assumption, so Σ is consistent, as required. $\square$

Lemma 4.3.5. *Every finitely consistent theory T can be extended to a finitely consistent Henkin theory T^* such that for all L -sentences ϕ , either $\phi \in T^*$ or $\neg\phi \in T^*$.*

Proof. Set $C_0 = \emptyset$. Now given C_i , we define:

$$C_{i+1} = C_i \cup \{c_{\phi(x)} \mid \phi(x) \text{ is some } L(C_i) \text{ formula}\}$$

Then define our set of witnesses $C = \bigcup_{i < \omega} C_i$. We can define the theory:

$$S = \{((\exists x)\phi(x)) \rightarrow \phi(c_{\phi(x)}) \mid \phi(x) \text{ is some } L(C) \text{ formula}\}$$

The theory S contains constants from C in each of its sentences, so is disjoint from T . We now show $S \sqcup T$ is finitely consistent. Let Σ be some finite subset of $S \sqcup T$. We write $\Sigma = \Sigma_S \sqcup \Sigma_T$. By assumption, T is finitely consistent, so we may find some L -structure A with $A \models \Sigma_T$. Fix $a \in A$ here. We will assign $c^A \in A$ for all $c \in C$ such that A is an $L(C)$ -structure with $A \models S$, and hence $A \models \Sigma_S$. Consider $\phi(x)$ some $L(C)$ -formula; if $A \models (\exists x)\phi(x)$ then by definition there is some $a_{\phi(x)} \in A$ such that $A \models \phi(a_{\phi(x)})$. Otherwise set $c_{\phi(x)}^A = a$. Now $A \models \Sigma_S \sqcup \Sigma_T$, so Σ is consistent, as required.

Now consider X , the set of finitely consistent theories containing $S \sqcup T$, partially ordered by inclusion. Given some chain $(T_i)_{i \in I}$, we see that $\bigcup_{i \in I} T_i$ is an upper bound in X by Lemma 4.3.4. We can then apply Zorn's Lemma (3.0.1) to get some maximal element T^* .

The theory S witnesses that T^* is a Henkin theory, so we now show T^* has the other required property. To that end suppose, looking for a contradiction, there exists ϕ be some L -sentence such that $\phi \notin T^*$ and $\neg\phi \notin T^*$. As T^* is maximal up to inclusion for finitely consistent theories containing $S \sqcup T$, neither of $T^* \cup \{\phi\}$ and $T^* \cup \{\neg\phi\}$ are finitely consistent. We may then find $U_1, U_2 \subseteq T^*$ finite subsets such that $U_1 \cup \{\phi\}$ and $U_2 \cup \{\neg\phi\}$ are inconsistent. Then $U := U_1 \cup U_2$ is a finite subset of T^* , so must have some model $M \models U$. But $U \cup \phi$ and $U \cup \{\neg\phi\}$ are both inconsistent, so $M \not\models \phi$ and $M \not\models \neg\phi$. This is a contradiction, so one of $\phi \in T^*$ and $\neg\phi \in T^*$ must hold. $\square$

We have extended our theory to a Henkin theory with nice properties. We use this to build a model for our theory from the Henkin constants!

Lemma 4.3.6. *Let T^* be a finitely consistent $L(C)$ -Henkin theory such that for each L -sentence ϕ , either $\phi \in T^*$ or $\neg\phi \in T^*$. Then T^* has a countable model.*

Proof. Define a relation on C :

$$c \sim d \text{ if and only if the } L(C)\text{-sentence } c = d \text{ is in } T^*$$

It is clear from the maximality of T^* that $\sim$ is an equivalence relation. We will write $[c]_{\sim}$ for the equivalence class of c with respect to $\sim$. We define an $L(C)$ -structure A as follows:

- $\text{dom}(A) = \{[c]_{\sim} \mid c \in C\}$
- For all relation symbols $R \in L$, $R^A([c_1]_{\sim}, \dots, [c_n]_{\sim})$ if and only if $R(c_1, \dots, c_n) \in T^*$
- For all function symbols $f \in L$, set $f^A([c_1]_{\sim}, \dots, [c_n]_{\sim}) = [c]_{\sim}$ if and only if $f^A(c_1, \dots, c_n) = c \in T^*$

We must check our functions and relations are well-defined. We will do the check for functions, which has an extra step, and leave the check for relations to the reader.

Let $[c_1]_\sim, \dots, [c_n]_\sim \in A$ be given, and let f be an n -ary function symbol. We must show there exists some $c \in C$ such that $f^A([c_1]_\sim, \dots, [c_n]_\sim) = [c]_\sim$. We use that T^* is a Henkin theory. Write

$$\phi(x) : f(c_1, \dots, c_n) = x$$

Now it is clear $(\exists x)\phi(x)$ is true in any $L(C)$ -structure, so must be in T^* . Thus there is some $c_{\phi(x)} \in C$ such that $((\exists x)\phi(x)) \rightarrow \phi(c_{\phi(x)}) \in T^*$. Now

$$((\exists x)\phi(x)) \rightarrow \phi(c_{\phi(x)}), (\exists x)\phi(x) \in T^* \implies \phi(c_{\phi(x)}) \in T^*$$

as T^* is finitely consistent. Referring to the definition of $\phi(x)$, we see

$$f^A([c_1]_\sim, \dots, [c_n]_\sim) = [c_{\phi(x)}]_\sim,$$

as required.

Now we must show that for $[d_1]_\sim, \dots, [d_n]_\sim \in A$ with $[c_1]_\sim = [d_1]_\sim, \dots, [c_n]_\sim = [d_n]_\sim$, we have $f^A([c_1]_\sim, \dots, [c_n]_\sim) = f^A([d_1]_\sim, \dots, [d_n]_\sim)$.

Suppose, looking for a contradiction, that $\neg(f(c_1, \dots, c_n) = f(d_1, \dots, d_n)) \in T^*$. Then

$$\{c_1 = d_1, \dots, c_n = d_n, \neg(f(c_1, \dots, c_n) = f(d_1, \dots, d_n))\}$$

is an inconsistent finite subset of T^* , a finitely consistent theory, giving a contradiction.

Thus $f(c_1, \dots, c_n) \stackrel{(1)}{=} f(d_1, \dots, d_n) \in T^*$. Also certainly $f(d_1, \dots, d_n) \stackrel{(2)}{=} f(d_1, \dots, d_n) \in T^*$ (as T^* is finitely consistent). Now we have

$$f^A([c_1]_\sim, \dots, [c_n]_\sim) \stackrel{\text{by (1)}}{=} [f(c_1, \dots, c_n)]_\sim \stackrel{\text{by (2)}}{=} [f(d_1, \dots, d_n)]_\sim \stackrel{\text{by (2)}}{=} f^A([d_1]_\sim, \dots, [d_n]_\sim)$$

completing our proof that f is well-defined.

After all this trouble, we must now show A is a model of T^* . As T^* is maximal, this amounts to showing:

$$(\clubsuit) \quad A \models \phi \text{ if and only if } \phi \in T^*$$

We exploit the recursive definition of formulae (4.1.14), first proving $(\clubsuit)$ for atomic sentences, then showing that $(\clubsuit)$ is preserved under conjunction, negation, and existential quantification. This technique is commonly called *induction on the complexity of ϕ* .

(Atomic sentences) Here we induct on the number of occurrences of function symbols in ϕ . If ϕ is an atomic sentence with no function symbols, it is of the form $R(c_1, \dots, c_n)$ or $c = d$ for some constant symbols. By construction, $(\clubsuit)$ holds here.

Now let ϕ with at least one function symbol be given and suppose the claim holds for all sentences with less function symbols than ϕ . Then $\phi = \psi(f(c_1, \dots, c_n))$ where $\psi(x)$ is some formula containing less function symbols than ϕ . By our work above, there is some $c \in C$ such that $f(c_1, \dots, c_n) = c \in T^*$. Then

$$A \models \phi \iff A \models \psi(f(c_1, \dots, c_n)) \iff A \models \psi(c)$$

and $\psi(c)$ is a sentence with less function symbols than ϕ , the induction hypothesis gives:

$$A \models \psi(c) \iff \psi(c) \in T^*$$

Now $f(c_1, \dots, c_n) = c \in T^*$. As T^* is finitely consistent, we get

$$\psi(c) \in T^* \iff \psi(f(c_1, \dots, c_n)) \in T^* \iff \phi \in T^*$$

We have built a chain of two way implications witnessing $(\clubsuit)$, closing the induction. The next steps will be easier.

(Conjunctions) Suppose $\phi = \phi_1 \wedge \phi_2$ where $A \models \phi_i \iff \phi_i \in T^*$ for $i \in \{1, 2\}$. Then

$$A \models \phi \iff A \models \phi_1 \text{ and } A \models \phi_2 \iff \phi_1 \in T^* \text{ and } \phi_2 \in T^* \iff \phi_1 \wedge \phi_2 \in T^*$$

where the final two-way implication uses the maximality and finite consistency of T^* .

(Negations) Suppose $\phi = \neg\psi$ where $A \models \psi \iff \psi \in T^*$. Then

$$A \models \phi \iff A \not\models \psi \iff \psi \notin T^* \iff \neg\psi \in T^*$$

where the final two-way implication uses the maximality of T^* .

(Existential quantification) Suppose $\phi = (\exists x)\psi(x)$, where for all constants $c \in C$, $A \models \psi([c]_{\sim}) \iff \psi(c) \in T^*$. Then

$$A \models \phi \iff A \models \psi([c]_{\sim}) \text{ for some } c \in C \iff \psi(c) \in T^* \text{ for some } c \in C$$

and by the maximality and finite consistency of T^* , the final condition above occurs if and only if $(\exists x)\psi(x) \in T^*$, i.e. $\phi \in T^*$.

We have completed our induction on the complexity of ϕ . Thus $A \models T^*$, so T^* is consistent. $\square$

Proof of Theorem 4.3.2. We noted that one direction is immediate. Now suppose T is a finitely consistent L -theory. Use Lemma 4.3.5 to enlarge T to some Henkin theory T^* in an extended language $L(C)$. Now by Lemma 4.3.6, T^* has some model A , and the L -reduct of A is a model of T . Thus T is consistent. $\square$

The compactness theorem is absolutely vital! Taking the contrapositive of one of the directions gives that an inconsistent theory has some finite inconsistent subset, which we use in proofs later on. Compactness admits many nice corollaries similar to the following.

Corollary 4.3.7. *Every group of unbounded exponent is elementarily equivalent to a group with elements of infinite order. ([8], Exercise 2.1.6)*

Proof. Let G be a group with unbounded exponent, viewed as an L -structure in L the standard language of groups. Add some new constant c , and define the $L(c)$ theory:

$$T = Th_L(G) \cup \{c^n \neq 1 \mid n \in \mathbb{N}\}$$

Now for every finite subset $\Sigma \subset T$ we have some assignment of the constant c to an element in G such that $G \models \Sigma$, so T is finitely consistent. By the Compactness Theorem, there exists some $L(C)$ -structure H with $H \models T$. Then $c^H \in H$ has $(c^H)^n \neq 1$ for all $n \in \mathbb{N}$, so must have infinite order. Now consider the L -reduct of H and note $Th_L(G) \subseteq Th_L(H)$ by construction, so by Lemma 4.2.5 $G \equiv_L H$ as required. $\square$

One can show by very similar means that if any theory has arbitrarily large finite models, it has an infinite model ([27], Exercise 2.5.2). In Theorem 4.4.1, we show that when we restrict our attention to abelian groups we can get even stronger results.

Theorem 4.3.8. *(Downward Löwenheim-Skolem Theorem) If a consistent L -theory T has an infinite model, then it has a countable model.*

If L is uncountable, this proof is a bit more involved. We will give a proof where L is at most countable:

Proof. In the proof of Compactness Theorem (4.3.2), we built a model from some set of constants in bijection with the set of L -formulae. If L is countable, from the recursive definition of formulae (see Definition 4.1.14) so is the set of L -formulae. Thus if L is countable, we built a countable model. $\square$

With more work one can show something even stronger than above, i.e. that every infinite model of T is an elementary extension of some countable model of T .

Theorem 4.3.9. (*Upward Löwenheim-Skolem Theorem*) *If a consistent L -theory T has an infinite model, it has an arbitrarily large model.*

Proof. Let κ be some infinite cardinal, then choose C some set of constant symbols not in the language L such that $|C| = \kappa$, and extend to $L(C)$. Now consider the theory

$$T' := T \cup \{c \neq d \mid c, d \in C\}$$

Now let A be some infinite model for T . For any finite subset $\Sigma \subseteq T'$, the theory Σ contains only finitely many constant symbols $c_1, \dots, c_n$. Assigning $c_1^A, \dots, c_n^A$ to be different elements of A (which we may always do as A is infinite), we see $A \models \Sigma$. We have shown T' is finitely consistent, so it has some model $M \models T'$. The sentences we added to T witness that M has cardinality at least κ . $\square$

Both Löwenheim-Skolem theorems warrant serious reflection. In fact, Skolem didn't believe in the upward direction for philosophical reasons [20].

4.4. A Group Theoretic Corollary of Compactness. We continue some of our work on elementary equivalence we began in Section 4.2 but with the new firepower of the compactness theorem, and some other logical techniques which we will introduce along the way.

Each result we prove will attest to the weakness of elementary equivalence and first-order logic in distinguishing between very different groups. Then in Chapter 6 we turn our attention to the special case where elementary equivalence is strong.

This section is devoted to the proof of following interesting theorem, which extends Corollary 4.3.7 in the case of abelian groups.

Theorem 4.4.1. *Let A be an abelian group. The following are equivalent:*

- (1) A has unbounded exponent
- (2) $A \equiv A \oplus \mathbb{Q}$
- (3) $A \equiv A \oplus (\bigoplus_{i < \omega} \mathbb{Q})$

We are following a proof of Hodges [20] and filling in some extra details. We need some technology. First, we note direct sums behave well with respect to elementary equivalence.

Lemma 4.4.2. *Let G_1, G_2, H be groups with $G_1 \equiv G_2$. Then $G_1 \oplus H \equiv G_2 \oplus H$*

It would be too far a diversion to prove the lemma here. Hodges [20] uses Ehrenfeucht-Fraïssé games, but it is also a nice corollary of the Keisler-Shelah Theorem for ultraproducts (see Chang and Keisler [8]).

We will also need a result on the divisible subgroups of abelian groups, which is Theorem 21.2 in Fuchs [17].

Theorem 4.4.3. *A divisible subgroup D of a group A is a direct summand, i.e.*

$$A = D \oplus C$$

for some $C \leq A$

Finally we need a basic result giving us solutions to inequations in $\mathbb{Z}$, which can be proved by induction.

Lemma 4.4.4. *Let $l \in \mathbb{N}$, I be some finite set, and $\alpha_{i,j} \in \mathbb{Z}$ for all $i \in I$ and $1 \leq j \leq l$. Then the set of inequations*

$$\left\{ \sum_{j=1}^l \alpha_{i,j} x_j \neq 0 \mid i \in I \right\}$$

has some integer solutions for $x_1, \dots, x_l$.

The three bits of technology we have noted so far are very different. We are now ready to piece them together into a proof of the theorem.

Proof of Theorem 4.4.1. Certainly (3) and (2) each individually imply (1). We first show that (1) implies (3)

Write L for the language of group theory. For each element of $\mathbb{Q}^{(\omega)}$, add a new constant to the language to get $L(\mathbb{Q}^{(\omega)})$. Now define

$$T_{\mathbb{Q}^{(\omega)}} = \{ \phi \mid \phi \text{ is a closed literal in } L(\mathbb{Q}^{(\omega)}) \text{ such that } \mathbb{Q}^{(\omega)} \models \phi \}.$$

Similarly, for each element of A , add a new constant to the language L to get $L(A)$, and recall $\text{eldiag}(A)$ from the Elementary Diagram Lemma (4.2.10). Then writing $L' := L(A \sqcup \mathbb{Q}^{(\omega)})$, define the L' -theory

$$T := T_{\mathbb{Q}^{(\omega)}} \sqcup \text{eldiag}(A).$$

We will show that T has a model, and then show that any model of T is isomorphic to $B \oplus \mathbb{Q}^{(\omega)}$ for some group B .

Suppose, looking for a contradiction T is inconsistent (the argument could be done without the contradiction, but it would not flow by so nicely). Then by the Compactness Theorem (4.3.2), we may find Σ some finite inconsistent subset of T , and write

$$\Sigma = \Sigma_{\mathbb{Q}^{(\omega)}} \sqcup \Sigma_A$$

where $\Sigma_{\mathbb{Q}^{(\omega)}} \subseteq T_{\mathbb{Q}^{(\omega)}}$ and $\Sigma_A \subseteq \text{eldiag}(A)$. Now $\Sigma_{\mathbb{Q}^{(\omega)}}$ is finite, so features only finitely many elements of $u_1, \dots, u_k \in \mathbb{Q}^{(\omega)}$. By Theorem 3.0.3, these elements form some subgroup isomorphic to $\mathbb{Z}^n$, generated by $q_1, \dots, q_n$ (write $\bar{q}$ for the tuple of these elements). Note for all $1 \leq i \leq k$, we can write $u_i = t_i(\bar{q})$ for some term $t_i(\bar{x})$ over L . Consider the theory

$$T_{\bar{q}} = \{ \phi(\bar{q}) \mid \phi(\bar{q}) \text{ is a closed literal in } L(\bar{q}) \text{ and } \mathbb{Q}^{(\omega)} \models \phi \}.$$

If the L' -theory $\Sigma_A \sqcup T_{\bar{q}}$ were consistent with a model M , then the corresponding assignment of constants would have

$$u_i^M = t_i(q_1^M, \dots, q_n^M),$$

and would force M to be a model for $\Sigma_A \sqcup \Sigma_{\mathbb{Q}^{(\omega)}}$, an inconsistent theory.

Thus $\Sigma_A \sqcup T_{\bar{q}}$ must be inconsistent. Applying compactness once more, we may find some finite subset $\Sigma_{\bar{q}} \subseteq T_{\bar{q}}$ such that $\Sigma_A \sqcup \Sigma_{\bar{q}}$ is inconsistent.

We will show that $A \models \Sigma_A \sqcup \Sigma_{\bar{q}}$, which will give a contradiction. We must choose some assignment of the constants $a \in A$ and $x \in \mathbb{Q}^{(\omega)}$ which witnesses the above satisfaction.

As $\Sigma_{\bar{q}}$ contains only closed literals, we know up to equivalence it has the form:

$$\Sigma_{\bar{q}} = \left\{ \sum_{j=1}^n \alpha_{i,j} q_j \neq 0 \mid i \in I_1 \right\} \sqcup \left\{ \sum_{j=1}^n \beta_{i,j} q_j = 0 \mid i \in I_2 \right\}$$

where I_1 and I_2 are some finite sets. We shall see below that we need only worry about satisfying the inequations. By Lemma 4.4.4, we may find some integers $z_1, \dots, z_n$ such

that $\sum_{j=1}^n \alpha_{i,j} z_j \neq 0$ is a valid inequation in $\mathbb{Z}$. Then as A has unbounded exponent, we may find some $c \in A$ with order greater than

$$\max\left\{\left|\sum_{j=1}^n \alpha_{i,j} z_j\right| \mid i \in I\right\}.$$

Now by our choice of $c \in A$, in the group A we have, for all $i \in I$:

$$\sum_{j=1}^n \alpha_{i,j} (z_j c) = \left(\sum_{j=1}^n \alpha_{i,j} z_j\right) c \neq 0.$$

We are now ready to assign our constants to elements of A . Set $q_j^A = z_j c$ for each $1 \leq j \leq n$, set $a^A = a$ for all $a \in A$, and for all $x \in \mathbb{Q}^{(\omega)} \setminus \{q_1, \dots, q_n\}$, set $x^A = 0$. Now certainly $A \models \Sigma_A$, as all of the constants in A were assigned in the usual way. By our work above, A satisfies all of the inequations in $\Sigma_{\bar{q}}$.

Finally, the equations in $\Sigma_{\bar{q}}$ are those true of $\bar{q}$, which generate $\mathbb{Z}^n$, a free abelian group. Some basic combinatorial group theory tells us that any equation involving some minimal generating set for a free abelian group is true in all abelian groups. As A is an abelian group, we then get A satisfies all of the equations in $\Sigma_{\bar{q}}$. We have shown $A \models \Sigma_A \sqcup \Sigma_{\bar{q}}$, which gives the required contradiction. This contradiction stemmed from the assumption T is inconsistent, so T must be consistent.

As T is consistent we may find some $M \models T$. As $T \supseteq \text{eldiag}(A)$, we have

$$(*) \quad A \equiv M.$$

Applying Lemma 4.4.2, we get

$$(**) \quad M \oplus \mathbb{Q}^{(\omega)} \equiv A \oplus \mathbb{Q}^{(\omega)}.$$

Recalling Example 4.1.23, as $M \models T_{\mathbb{Q}^{(\omega)}}$, there is some subgroup of M isomorphic to $\mathbb{Q}^{(\omega)}$. Now applying Theorem 4.4.3, we see

$$M \cong K \oplus \mathbb{Q}^{(\omega)}$$

for some $K \leq M$. Now noting $\mathbb{Q}^{(\omega)} \cong \mathbb{Q}^{(\omega)} \oplus \mathbb{Q}^{(\omega)}$ (a standard property of infinite direct sums), we put everything into one slick chain of equivalences

$$\begin{aligned} A &\stackrel{(*)}{\equiv} M \cong K \oplus \mathbb{Q}^{(\omega)} \cong K \oplus \mathbb{Q}^{(\omega)} \oplus \mathbb{Q}^{(\omega)} \\ &\cong M \oplus \mathbb{Q}^{(\omega)} \stackrel{(**)}{\equiv} A \oplus \mathbb{Q}^{(\omega)}. \end{aligned}$$

We have completed the proof that (1) implies (3).

Now (3) implies (2) quickly: Assume $A \equiv A \oplus \mathbb{Q}^{(\omega)}$. Taking a similar strategy to above, we see

$$A \equiv A \oplus \mathbb{Q}^{(\omega)} \equiv (A \oplus \mathbb{Q}^{(\omega)}) \oplus \mathbb{Q} \equiv A \oplus \mathbb{Q}$$

which gives (2). □

This is a striking result, giving for example that

$$\mathbb{Z} \equiv \mathbb{Z} \oplus \mathbb{Q}^{(\omega)}.$$

These two groups have a very different subgroup structure. In $\mathbb{Z}$ any two nontrivial subgroups have infinite intersection, while in $\mathbb{Z} \oplus \mathbb{Q}$ we have an infinite family of trivially intersecting normal subgroups. We can state the following as corollary.

Corollary 4.4.5. *Being a direct sum of a set of subgroups is not first-order expressible in the natural language of group theory.*

The natural language of group theory can always detect the centre of a group H , as this is exactly those elements satisfying the formula $\phi(x) : (\forall y)(xy = yx)$. If we added an element $h \in H$ to the language, we could similarly define the centralizer of h in H . One of the main research programmes in model theory is, given a class of structures, find the substructures which are definable (with some parameters) in a given language.

4.5. Quantifier Elimination. We begin with a motivating example. Consider the conjugacy problem in a group G . It asks, given $g, h \in G$, is it true that

$$G \models (\exists x)g^x = h?$$

This has only one quantifier, and for some groups (even some well-behaved ones [4]) this problem is unsolvable. However, if we know some things about the group, this question can become easy. For example:

$$Q_8 \models (\exists x)g^x = h \iff Q_8 \models (g = h) \vee (g^3 = h),$$

$$\text{If } G \text{ is abelian, then } G \models (\exists x)g^x = h \iff G \models g = h.$$

Roughly speaking, in these cases as we know some of the theory of the group, the conjugacy problem can be resolved 'locally'. If in all models of some theory we could resolve all such problems locally, the theory would be easy to understand.

Definition 4.5.1. We say an L -theory T has *quantifier elimination* if each L -formula $\phi(\bar{x})$ is equivalent to some quantifier-free L -formula $\psi(\bar{x})$ modulo T

One of the earliest quantifier elimination results is that in the natural language of fields

$$\{0, 1, +, -, \cdot, ^{-1}\},$$

the theory of algebraically closed fields has quantifier elimination (see [27], Chapter 3). The following lemma is one of the reasons quantifier elimination is a sought-after property. An analogue for propositional logic was proved in MT5867, so we omit a proof here.

Lemma 4.5.2. (*Disjunctive Normal Form*) Let ϕ be some L -sentence. Then ϕ is equivalent to some formula

$$\psi : \bigvee_{i=1}^n \bigwedge_{j=1}^m \theta_{i,j},$$

where each $\theta_{i,j}$ is a closed literal.

The following result says (hopefully intuitively) that to eliminate all quantifiers, we can proceed one quantifier at a time. The proof is an induction which is better written than read.

Lemma 4.5.3. Let T be an L -theory. Suppose for each quantifier-free L -formula $\phi(\bar{x}, y)$ that there exists some $\psi(\bar{x})$ such that $(\exists y)\phi(\bar{x}, y) \equiv_T \psi(\bar{x})$. Then T has quantifier elimination.

For this section, we will need some notation for a sentence which is always true and a sentence which is always false. Define

$$\top : (\forall x)x = x \text{ and } \perp : \neg \top.$$

Very importantly, quantifier elimination applies to sentences. If T has quantifier elimination, then every sentence is equivalent to some sentence with no variables modulo T . In a relational structure, this means every sentence is equivalent to either $\top$ or $\perp$, so any theory with quantifier elimination is complete.

In a structure with constants, things are more complicated; for example in the language of fields the sentence describing characteristic n :

$$\sum_{i=1}^n 1 = 0$$

is a quantifier-free sentence. While the theory of algebraically closed fields has quantifier elimination, those sentences describing characteristic n witness that the theory is not complete.

The following lemma says we do not have the same worries in the language of groups.

Lemma 4.5.4. *Suppose L is the language of groups. If T is an L -theory with quantifier elimination, then T is complete.*

Proof. As T has quantifier elimination, we need only show $T \vdash \psi$ or $T \vdash \neg\psi$ for all quantifier-free sentences ψ . By Lemma 4.5.2, we may assume ψ is in disjunctive normal form:

$$\psi : \bigvee_{i \in I} \bigwedge_{j \in J} \theta_{i,j}$$

where $\theta_{i,j}$ is an atomic formula or the negation of an atomic formula. As we are in the language of groups, and $\theta_{i,j}$ must be quantifier free, we know

$$\theta_{i,j} : 1^n = 1^m \quad \text{or} \quad \theta_{i,j} : \neg(1^n = 1^m)$$

for some $n, m \in \mathbb{Z}$. Then immediately $\theta_{i,j} \equiv_T \perp$ or $\theta_{i,j} \equiv_T \top$. Using the recursive definition of valuations (4.1.16) we see $\psi \equiv_T \perp$ or $\psi \equiv_T \top$. If $\psi \equiv_T \perp$, then $T \vdash \neg\psi$ and if $\psi \equiv_T \top$ then $T \vdash \psi$. $\square$

Theorem 4.5.5. *Any two torsion-free divisible abelian groups are elementarily equivalent*

In particular this will give us that $\mathbb{R} \equiv \mathbb{Q} \equiv \mathbb{Q}^{(\omega)}$. Our strategy is to build a theory whose models are exactly the torsion-free divisible abelian groups, then show it is complete. For all $n \in \mathbb{N}$ define

$$\text{div}_n : (\forall x)(\exists y)(x = ny),$$

$$\text{tfree}_n : (\forall x)(nx = 0 \rightarrow x = 0),$$

$$T := \{\text{div}_n \mid n \in \mathbb{N}\} \cup \{\text{tfree}_n \mid n \in \mathbb{N}\} \cup \{(\forall x)(\forall y)(x + y = y + x)\}.$$

The models for T are exactly the torsion-free divisible abelian groups. Recalling Lemma 4.2.4, we will be done if we can show the theory T is complete. We will apply the method of quantifier elimination. This proof is original. Marker [27] has a different proof using a lot of technology.

Lemma 4.5.6. *T as defined above has quantifier elimination*

Proof. We will show T satisfies the conditions in Lemma 4.5.3. Let $\bar{x} = (x_1, \dots, x_n)$. To apply Lemma 4.5.3 we must also consider the case where $\bar{x}$ is some empty tuple. This case is identical, but would make our notation more laboured, so we will assume $\bar{x}$ is non-empty. Let $\psi(\bar{x}) = (\exists y)\phi(\bar{x}, y)$ where $\phi(\bar{x})$ is a quantifier-free formula. Now by Lemma 4.5.2, $\phi(\bar{x}, y)$ is equivalent to some formula in disjunctive normal form

$$\bigvee_{i \in I} \bigwedge_{j \in J} \theta_{i,j}(\bar{x}, y),$$

where $\theta_{i,j}$ is a term or the negation of a term, and where I and J are finite index sets. Then one can check

$$(\exists y) \bigvee_{i \in I} \bigwedge_{j \in J} \theta_{i,j}(\bar{x}, y) \equiv \bigvee_{i \in I} (\exists y) \left(\bigwedge_{j \in J} \theta_{i,j}(\bar{x}, y) \right)$$

As we are working $\{0, +, -\}$, the language of abelian groups, we know what terms look like. Up to equivalence we can write

$$\theta_{i,j}(\bar{x}, y) : \left(\sum_{k=1}^n \alpha_{i,j,k} x_k = \beta_{i,j} y \right) \wedge \left(\sum_{k=1}^n \alpha'_{i,j,k} x_k \neq \beta'_{i,j} y \right)$$

We will be done if we can show that for each i the formula $(\exists y) \bigwedge_{j \in J} \theta_{i,j}(\bar{x}, y)$ is equivalent to some quantifier-free formula modulo T . Fix i here. There is some casework to be done.

(Case 1: If $\beta_{i,q} \neq 0$ for some $q \in J$) then $\theta_{i,j}(\bar{x}, y)$ forces $\beta_{i,q} y \stackrel{\spadesuit}{=} \sum_{k=1}^n \alpha_{i,q,k} x_k$. As $\beta_{i,q} \neq 0$, and as T encodes that elements are torsion-free, we can multiply both sides of each term by $\beta_{i,q}$ while preserving equivalence modulo T .

$$\begin{aligned} (\exists y) \left(\bigwedge_{j \in J} \theta_{i,j}(\bar{x}, y) \right) &\equiv_T \bigwedge_{j \in J} \left(\beta_{i,q} \sum_{k=1}^n \alpha_{i,j,k} x_k = \beta_{i,q} \beta_{i,j} y \right) \wedge \left(\beta_{i,q} \sum_{k=1}^n \alpha'_{i,j,k} x_k \neq \beta_{i,q} \beta'_{i,j} y \right) \\ &\stackrel{\spadesuit}{=} \bigwedge_{j \in J} \left(\beta_{i,q} \sum_{k=1}^n \alpha_{i,j,k} x_k = \beta_{i,j} \sum_{k=1}^n \alpha_{i,q,k} x_k \right) \wedge \left(\beta_{i,q} \sum_{k=1}^n \alpha'_{i,j,k} x_k \neq \beta'_{i,j} \sum_{k=1}^n \alpha_{i,q,k} x_k \right) \end{aligned}$$

which is a quantifier-free formula over $\bar{x}$.

(Case 2: If $\beta_{i,j} = \beta'_{i,j} = 0$ for all $j \in J$) Consider Z , the set of indexes $j \in J$ where $\beta'_{i,j} = 0$. Reorganising our formulae, we write:

$$\begin{aligned} \bigwedge_{j \in J} \theta_{i,j}(\bar{x}, y) &\equiv_T \\ &\bigwedge_{j \in J} \left(\sum_{k=1}^n \alpha_{i,j,k} x_k = 0 \right) \wedge \bigwedge_{j \in Z} \left(\sum_{k=1}^n \alpha'_{i,j,k} x_k \neq 0 \right) \wedge \bigwedge_{j \in J \setminus Z} \left(\sum_{k=1}^n \alpha'_{i,j,k} x_k \neq \beta'_{i,j} y \right) \end{aligned}$$

As many of the terms no longer feature y we can pass the quantifier through some of the conjunctions:

$$\begin{aligned} (\exists y) \left(\bigwedge_{j \in J} \theta_{i,j}(\bar{x}, y) \right) &\equiv_T \\ &\bigwedge_{j \in J} \left(\sum_{k=1}^n \alpha_{i,j,k} x_k = 0 \right) \wedge \bigwedge_{j \in Z} \left(\sum_{k=1}^n \alpha'_{i,j,k} x_k \neq 0 \right) \wedge (\exists y) \left(\bigwedge_{j \in J \setminus Z} \left(\sum_{k=1}^n \alpha'_{i,j,k} x_k \neq \beta'_{i,j} y \right) \right) \end{aligned}$$

So we have reduced our task to showing that the following is equivalent to some quantifier-free formula modulo T :

$$(\heartsuit) \quad (\exists y) \left(\bigwedge_{j \in J \setminus Z} \left(\sum_{k=1}^n \alpha'_{i,j,k} x_k \neq \beta'_{i,j} y \right) \right)$$

But any model $G \models T$ is an infinite group! For some fixed assignment $g_m \mapsto x_m$ for each $1 \leq m \leq n$, as J is finite, we may choose $g \notin \left\{ \sum_{k=1}^n \frac{\alpha'_{i,j,k}}{\beta_{i,j}} g_k \mid j \in J \right\}$, giving:

$$\begin{aligned} G &\models \bigwedge_{j \in J \setminus Z} \left(\sum_{k=1}^n \alpha'_{i,j,k} g_k \neq \beta'_{i,j} g \right) \\ &\implies G \models (\exists y) \left(\bigwedge_{j \in J \setminus Z} \left(\sum_{k=1}^n \alpha'_{i,j,k} g_k \neq \beta'_{i,j} y \right) \right) \end{aligned}$$

But $g_m \mapsto x_m$ was some arbitrary assignment of elements of G , so

$$G \models (\forall x_1) \dots (\forall x_n) (\exists y) \left(\bigwedge_{j \in J \setminus Z} \left(\sum_{k=1}^n \alpha'_{i,j,k} x_k \neq \beta'_{i,j} y \right) \right)$$

But G was an arbitrary model of T , so in fact

$$T \vdash (\forall x_1) \dots (\forall x_n) (\exists y) \left(\bigwedge_{j \in J \setminus Z} \left(\sum_{k=1}^n \alpha'_{i,j,k} x_k \neq \beta'_{i,j} y \right) \right)$$

So up to equivalence modulo T , the formula in $(\heartsuit)$ can be removed, giving:

$$(\exists y) \left(\bigwedge_{j \in J} \theta_{i,j}(\bar{x}, y) \right) \equiv_T \bigwedge_{j \in J} \left(\sum_{k=1}^n \alpha_{i,j,k} x_k = 0 \right) \wedge \bigwedge_{j \in Z} \left(\sum_{k=1}^n \alpha'_{i,j,k} x_k \neq 0 \right)$$

which is quantifier-free. We have shown Lemma 4.5.3 applies to T , so T has quantifier elimination. $\square$

Proof of Theorem 4.5.5. By construction, the models of T are exactly the torsion-free divisible abelian groups. By combining Lemma 4.5.4 and Lemma 4.5.6, we see T is complete, so all models of T (i.e. all torsion-free divisible abelian groups) are elementarily equivalent by Lemma 4.2.4. $\square$

Divisible abelian groups will return throughout the project as useful examples for many of our techniques. Quantifier elimination will also return in Chapter 7.

We finish the chapter with of the great recent applications of the method quantifier elimination. The theorem below was originally conjectured by Tarski in 1945, and proved by Sela [37] in 2006. The proof is well beyond the scope of this project, but the main idea is showing that for T the largest theory shared by all finitely generated non-abelian free groups, every formula $\phi(\bar{x}, \bar{y}, \bar{z})$ is equivalent modulo T to some formula $(\forall \bar{x})(\exists \bar{y})\psi(\bar{x}, \bar{y}, \bar{z})$ where $\psi(\bar{x}, \bar{y}, \bar{z})$ is a quantifier free formula.

Theorem 4.5.7. *Let F, G be two finitely generated non-abelian free groups, then $F \equiv G$.*

5. FRAÏSSÉ THEORY

So far our main invariant of structures we have studied is the first-order theory, which naturally lead to some logic-heavy inquiries.

In this chapter, we proceed without much first-order logic at all; We look at the finitely generated substructures of a given structure, and the control they exert over the structure. We will then see in Chapter 7 that there are plenty of interesting logical properties beneath the surface in Fraïssé theory.

5.1. Age and Homogeneity.

Definition 5.1.1. For an L -structure D we define the *age* of D , write $\text{age}(D)$, for the class of finitely generated structures which embed into D .

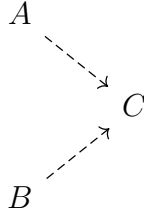
In practice, we will consider the age as those isomorphism classes of structures which embed into D . For example, we have no qualms with saying the quaternion group Q_8 has age $\{1, C_2, C_4, Q_8\}$.

In relational languages, finitely generated structures are finite. Some authors, such as Macpherson [26], will consider the age of D as the class of finite structures which embed into D , even when working on groups!

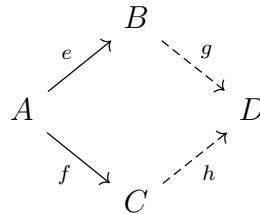
It will turn out that a class of structures being the age of another structure is determined by only two properties, and if we force only one more property (which we call the amalgamation property) then age becomes a rigid property for countable structures.

Definition 5.1.2. Let $\mathbf{K}$ be a class of L -structures. We say $\mathbf{K}$ has the:

- Hereditary property (HP) if for all $A \in \mathbf{K}$, B a finitely generated substructure of A , we have $B \in \mathbf{K}$
- Joint embedding property (JEP) if for all $A, B \in \mathbf{K}$ there exists $C \in \mathbf{K}$ such that $A \hookrightarrow C, B \hookrightarrow C$.



- Amalgamation property (AP) if for all $A, B, C \in \mathbf{K}$ with $e : A \hookrightarrow B, f : A \hookrightarrow C$ embeddings, there exists $D \in \mathbf{K}$ and $g : B \hookrightarrow D, h : C \hookrightarrow D$ embeddings such that $ge = hf$



We will call the above setup involving (A, B, C, e, f) an amalgam, and say (D, g, h) completes the amalgam.

If $\mathbf{K}$ has (HP), (JEP), and (AP) we will call $\mathbf{K}$ a Fraïssé class. It will become clear shortly why we give such a class this name.

There is some subtlety to these properties. We fill out Figure 2 and give examples isolating (HP), (JEP), (AP). We will need some lemmas along the way. At first glance, one might think (AP) implies (JEP), but this is not quite true.

Lemma 5.1.3. *The class of finite fields $\mathbf{FF}$ has (AP), (HP), but not (JEP)*

Proof. In the natural language of fields, the theory of a field of characteristic p contains the atomic sentence

$$\text{char}_p : \sum_{i=1}^p 1 = 0$$

and the atomic sentences $\neg \text{char}_q$ for all primes $q \neq p$. Considering Lemma 4.1.19, embeddings of finite fields can only be between fields of the same characteristic, so $\mathbb{F}_2$ and $\mathbb{F}_3$ cannot jointly embed into another finite field. We have shown $\mathbf{FF}$ does not have (JEP).

The finitely generated substructures of a finite field are its subfields, which must also be finite, so $\mathbf{FF}$ has (HP).

To see $\mathbf{FF}$ has (AP), let some amalgam (A, B, C, e, f) be given where A, B, C are finite fields and $e : A \hookrightarrow B$, $f : A \hookrightarrow C$ are embeddings. Now by our discussion above, A, B , and C must have the same characteristic p . Finite fields have a unique subfield of a given order.

Choose some k such that $p^k \geq |B|, |C|$, and define the embeddings:

$$g : B \hookrightarrow \mathbb{F}_{p^k} \text{ where } g(B) \text{ is the unique subfield of } \mathbb{F}_{p^k} \text{ of order } |B|,$$

$$h : C \hookrightarrow \mathbb{F}_{p^k} \text{ where } h(C) \text{ is the unique subfield of } \mathbb{F}_{p^k} \text{ of order } |C|.$$

Now $ge(A)$ and $fh(A)$ are both subfields of $\mathbb{F}_{p^k}$ of order $|A|$, so must be equal. We have completed the arbitrary amalgam (A, B, C, e, f) , so $\mathbf{FF}$ has (AP) $\square$

With one condition, we can make (AP) imply (JEP). In fact, this is similar to the condition which makes quantifier elimination of a theory imply completeness.

Lemma 5.1.4. ([20], Section 7.1, Exercise 1) *If $\mathbf{K}$ is a class of L -structures with $A \in \mathbf{K}$ such that A embeds into every element of $\mathbf{K}$. Then $\mathbf{K}$ has (AP) implies $\mathbf{K}$ has (JEP).*

Proof. Suppose $\mathbf{K}$ has (AP). For all $B, C \in \mathbf{K}$, we have the embeddings $e : A \hookrightarrow B$ and $f : A \hookrightarrow C$. But then by (AP), some $D \in \mathbf{K}$ completes this amalgam with embeddings $g : B \hookrightarrow D$ and $h : C \hookrightarrow D$. Thus B and C jointly embed into some element of $\mathbf{K}$. $\square$

Then $\mathbf{FF} \setminus \{\mathbb{F}_2\}$ retains (AP), but as $\mathbb{F}_2$ embeds in $\mathbb{F}_4$ this does not have (HP).

We show (HP) need not follow from the other conditions: Consider the rather trivial set of groups $\mathbf{C}_2 = \{C_2\}$. That there is only one element forces (AP), (JEP) to hold, but C_2 has the substructure 1 which is not in $\mathbf{C}_2$, so (HP) does not hold.

We will show later in Lemma 5.4.1 that the class of finite groups $\mathbf{FG}$ is a Fraïssé class. When we start putting restrictions on our finite groups, we may lose amalgamation. We follow Leinen [25] in the following proof.

Lemma 5.1.5. *The class of finite soluble groups $\mathbf{FS}$ has (HP), (JEP), but not (AP).*

Proof. Solubility is preserved by subgroups, giving (HP). Two finite soluble groups jointly embed into their direct product, which is also soluble, giving (JEP).

To show the class of finite soluble groups doesn't have AP, we exhibit some amalgam which cannot be completed. First note the general linear group $GL(3, 2)$ is a non-abelian simple group (so insoluble), and is generated by two elements α, β . Now consider

$$U = C_2 \oplus C_2 \oplus C_2,$$

which admits a natural action from its automorphism group (which we identify with $GL(3, 2)$ by an MT5864 exercise). Consider the following amalgam of inclusions:

$$\begin{array}{ccc} & & U \rtimes \langle \alpha \rangle \\ & \nearrow & \\ U & & \\ & \searrow & \\ & & U \rtimes \langle \beta \rangle \end{array}$$

To see $U \rtimes \langle \alpha \rangle$ is soluble, consider the commutator:

$$\begin{aligned} [u\alpha^i, v\alpha^j] &= (u\alpha^i)^{-1}(v\alpha^j)^{-1}u\alpha^i v\alpha^j = \alpha^{-i}u^{-1}\alpha^{-j}v^{-1}u\alpha^i v\alpha^j \\ &= u\alpha^i \alpha^{-i-j}v^{-1}uv\alpha^{-i} \alpha^{i+j} = u\alpha^i (v^{-1}uv\alpha^{-i})^{\alpha^{i+j}} \in U \end{aligned}$$

so $(U \rtimes \langle \alpha \rangle)' \subseteq U$. As U is abelian this gives $U \rtimes \langle \alpha \rangle$ soluble of derived length at most 2. A similar argument shows $U \rtimes \langle \beta \rangle$ is soluble.

Now suppose, looking for a contradiction, that some soluble group G completes the amalgam with embeddings $U \rtimes \langle \alpha \rangle \rightarrow G$ and $U \rtimes \langle \beta \rangle \rightarrow G$. To save ink, we may assume these are inclusions. We claim $N_G(U)/C_G(U)$ is isomorphic to $GL(3, 2)$, and hence simple. There is a natural map:

$$f : N_G(U) \rightarrow GL(3, 2) \text{ where } w \text{ is sent to the map } u \mapsto u^w$$

This map f has kernel $C_G(U)$. As $\alpha, \beta \in G$, we have that for every $\gamma \in GL(3, 2)$, we can write γ as some term $t(\alpha, \beta)$. Then for all $u \in U$, $u^{t(\alpha, \beta)} = u^\gamma$, and $\alpha, \beta \in N_G(U)$, so $t(\alpha, \beta) \in N_G(U)$. Then $(t(\alpha, \beta))f = \gamma$, and as γ was arbitrary our map f is surjective. By the first isomorphism theorem $N_G(U)/C_G(U) \cong GL(3, 2)$, as required.

Now solubility is closed under taking subgroups, so $N_G(U) \leq G$ must be soluble. Solubility is also closed under taking quotients, so $N_G(U)/C_G(U)$ is soluble. But we showed $N_G(U)/C_G(U)$ is a non-abelian simple group, so cannot be soluble. We arrived at a contradiction, so there can be no soluble group G completing the diagram above, and the class of finite soluble groups cannot have (AP). $\square$

Then considering $\mathbf{FS} \setminus \{1\}$, we see (JEP) is preserved but (HP) is lost.

To get (HP) but neither (JEP) nor (AP), we consider

$$\mathbf{X} = \{1, C_2, C_2 \oplus C_2, C_4\}.$$

This has (HP), but C_4 and $C_2 \oplus C_2$ cannot jointly embed into any element of $\mathbf{X}$, so (JEP) does not hold. As groups have the identity embedding into X has the identity embedding into every element of $\mathbf{X}$, Lemma 5.1.4 tells us (AP) does not hold in $\mathbf{X}$ either. We have done all the work required to fill in Figure 2.

It turns out (HP) and (JEP) completely capture whether or not a class is an age.

Lemma 5.1.6. *Let L be some countable or finite language. A countable/finite class $\mathbf{K}$ of finitely generated L -structures has (HP) and (JEP) if and only if there exists some countable L -structure A such that $\text{age}(A) = \mathbf{K}$.*

Proof. First let A be an L -structure. We know by Lemma 4.1.7 that substructures are inherited, so $\text{age}(A)$ has (HP). Also, given $B = \langle b_1, \dots, b_n \rangle$ and $C = \langle c_1, \dots, c_m \rangle$ in $\text{age}(A)$, B and C jointly embed in $\langle b_1, \dots, b_n, c_1, \dots, c_m \rangle$. Thus (JEP) holds, and we have proven one direction.

For the other direction, suppose $\mathbf{K}$ is some class with (HP) and (JEP). We assume $\mathbf{K}$ is countable, and enumerate its elements $A_1, A_2, \dots$. Now we inductively define

$$B_1 \leq B_2 \leq \dots$$

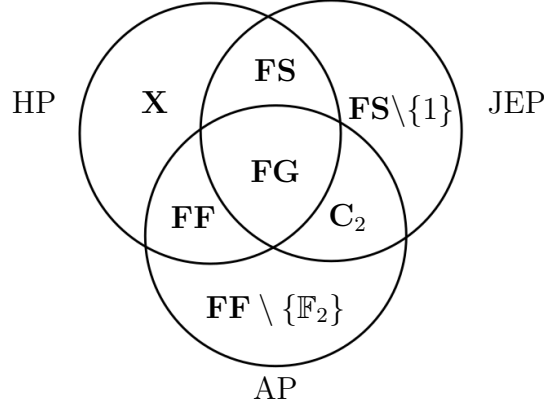


FIGURE 2. Venn diagram of HP, JEP, and AP

such that $B_i \in \mathbf{K}$ and $\{A_1, \dots, A_i\} \subseteq \text{age}(B_i) \subseteq \mathbf{K}$ for each $i < \omega$: Set $B_1 = A_1$. Now given B_i such that $A_1, \dots, A_i$ all embed into B_i , use (JEP) to find some $B_{i+1} \in \mathbf{K}$ which admits an embedding from both B_i and A_{i+1} . Without loss of generality we may assume B_i is a genuine substructure of B_{i+1} . Then by (HP), each of $A_1, \dots, A_i$ embed into B_{i+1} , and A_{i+1} does so by construction. The induction loop is now closed.

Set $B = \bigcup_{i < \omega} B_i$. By construction, $\mathbf{K} \subseteq \text{age}(B)$. For the other direction, if $b_1, \dots, b_n \in B$, then there exists some $k < \omega$ such that $b_1, \dots, b_n \in B_k \in \mathbf{K}$, giving that $\langle b_1, \dots, b_n \rangle \in \mathbf{K}$ by (HP). Thus $\text{age}(B) = \mathbf{K}$, and B is a countable union of finitely generated substructures (which are countable by Lemma 4.1.12) so is countable. $\square$

This theory begins to break down when we consider uncountable structures.

Lemma 5.1.7. ([20], Section 7.1, Exercise 16)

If $\mathbf{K}$ is some countable/finite set of finitely generated L -structures with (HP), (JEP), (AP), there need not be an uncountable structure A with $\text{age}(A) = \mathbf{K}$

Proof. Let L be the language of group theory and define $\mathbf{K} = \text{age}(\mathbb{Z}) = \{1, \mathbb{Z}\}$. Suppose A is some group with $\text{age}(A) = \mathbf{K}$. Now A must be torsion-free, as any torsion element would generate a finite cyclic subgroup (which would then be in $\text{age}(A)$). Also A must be abelian, as any two elements generate an abelian subgroup. Write A additively. We claim that A injects into $\mathbb{Q}$.

Fix $a \in A$. Define $f : A \rightarrow \mathbb{Q}$ as follows. Set $f(0_A) = 0$. For all $b \in A \setminus \{0_A\}$ we have $\langle a, b \rangle \cong \mathbb{Z}$, so there are some unique nonzero integers p, q with $\gcd(p, q) = 1$ such that $pa = qb$. Define $f(b) = \frac{p}{q}$. In particular, $f(a) = 1$. This is certainly a well-defined function, so we need only check injectivity.

It is clear that the identity 0_A is the only element sent to $0_{\mathbb{Q}}$. Suppose $b_1, b_2 \in A \setminus \{0_A\}$ have $f(b_1) = f(b_2) = \frac{p}{q}$. Now $\langle a, b_1, b_2 \rangle \cong \mathbb{Z}$, and in $\mathbb{Z}$ elements have unique roots, so pa has a unique q th root $b_1 = b_2$.

Thus f is an injection from A into $\mathbb{Q}$, so A is countable. $\square$

Jónsson [23] generalised some of this chapter for uncountable relational structures, but it requires some extra set theory which makes it outside the scope of this project.

If the age of a structure has (AP), the structure will be 'nice'. This is made more precise with the following property.

Definition 5.1.8. We say an L -structure D is *homogeneous* if every isomorphism between finitely generated substructures $f : A \rightarrow B$ there is an automorphism $\bar{f} : D \rightarrow D$

extending f . This is summarised in the following diagram.

$$\begin{array}{ccc} A & \xrightarrow{\subseteq} & D \\ f \downarrow & & \downarrow \bar{f} \\ B & \xrightarrow{\subseteq} & D \end{array}$$

One should think of homogeneous structures as having very large automorphism groups. For example in a homogeneous group, any two elements of the same order are in the same orbit. Let us first see an example, then a non-example.

In the setup of the above definition, it is sometimes nice to refer to $f : A \rightarrow B$ as a *partial automorphism*.

Example 5.1.9. Let $L = \{0, +, -\}$. Let F be a field and $L(F)$ be the language where we add the unary function symbol $\hat{x}$ to L for scalar multiplication by x for all $x \in F$. Now consider V a vector space over F as some $L(F)$ -structure. We show V is homogeneous.

let U_1, U_2 be finitely generated substructures of V , with some partial automorphism $f : U_1 \rightarrow U_2$. Now U_1, U_2 are each closed under scalar multiplication and addition, so are in fact vector subspaces of V . As U_1, U_2 are isomorphic, they have the same dimension. They are finitely generated, so we may find bases $\{u_1, \dots, u_n\}$ and $\{w_1, \dots, w_n\}$. Now by Zorn's Lemma (3.0.1) we can extend these to bases for V , which we can write as $\{u_i \mid i \in I\}$ and $\{v_i \mid i \in I\}$, where I is some index set containing $\{1, \dots, n\}$. Then $u_i \mapsto v_i$ induces an automorphism $\bar{f} : V \rightarrow V$ which extends f .

Example 5.1.10. Consider $C_4 \oplus C_2$, writing x for the generator of the C_4 component and y for the generator of the C_2 component. Now

$$\langle 2x \rangle \cong \langle y \rangle \quad \text{under the map } 2x \mapsto y.$$

This cannot be extended to some automorphism of $C_4 \oplus C_2$, as $2x$ is divisible by two, but y is not, and divisibility by two can be written as a formula in the first-order language of groups. We have shown $C_4 \oplus C_2$ is not homogeneous.

One should consider that this forces a failure in the amalgamation property for $\text{age}(C_4 \oplus C_2)$, as the following amalgam cannot be completed.

$$\begin{array}{ccc} & & \langle x, y \rangle \\ & \nearrow^{2x \mapsto y} & \\ \langle 2x \rangle & & \\ & \searrow_{2x \mapsto 2x} & \\ & & \langle x, y \rangle \end{array}$$

We now focus on homogeneous groups and investigate closure properties. Some of the following results were stated by Cherlin and Felgner in their classifications of soluble homogeneous groups [10] and finite homogeneous groups [9], but were left unproven. These proofs are original.

As homogeneity is a condition about automorphisms of a group, one might expect characteristic subgroups behave nicely with respect to homogeneity.

Lemma 5.1.11. *Let G be a homogeneous group and $K \text{ char } G$. Then K is homogeneous. Furthermore, if K contains an element of order n (or infinity), then K contains all the elements of order n (or infinity) in G .*

Proof. Let $A, B \leq K$ be finitely generated subgroups with the isomorphism $f : A \rightarrow B$. As G is homogeneous, there is some $\bar{f} : G \rightarrow G$ an automorphism extending f . As $K \text{ char } G$, $\bar{f}|_K$ is an automorphism of K which extends f , witnessing that K is homogeneous.

Now if $x \in K$, note that every element of order equal to $|x|$ has must be in the automorphic orbit of x . As $x \in K \text{ char } G$, the orbit of x is contained in K , so K contains all elements of order $|x|$. $\square$

In particular, if a homogeneous group is of finite exponent, it can have only finitely many characteristic subgroups. Unfortunately, homogeneity is not necessarily preserved under other constructions on groups.

Lemma 5.1.12. *Homogeneity is not closed under taking normal subgroups or quotients.*

Proof. Consider the group $Q_8 \oplus C_2$, where we write $C_2 = \langle x \rangle$, $Q_8 = \langle i, j, k \rangle$. We can show by exhaustion that this group is homogeneous. This group has the index 2 subgroup $G\langle i, x \rangle \cong C_4 \oplus C_2$, which is not homogeneous by Example 5.1.10. Index 2 subgroups are normal, so we have shown homogeneity is not closed under taking normal subgroups.

To show homogeneity is not closed under taking quotients, consider $Q_8 \oplus C_4$. Again, we can show by exhaustion this group is homogeneous. Now the normal subgroup $\langle i \rangle$ gives the quotient group $(Q_8 \oplus C_4)/\langle i \rangle \cong C_4 \oplus C_2$, which is not homogeneous. $\square$

The $C_{2n} \oplus C_n$ structure really is a problem for homogeneity. The idea from Example 5.1.10 is all that is needed to prove the following.

Lemma 5.1.13. *For p some prime, a homogeneous abelian p -group of bounded exponent must be homocyclic.*

Lemma 5.1.14. *A homogeneous abelian group with an element of infinite order is divisible.*

Proof. Let G a homogenous abelian group with an element of infinite order. Consider

$$\text{div}(G) = \{g \in G \mid \text{for all } n \in \mathbb{N} \text{ there exists } h \in G \text{ such that } g = nh\}$$

i.e. the subgroup of all the divisible elements of G . By Theorem 4.4.3, we may find some $H \leq G$ such that

$$G = \text{div}(G) \oplus H.$$

Also $\text{div}(G)$ is characteristic so by Lemma 5.1.11, if g is an element of $\text{div}(G)$ then so are all elements in G of the same order as g . By assumption there is some $x \in G$ with infinite order. We claim $x \in \text{div}(G)$.

Suppose, looking for a contradiction, that $x \notin \text{div}(G)$. Then we may find some $n \in \mathbb{N}$ such that x is not divisible by n . But then

$$\langle x \rangle \cong \langle nx \rangle \cong \mathbb{Z}$$

so the homogeneous group G admits some automorphism $f : G \rightarrow G$ such that $f(x) = nx$. But certainly automorphisms preserve whether an element is divisible by n , so f cannot be an automorphism, a contradiction.

Thus $x \in \text{div}(G)$ and by our discussion above $\text{div}(G)$ contains every element of infinite order. We now claim H is trivial. Let $h \in H$ be given. Then xh has infinite order, so $xh, x \in \text{div}(G)$, giving

$$h \in \text{div}(G) \cap H = \{0\}$$

noting that components of a direct sum intersect trivially. The element $h \in H$ was arbitrary, so $H = \{0\}$. We have $G = \text{div}(G)$ which is divisible, as required. $\square$

We now step back to more general model theory. Showing that a structure is homogeneous can be hard, so we look for something more flexible.

Definition 5.1.15. We will say D is *weakly homogeneous* if for all A, B finitely generated substructures of H with $A \subseteq B$ and $f : A \hookrightarrow D$ an embedding of A into D , there is some $\bar{f} : B \rightarrow D$ extending f . This is summarised in the following diagram.

$$\begin{array}{ccc} A & \xhookrightarrow{f} & D \\ \downarrow \subseteq & \nearrow \bar{f} & \\ B & & \end{array}$$

Homogeneity is the nicer property to use, but showing that a structure is weakly homogeneous is usually easier. This makes the following lemma particularly useful.

Lemma 5.1.16. *A countable L -structure D is homogeneous if and only if it is weakly homogeneous*

To show weak homogeneity implies homogeneity, we will employ a back-and-forth argument. This is the best way of explicitly building isomorphisms in model theory!

Proof. ($\Rightarrow$) First suppose D is homogeneous. Let A, B be finitely generated substructures of D with $A \subseteq B$ and $f : A \hookrightarrow D$ an embedding of A into D . Then, as $B \in \text{age}(D)$, we have B embeds into D , say under $g : B \hookrightarrow D$. Now $g^{-1} : g(B) \hookrightarrow B$ is an embedding. Putting things together we get the isomorphism $fg^{-1} : g(A) \hookrightarrow f(A)$.

We are now ready to apply homogeneity and extend fg^{-1} to an automorphism of D , say $h : D \rightarrow D$. Then $hg : B \hookrightarrow D$ has

$$hg(A) = h(g(A)) = fg^{-1}(g(A)) = f(A).$$

So $\bar{f} := hg$ extends f , and we are done with this direction.

($\Leftarrow$) Now suppose D is weakly homogeneous, and let A, B be isomorphic finitely generated substructures of D with the partial automorphism $f : A \rightarrow B$. Write $A = \langle a_1, \dots, a_n \rangle$ and $B = \langle b_1, \dots, b_n \rangle$. D is countable, so we may write $D = \{d_i \mid i \in \mathbb{N}\}$, and for all $i \in \mathbb{N}$ define

$$A_i = \langle a_1, \dots, a_n, d_1, \dots, d_i \rangle, B_i = \langle b_1, \dots, b_n, d_1, \dots, d_i \rangle$$

giving the ascending chains

$$A_0 \leq A_1 \leq \dots \quad \text{and} \quad B_0 \leq B_1 \leq \dots$$

where $D = \bigcup_{i \in \mathbb{N}} A_i = \bigcup_{i \in \mathbb{N}} B_i$.

We are now ready to employ a back-and-forth argument to extend f to some automorphism $\bar{f} : D \rightarrow D$. We build embeddings into D :

$$f_0 \subseteq f_1 \subseteq f_2 \subseteq \dots$$

such that $A_i \leq \text{dom}(f_{2i})$ and $B_i \in \text{im}(f_{2i})$. These properties will give that

$$\bar{f} = \bigcup_{i < \omega} f_i$$

is an automorphism of D extending f , witnessing the homogeneity of D . Begin by setting $f_0 := f$. We build $\{f_i \mid i < \omega\}$ inductively.

(Odd step) Suppose we have $f_{2i} : \text{dom}(f_{2i}) \hookrightarrow D$ an embedding where

$$A_i \leq \text{dom}(f_{2i}) \leq D,$$

where $B_i \leq \text{im}(f_{2i})$, and where $\text{dom}(f_{2i})$ is finitely generated. We define some extension f_{2i+1} whose domain contains A_{i+1} . We may find $\theta(2i+1) \in \mathbb{N}$ such that $A_{i+1}, \text{dom}(f_{2i}) \leq A_{\theta(2i+1)}$, and define f_{2i+1} to be the extension guaranteed by weak homogeneity in the following diagram:

$$\begin{array}{ccc} \text{dom}(f_{2i}) & \xhookrightarrow{f_{2i}} & D \\ \downarrow \subseteq & \nearrow f_{2i+1} & \\ A_{\theta(2i+1)} & & \end{array}$$

(Even step) Now we have $f_{2i+1} : A_{\theta(2i+1)} \hookrightarrow D$, we define some extension $f_{2(i+1)}$ whose image contains B_{i+1} . As f_{2i+1} is an embedding, its image $\text{im}(f_{2i+1})$ is finitely generated, so we can use it in our weak homogeneity constructions. Also

$$g_{2i+1} : \text{im}(f_{2i+1}) \rightarrow D \quad \text{where} \quad g_{2i+1} = (f_{2i+1})^{-1}$$

is an embedding. Choose $\sigma(2i+1) \in \mathbb{N}$ such that $B_{i+1}, \text{dom}(f_{2i+1}) \leq B_{\sigma(2i+1)}$. Now define $g_{2(i+1)}$ to be the extension guaranteed by weak homogeneity in the following diagram:

$$\begin{array}{ccc} \text{im}(f_{2i+1}) & \xhookrightarrow{g_{2i+1}} & D \\ \downarrow \subseteq & \nearrow g_{2(i+1)} & \\ B_{\sigma(2i+1)} & & \end{array}$$

Now defining $f_{2(i+1)} = (g_{2(i+1)})^{-1}$, we see $f_{2(i+1)}$ is an embedding such that

$$A_{i+1} \leq \text{dom}(f_{2(i+1)}) \leq D,$$

where $B_{i+1} \leq \text{im}(f_{2(i+1)})$, and where $\text{dom}(f_{2(i+1)})$ is finitely generated. This completes the induction loop, so we are done by the discussion above. $\square$

Before closing the section we mention an interesting generalisation of homogeneity, normally only considered for relational signatures, introduced by Cameron and Nešetřil [7]. Let D be an L -structure, we say D is HH (homomorphism homogeneous) if for all finitely generated substructures $A, B \leq D$, every homomorphism $f : A \rightarrow B$ can be extended to a homomorphism $\bar{f} : D \rightarrow D$.

For example, our old friend $C_4 \oplus C_2$ is HH but not homogeneous.

We have a very good handle on countable HH relational structures (we have full classifications for tournaments [14] and oriented graphs [32]) but not much has been said about HH groups. This could be a line of enquiry for a sequel to this project.

5.2. Fraïssé's Theorem. We can now prove the main theorem of the chapter, following Hodges [20].

Theorem 5.2.1. *Let L be a countable signature and $\mathbf{K}$ a non-empty Fraïssé class with at most countably many isomorphism types. Then there is a countable L -structure H , unique up to isomorphism such that:*

- (1) $\mathbf{K}$ is the age of H .
- (2) H is homogeneous.

Proof. We will prove the theorem for $\mathbf{K}$ containing countably many isomorphism types. The proof for the finite case is the same, but would require different notation. The proof will be in two parts: Existence and uniqueness.

Existence

We construct $\{D_i \mid i \in \mathbb{N}\}$ an ascending chain of finitely generated substructures, each in $\mathbf{K}$, with the following property:

- (*) For all $A, B \in \mathbf{K}$ with $A \subseteq B$ and $f : A \hookrightarrow D_i$ an embedding for some $i \in \mathbb{N}$, there exists $\bar{f} : B \hookrightarrow D_j$ an embedding extending f for some $j \in \mathbb{N}_{>i}$.

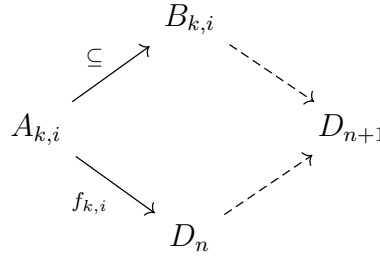
We are trying to build a structure which allows every possible embedding from finitely generated substructures! Let us construct the set $\{D_i \mid i \in \mathbb{N}\}$. Define

$$P = \{(A, B) \mid A, B \in \mathbf{K}, A \subseteq B\}$$

and let $\pi : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ be a bijection such that $\pi(i, j) \geq i$. Let $D_0 \in \mathbf{K}$. Now suppose we are given D_n . As $\mathbf{K}$ is countable, we may enumerate the following triples:

$$\{(f_{n,i}, A_{n,i}, B_{n,i}) \mid f_{n,i} : A_{n,i} \rightarrow D_n, A \text{ and } (A_{n,i}, B_{n,i}) \in P, i \in \mathbb{N}\}$$

Then for $n = \pi(k, i)$, we may define D_{n+1} to be the element of $\mathbf{K}$ completing the following amalgam:

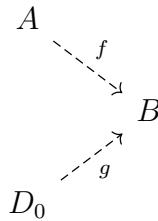


To show that $\{D_i \mid i \in \mathbb{N}\}$ has property (*), let $A, B \in \mathbf{K}$ with $A \subseteq B$ and $f : A \hookrightarrow D_k$ for some $k \in \mathbb{N}$ be given. Then $(f, A, B) = (f_{k,j}, A_{k,j}, B_{k,j})$ for some $j \in \mathbb{N}$. Then take $m = \pi(k, j)$, and then by (AP) as above we get an embedding from B into D_m , as required.

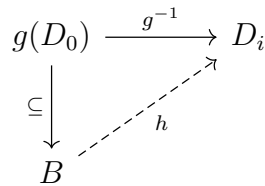
Now take $D = \bigcup_{i \in \mathbb{N}} D_i$. We show that D satisfies (1) and (2):

First we will show $\text{age}(D) \subseteq \mathbf{K}$. Let $A = \langle a_1, \dots, a_n \rangle$ be some finitely generated substructure of D , then there exists some $i \in \mathbb{N}$ such that $a_1, \dots, a_n \in D_i$, so A is a finitely generated substructure of $D_i \in \mathbf{K}$. By (HP), $A \in \mathbf{K}$.

For the reverse, let $A \in \mathbf{K}$ be given. By (JEP) we have the following diagram of embeddings:



We can then apply property (*), noting $g(D_0) \subseteq B$, and $g^{-1} : g(D_0) \hookrightarrow D_0$ is an embedding. This gives us some $i \in \mathbb{N}$ so that we get the following diagram of embeddings:



So $hf(A)$ is an embedding of A into D_i , so $A \in \text{age}(D_i) \subseteq \text{age}(D)$. We are now happy that D satisfies (1). For (2), let us note that property (*) immediately implies weak

homogeneity, which is equivalent to homogeneity by Lemma 5.1.16. So D is homogeneous and its age is $\mathbf{K}$.

Uniqueness

Suppose C, D are both countable homogeneous structures with age $\mathbf{K}$. Then both are also weakly homogeneous by Lemma 5.1.16. The proof that C and D are isomorphic is very similar to the proof of the leftward direction in Lemma 5.1.16.

We first write $C = \{c_n \mid n \in \mathbb{N}\}, D = \{d_n \mid n \in \mathbb{N}\}$. We build a countable chain of finitely generated substructures with which we can do a back-and-forth argument. The class $\mathbf{K}$ is non-empty by assumption, so we may find $C_0 \in \mathbf{K}$ with associated embeddings $g : C_0 \rightarrow C, h : C_0 \rightarrow D$. Define $f_0 = hg^{-1}$ an embedding from $g(C_0) \subseteq C$ into $h(C_0) \subseteq D$. Also for all $n \in \mathbb{N}$, define

$$C_n = \langle g(C_0), c_1, \dots, c_n \rangle, D_n = \langle h(C_0), d_1, \dots, d_n \rangle$$

Then each of the C_n and D_n are finitely generated, so are in the age of C and D . Also, $C = \bigcup_{n \in \mathbb{N}} C_n$ and $D = \bigcup_{n \in \mathbb{N}} D_n$.

Now similar to our proof of Lemma 5.1.16, we use the back-and-forth technique to define $\{f_n \mid n \in \mathbb{N}\}$ such that $C_n \subseteq \text{dom}(f_{2n})$ and such that $D_n \subseteq \text{im}(f_{2n-1})$. Then $f = \bigcup_{n \in \mathbb{N}} f_n : C \rightarrow D$ is an isomorphism. $\square$

The converse also holds!

Theorem 5.2.2. *Let L be some countable or finite language and A be some countable or finite homogeneous L -structure. Then $\text{age}(D)$ has (HP), (JEP), (AP).*

Proof. By Lemma 5.1.6, we need only show $\text{age}(D)$ has (AP). Let (A, B, C, e, f) some amalgam be given. Write $A = \langle \bar{a} \rangle, B = \langle \bar{b} \rangle, C = \langle \bar{c} \rangle$ for $\bar{a}, \bar{b}, \bar{c}$ some finite tuples of elements of D .

Now $e(A) \cong f(A)$ under the map given by $e(\bar{a}) \mapsto f(\bar{a})$. By the homogeneity of D , this may be extended to some automorphism $g \in \text{Aut}(D)$. So in particular

$$ge(\bar{a}) = gf(\bar{a}) \text{ giving } ge(A) = gf(A)$$

Now consider the finitely generated substructure of D

$$F = \langle g(\bar{b}), g(\bar{c}) \rangle.$$

In particular $F \in \text{age}(D)$. The restriction maps

$$g|_B : B \rightarrow F \text{ and } g|_C : C \rightarrow F$$

are embeddings with $g|_B e(A) = g|_C f(A)$, so $(F, g|_B, g|_C)$ completes the amalgam. $\square$

5.3. First Examples of Fraïssé Limits. Let us build some intuition as to what Fraïssé limits should look like. Up to isomorphism, the class of finitely generated torsion-free abelian groups is

$$\{\mathbb{Z}^n \mid n \in \mathbb{N}\}.$$

Now $\mathbb{Z}^{(\omega)}$ is some countable group whose age is the class of torsion-free abelian groups, along with the trivial group, but $\mathbb{Z}^{(\omega)}$ doesn't admit a rich enough structure of embeddings to be homogeneous. Recalling Lemma 5.1.14, to get a homogeneous structure of this age, we need divisibility.

Proposition 5.3.1. ([20] Section 7.1, Exercise 3)

Let $\mathbf{K}$ be the class of finitely generated torsion-free abelian groups, along with the trivial group. Then $\mathbf{K}$ has (HP), (JEP), (AP), and has Fraïssé limit $\mathbb{Q}^{(\omega)}$.

Proof. We know from the Example 5.1.9 that $\mathbb{Q}^{(\omega)}$ is homogeneous as a vector space over $\mathbb{Q}$. We use this to show $\mathbb{Q}^{(\omega)}$ is homogeneous as an abelian group, and will then apply Theorem 5.2.2, the converse of Fraïssé's Theorem.

Consider $A, B \leq \mathbb{Q}^{(\omega)}$ with $f : A \rightarrow B$ some isomorphism. By Theorem 3.0.3, we see $A \cong B \cong \mathbb{Z}^n$ for some $n \in \mathbb{N}$. In particular taking minimal generating sets such that

$$A = \langle a_1, \dots, a_n \rangle, B = \langle b_1, \dots, b_n \rangle, f(a_i) = b_i \text{ for all } 1 \leq i \leq n.$$

We show these generating sets are linearly independent over $\mathbb{Q}$. Suppose we have

$$\sum_{i=1}^n k_i a_i = 0$$

for $k_1 = \frac{p_1}{q_1}, \dots, k_n = \frac{p_n}{q_n} \in \mathbb{Q}$ (we can take $q_i \neq 0$ for all $1 \leq i \leq n$). Writing, $d = \prod_{i=1}^n q_i$, we have

$$\sum_{i=1}^n dk_i a_i = 0$$

which is an expression in A . We must have $k_1 = k_2 = \dots = k_n = 0$, as otherwise the above expression violates our assumption that $\{a_1, \dots, a_n\}$ is a minimal generating set for A . We have shown $\{a_1, \dots, a_n\}$ is linearly independent over $\mathbb{Q}$, and the same argument works for $\{b_1, \dots, b_n\}$. We have that $f : A \rightarrow B$ induces a vector space isomorphism $f_{\mathbb{Q}} : \langle A \rangle_{\mathbb{Q}} \rightarrow \langle B \rangle_{\mathbb{Q}}$ between n -dimensional vector subspaces of $\mathbb{Q}^{(\omega)}$. As $\mathbb{Q}^{(\omega)}$ is homogeneous as a vector space, $f_{\mathbb{Q}}$ induces a vector space automorphism $\overline{f_{\mathbb{Q}}} : \mathbb{Q}^{(\omega)} \rightarrow \mathbb{Q}^{(\omega)}$. This is also a group isomorphism which extending f .

Now by Theorems 5.2.1 and 5.2.2 $\mathbb{Q}^{(\omega)}$ is the Fraïssé limit of its age, which has (HP), (JEP), (AP). It is clear from Theorem 3.0.3 that $\text{age}(\mathbb{Q}^{(\omega)})$ is the class of finitely generated torsion-free abelian groups, along with the trivial group, so we are done. $\square$

The group $\mathbb{Q}^{(\omega)}$ will show up as a nice example again in Chapter 6

We turn to the Rado graph, the classical example of a Fraïssé limit, offering a different construction to the standard BIT predicate [33].

Definition 5.3.2. We say a graph G has the extensional property (EP) if for all U, W finite disjoint sets of vertices of G , there exists some $x \in V(G)$ such that x is adjacent to all the vertices in U and adjacent to none of the vertices in W .

Proposition 5.3.3. For a countable graph $G = (V, E)$, (EP) implies homogeneity.

Proof. Let G be a countable graph with (EP). We may write $G = \{g_1, g_2, \dots\}$. We will employ a back-and-forth argument.

Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be finite subgraphs of G isomorphic under the map $\phi_0 : G_1 \rightarrow G_2$. We inductively define isomorphisms $(\phi_i)_{i < \omega}$, at even stages extending the domain to include V_1 , at odd stages extending the image to include V_2 . We show the even step here, and the odd step is nearly identical.

(Even Step) Suppose the isomorphism ϕ_{2i} has been defined, with $g_1, \dots, g_i \in \text{dom}(\phi_{2i})$. If also $g_{i+1} \in \text{dom}(\phi_{2i})$, set $\phi_{2i+1} = \phi_{2i}$. Otherwise, set $B_{i+1} = \{y \in \text{dom}(\phi_{2i}) \mid y \sim g_{i+1}\}$. Now by (EP), there exists $z \in V$ such that $z \sim \phi_{2i}(B_{i+1})$ and $z \parallel (\text{im}(\phi_{2i}) \setminus \phi_{2i}(B_{i+1}))$. Then define:

$$\phi_{2i+1} : \text{dom}(\phi_{2i}) \cup \{g_{i+1}\} \rightarrow \text{im}(\phi_{2i}) \cup \{z\}$$

as the extension of ϕ_{2i} , where $g_{i+1} \mapsto z$. As z was chosen to have the correct edge relations, ϕ_{2i+1} is an isomorphism between induced subgraphs of G , closing the induction.

We have inductively defined

$$\phi_0 \subseteq \phi_1 \subseteq \phi_2 \subseteq \dots$$

a countable chain of isomorphisms. Now $\phi := \bigcup_{i < \omega} \phi_i$ has $\text{dom}(\phi) = G = \text{im}(\phi)$, so ϕ is an automorphism of G , as required. $\square$

So to build a countable homogeneous graph, we need only build a countable graph with (EP). To that end, let $G_0 = (V_0, E_0)$ be the graph with a single vertex. Given $G_i = (V_i, E_i)$, we will add vertices x_U for all $U \subseteq V_i$, where $x_U \sim v$ if and only if $v \in U$. Now set $R = \bigcup_{i < \omega} G_i$. We call R the Rado graph. Then the vertices added at each stage witness that R has (EP).

We now show R is universal for the class of finite graphs. Let $K = (V, E)$ be a finite graph, and enumerate its vertices $k_1, \dots, k_n$. We want to show K embeds into R , and do so by listing the vertices $v_1, \dots, v_k$ of R so that $k_1 \mapsto v_1, \dots, k_n \mapsto v_n$ is an embedding. Let v_1 be the vertex corresponding to the lone vertex in G_0 . Given $v_1, \dots, v_i$ which are the limits of points found in G_i , define $N := \text{Nbd}(k_{i+1}) \cap \{k_1, \dots, k_i\}$. Then set v_{i+1} to be the limit of the point x_N found in G_{i+1} . Then by construction $k_1 \mapsto v_1, \dots, k_n \mapsto v_n$ is the embedding we require, so $K \hookrightarrow R$.

By Proposition 5.3.3 along with Fraïssé's Theorem (Theorem 5.2.1) and its converse, we have that R is the Fraïssé limit of the class of finite graphs, which has (HP), (JEP), (AP).

If the reader has enjoyed the back-and-forth arguments from this chapter, they should seek out Madhav Singh's work [38], which deals with higher cardinality analogues of the Rado graph.

5.4. Hall's group. In 1959, Hall [19] constructs a countable homogeneous group H with age **FG**, and shows H is unique with respect to these properties. We will call H Hall's universal group.

Hall constructs H without any reference to Fraïssé's Theorem (5.2.1). We will build H as a Fraïssé limit and show H is nice in many different ways, including simplicity, order conjugacy, and verbal completeness.

Lemma 5.4.1. *The class of finite groups **FG** is a Fraïssé class.*

Proof. We must show **FG** has (HP), (JEP), and (AP). A finitely generated subgroup of a finite group is finite, so **FG** has (HP). By Lemma 5.1.4, as the trivial group embeds into every finite group, we are done if we can show **FG** has (AP). This is where the real work lies.

Let us be given some amalgam (A, B, C, e, f) . To save ink, let us assume $A \subseteq B$ and $A \subseteq C$, so that e and f are inclusions (we will see below that we do not want many more functions kicking around). We will complete this amalgam using Neumann's permutational products [30].

Let S be a left transversal of B by A . Then $S \subseteq B$ such that $B = \bigsqcup_{s \in S} sA$. Thus every element $b \in B$ has

$$b = sa \text{ for some unique } s \in S, a \in A.$$

We can thus define π_S, π_A as the projection maps onto S and A respectively. Because these need not be homomorphisms, they are written on the left. We have the immediate fact:

$$(\dagger) \quad \pi_S(b)\pi_A(b) = b \text{ for all } b \in B.$$

Now similar to above, we let T be a left transversal of C by A . Then $T \subseteq C$ such that $C = \bigsqcup_{t \in T} tA$. Thus every element in $c \in C$ has

$$c = ta \text{ for some unique } t \in T, a \in A.$$

We can thus define σ_T, σ_A as the projection maps onto T and A respectively. As above:

$$\sigma_S(c)\sigma_A(c) = c \text{ for all } c \in C.$$

Now define

$$K = S \times T \times A.$$

We claim that $\text{Sym}(K)$, along with two embeddings will complete the amalgam. For each $b \in B$, we associate

$$\rho_b : K \rightarrow K \text{ given by } (s, t, a) \mapsto (\pi_S(sab), t, \pi_A(sab)).$$

For each $c \in C$, we associate

$$\tilde{\rho}_c : K \rightarrow K \text{ given by } (s, t, a) \mapsto (s, \sigma_B(tac), \sigma_A(tac)).$$

We first show $\rho_a = \tilde{\rho}_a$ for all $a \in A$. Let $a \in A, (s, t, x) \in K$ be given. Then noting A is a subgroup, so $xa \in A$, we have

$$(s, t, x)\rho_a = (\pi_S(sxa), t, \pi_A(sxa)) = (s, t, xa) = (s, \sigma_B(txa), \sigma_A(txa)) = (s, t, x)\tilde{\rho}_a$$

as required.

Then if each of the functions $b \mapsto \rho_b$ and $c \mapsto \tilde{\rho}_c$ are embeddings into $\text{Sym}(K)$ we will have that $(\text{Sym}(K), b \mapsto \rho_b, c \mapsto \tilde{\rho}_c)$ completes the amalgam. We show $b \mapsto \rho_b$ is an embedding into $\text{Sym}(K)$, and the proof for $c \mapsto \tilde{\rho}_c$ is very similar.

We first must show $\rho_b \in \text{Sym}(K)$ for all $b \in B$. We can do this by noting that

$$\begin{aligned} (s, t, a)\rho_{b^{-1}}\rho_b &= (\pi_S(sab), t, \pi_A(sab))\rho_{b^{-1}} = (\pi_S(sabb^{-1}), t, \pi_A(sabb^{-1})) \\ &= (\pi_S(sa), t, \pi_A(sa)) = (s, t, a) \end{aligned}$$

for all $(s, t, x) \in K$. Thus $\rho_b\rho_{b^{-1}} = id_K$, and by symmetry $\rho_{b^{-1}}\rho_b = id_K$, so ρ_b has an inverse and is therefore a permutation of K .

Now we must show the function $b \mapsto \rho_b$ is an injective homomorphism. Note

$$\begin{aligned} (s, t, a)\rho_{b_1}\rho_{b_2} &= (\pi_S(sab_1), t, \pi_A(sab_1))\rho_{b_2} \\ &= (\pi_S(\pi_S(sab_1)\pi_A(sab_1)b_2), t, \pi_A(\pi_S(sab_1)\pi_A(sab_1)b_2)) \\ &\stackrel{(\dagger)}{=} (\pi_S(sab_1b_2), t, \pi_A(sab_1b_2)) = (s, t, a)\rho_{b_1b_2} \end{aligned}$$

So we certainly have a homomorphism. Now to show injectivity, suppose $s \in S, a \in A$ such that $sa = 1$, and let $t \in T$ be arbitrary. Then $(s, t, a)\rho_{b_1} = (s, t, a)\rho_{b_2}$ forces $(\pi_S(b_1), t, \pi_A(b_1)) = (\pi_S(b_2), t, \pi_A(b_2))$ which gives $\pi_S(b_1) = \pi_S(b_2), \pi_A(b_1) = \pi_A(b_2)$, giving $b_1 = b_2$, as S is a transversal of A in B . $\square$

We will spend the remainder of this chapter studying the limit of the Fraïssé class **FG**.

Definition 5.4.2. We call the Fraïssé limit of **FG** *Hall's group*, write H .

By Fraïssé's Theorem (5.2.1) any two elements of H of the same order are in the same orbit of $\text{Aut}(H)$. We show this can be strengthened: any two elements of H of the same order are conjugate. We say a group with this property is *order conjugate*.

Lemma 5.4.3. Let G be a group and let $\rho : G \rightarrow \text{Sym}(G)$ be the right regular representation of G (i.e. $h(g\rho) = h \cdot g$ for all $g, h \in G$). If G_1, G_2 are isomorphic subgroups of G , then $G_1\rho$ and $G_2\rho$ are conjugate in $\text{Sym}(G)$.

Proof. The subgroups G_1, G_2 are isomorphic, say under a map $f : G_1 \rightarrow G_2$. Thus $|G : G_1| = |G : G_2| := i$. Let $x_1, \dots, x_i$ be a left transversal of G by G_1 , and let $y_1, \dots, y_i$ be a left transversal of G by G_2 . Thus

$$G = \bigsqcup_{j=1}^i x_j G_1 = \bigsqcup_{j=1}^i y_j G_2$$

This are about to be a lot of functions flying around, so we add in the symbol $\cdot$ for multiplication in the group G for visual clarity.

Define the function

$$\sigma : G \rightarrow G \text{ where } (x_j \cdot g_1)\sigma = y_j \cdot (g_1 f).$$

As f is a bijection, σ is a permutation of G . Now for all $1 \leq j \leq i$ and for all $g_1, h_1 \in G_1$, we see:

$$(x_j \cdot h_1 \cdot g_1)\sigma = y_j \cdot ((h_1 \cdot g_1)f) = (y_j \cdot (h_1 f)) \cdot (g_1 f) = ((x_j \cdot h_1)\sigma) \cdot (g_1 f).$$

In particular we have shown that for all $g \in G, g_1 \in G_1$, we have

$$(\star) \quad (g \cdot g_1)\sigma = (g\sigma) \cdot (g_1 f).$$

We now show $(G_1 \rho)^\sigma = G_2 \rho$. For all $g \in G, g_1 \in G_1$ we have:

$$g(\sigma^{-1}(g_1 \rho)\sigma) = ((g\sigma^{-1}) \cdot g_1)\sigma$$

$$\stackrel{(\star)}{=} (g\sigma^{-1}\sigma) \cdot (g_1 f) = g \cdot (g_1 f) = g((g_1 f)\rho)$$

Thus $(g_1 \rho)^\sigma = (g_1 f)\rho$. The element $g_1 \in G_1$ was arbitrary, so $(G_1 \rho)^\sigma = G_2 \rho$ and we are done. □

Proposition 5.4.4. Hall's group H is order conjugate.

Proof. Let $g, h \in H$ be given with $|g| = |h|$. Consider

$$\rho : \langle g, h \rangle \rightarrow \text{Sym}(\langle g, h \rangle), \text{ the right regular representation.}$$

Now by weak homogeneity (which is implied by homogeneity by Lemma 5.1.16) we may find some embedding γ making the following diagram commute:

$$\begin{array}{ccc} \langle g, h \rangle & \xrightarrow{\subseteq} & H \\ \rho \downarrow & \nearrow \gamma & \\ \text{Sym}(\langle g, h \rangle) & & \end{array}$$

By Lemma 5.4.3, there is some $x \in \text{Sym}(\langle g, h \rangle)$ such that $(g\rho)^x = h\rho$. Thus

$$g^{(x\gamma)} = h,$$

so g and h are conjugate in H . □

Above, we were able to find nice characterisations of some Fraïssé limits. We have a nice one for H here.

Proposition 5.4.5. Let G be any nontrivial finite group. Let $G_0 = G$. Given G_i define $G_{i+1} = \text{Sym}(G_i)$, and define an embedding $\rho_i : G_i \rightarrow G_{i+1}$ to be the right regular representation of G_i . Then $H \cong \bigcup_{i \in \mathbb{N}} G_i$.

Proof. Thanks to Fraïssé's Theorem (5.2.1), we need only show the group $\bigcup_{i \in \mathbb{N}} G_i$ is homogeneous, and has age **FG**. By Cayley's theorem every group embeds into a symmetric group, so the age condition is clear.

The proof of homogeneity is more involved, but neat. Consider A, B isomorphic finitely generated subgroups of $\bigcup_{n \in \mathbb{N}} G_i$, under isomorphism $\varphi : A \rightarrow B$. Write $Hol(A) = A \rtimes \text{Aut}(A)$ for the holomorph of A . Now $Hol(A)$ embeds into H , giving a subgroup D with a normal subgroup $A' \cong A$. Now by Lemma 5.4.3, there exist x, y such that $A^x = B$ and $B^y = A'$. Then conjugation by xy is an isomorphism from A to A' . Then $a^{xy} \mapsto (a\varphi)^y$ is an automorphism of A' , which can be realised as conjugation by some z in D . Now consider the map $\bar{\varphi} : g \mapsto g^{xyzy^{-1}}$, an automorphism of $\bigcup_{i \in \mathbb{N}} G_i$. For $a \in A$, we have

$$a \mapsto a^{xyzy^{-1}} = ((a^{xy})^z)^{y^{-1}} = ((a\varphi)^y)^{y^{-1}} = a\varphi$$

So $\bar{\varphi}$ does extend φ , and we are done. The diagram for this proof is below.

$$\begin{array}{ccc}
 & & a' \mapsto (a')^z \\
 & \curvearrowright & \\
 A & \xrightarrow{\varphi} & A' \\
 \downarrow a \mapsto a^x & \nearrow b \mapsto b^y & \\
 B & &
 \end{array}
 \qquad
 \begin{array}{c}
 \bigcup_{i \in \mathbb{N}} G_i \\
 \vdots \bar{\varphi} \\
 \bigcup_{i \in \mathbb{N}} G_i
 \end{array}$$

□

This is the characterization of H that Hall gave in his paper [19].

Lemma 5.4.6. *For all $n, m \in \mathbb{N}_{>1}$, there exists a finite group G with elements g_1, g_2 such that $|g_1| = |g_2| = n$ and $|g_1 g_2| = m$.*

We present a streamlined proof from [19] which is all we need for considerations about H , but König [24] has an alternate proof where G can be taken to be the symmetric group S_{m+2} .

Proof. Let $P = C_n * C_n$ the free product of two cyclic groups, writing x and y for the generators of the left and right C_n terms respectively.

Consider P' the derived subgroup of P . The group P/P' is abelian, so elements of P' have zero multiplicity for each of x and y . We note x has nonzero multiplicity, so $x \notin P'$. Now the abelianisation of a free product is the direct product of the abelianisations by Theorem 3.0.2, thus

$$(\odot) \qquad P/P' \cong C_n \oplus C_n.$$

Now using the commutator identities $[ab, c] = [a, c]^b [b, c]$ and $[a, bc] = [a, c][a, b]^c$ we can note that

$$P' = \langle [x, y]^p \mid p \in P \rangle = \langle [x, y]^p \mid p \in \{x, y\} \rangle,$$

so is finitely generated.

Now writing $P'_m = \{a^m \mid a \in P'\}$, we see $N = \langle P'', P'_m \rangle$ is a normal subgroup of P , as the sets P'_m and P'' are both closed under conjugation.

Now by work above P'/N is a finitely generated abelian group of exponent n , so $|P' : N|$ is finite. Recall from $(\odot)$ that $|P : P'| = n^2$. Then we have

$$|P : N| \leq |P : P'| \cdot |P' : N| < \infty,$$

so P/N is a finite group. Then as x^i has nonzero multiplicity (modulo n) for all $1 \leq i < n$ so $Nx^i \neq N$ for all $1 \leq i < n$. Thus Nx^{-1} and Nx^y have order n , but their product $N[x, y]$ has order m , so we are done. □

Proposition 5.4.7. The group H is simple.

We will in fact prove that H is generated by any nontrivial conjugacy class, and that the generation of any element of H requires only two elements from the conjugacy class.

Proof. Let C be some nontrivial conjugacy class in H . By the order conjugacy of H (Lemma 5.4.4) we know C contains every element of order n for some $n \in \mathbb{N}$. Now let $x \in H$ be given. By Lemma 5.4.6, we may find some finite group G with $g_1, g_2 \in G$ such that $|g_1| = |g_2| = n$ and $|g_1 g_2| = |x|$. As H is universal for the class of finite groups, we may identify G with some subgroup of H to save ink. Again applying the order conjugacy of H , we may find some $y \in H$ such that

$$g_1^y g_2^y = (g_1 g_2)^y = x.$$

So x is generated by g_1^y and g_2^y , which have order n so must be in C . $\square$

In view of the property from the above proof, Hall posed the following question.

Question 5.4.8. Does there exist a finite simple group G such that $G = CC^{-1}$ for all nontrivial conjugacy classes C of G ?

Certainly we are in a better position to answer this now we have the classification of finite simple groups. We can provide a proof in the negative here for alternating groups: Note conjugacy classes in A_n are subsets of the partition of A_n into sets corresponding to cycle structures. Now for $n \geq 6$, we can consider $(1\ 2\ 3\ 4)(5\ 6)$, and note this has a support of size 6. Write C for the conjugacy class of 3-cycles. Let

$$\sigma = (i\ j\ k)(l\ m\ n) \in CC = CC^{-1},$$

then support of σ has size 6 if and only if $|\{i, j, k, l, m, n\}| = 6$, in which case σ is a product of two disjoint 3-cycles. Thus $(1\ 2\ 3\ 4)(5\ 6) \notin CC^{-1}$. For A_5 , we can exhaustively check that the conjugacy class containing $(1\ 2\ 3\ 4\ 5)$ does not have the generation property.

In our earlier discussion, Lemma 5.1.14 showed homogeneity (plus some other conditions) can imply divisibility. With Hall's group, we can get something even stronger.

Definition 5.4.9. We say a group G is *verbally complete* if for all $g \in G$ and for all finite words $w(x_1, \dots, x_n)$, there exist some $g_1, \dots, g_n \in G$ such that

$$w(g_1, \dots, g_n) = g.$$

Any verbally group is certainly divisible, witnessed by the words

$$\{x^n \mid n \in \mathbb{N}\}.$$

Before 1959, we had some examples of verbally complete groups, but none were locally finite, and all were constructed using HNN-extensions.

Theorem 5.4.10. *Hall's group H is verbally complete.*

We need the help of a technical lemma here.

Lemma 5.4.11. *For all $n, k \in \mathbb{N}$ and for all words $w(x_1, \dots, x_n)$ over an alphabet of n letters, there exists some finite group G with elements $g_1, \dots, g_n \in G$ such that $w(g_1, \dots, g_n)$ has order k .*

This lemma is proved by a study of the lower central series of a finitely generated free group, using some classical results from combinatorial group theory which would be too much of a diversion for us here.

Proof of Theorem 5.4.10. Let $g \in H$ and $w(x_1, \dots, x_n)$ be given. Applying Lemma 5.4.11, we find some finite group G (which we can take to be a subgroup of H) and $g_1, \dots, g_n \in G$ such that $w(g_1, \dots, g_n) \in G$ has order $|g|$. Now by the order conjugacy of H (Lemma 5.4.4), there is some $x \in H$ such that

$$g = w(g_1, \dots, g_n)^x = w(g_1^x, \dots, g_n^x).$$

The element $g \in H$ and the word $w(x_1, \dots, x_n)$ were arbitrary, so H is verbally complete. $\square$

A finite group cannot be divisible, so cannot be verbally complete. However, finite groups can satisfy some of the other properties of Hall's group we have noted, including homogeneity and order conjugacy, though these properties are restrictive. We consider this in the appendix, as the investigations require absolutely no model theory.

6. ω -CATEGORICITY

We saw in Theorem 4.2.7 that elementary equivalence is a rigid relation for finite structures. There were many group-theoretic examples showing that elementary equivalence is not so rigid for infinite structures. In this chapter we investigate categorical structures, which are those structures behaving rigidly under elementary equivalence. The Upward Löwenheim-Skolem Theorem (4.3.9) means we need to focus our efforts on a specific cardinality. In particular we will focus on ω -categorical structures:

Definition 6.0.1. We say a complete consistent theory T is κ -categorical if $A \models T$, $B \models T$, and $|A| = |B| = \kappa$ implies $A \cong B$.

We are happy to accept that finite structures are ω -categorical, as they have no countable models (recalling Proposition 4.2.7), so the condition above is trivially true

One of model theory's most celebrated results is Morley's Theorem [28], which states that κ_1 -categoricity and κ_2 -categoricity are equivalent for uncountable cardinals κ_1 and κ_2 .

This does not extend to all infinite cardinalities: We have seen all torsion-free divisible abelian groups are elementary equivalent, and that $\mathbb{Q} \not\cong \mathbb{Q}^{(\omega)}$. Now all torsion-free abelian divisible groups are vector spaces over $\mathbb{Q}$, and a standard linear algebra result tells us there is only one $\mathbb{Q}$ -vector space of a given dimension up to isomorphism. Thus $\text{Th}(\mathbb{Q})$, the theory of the group $\mathbb{Q}$, is uncountably categorical but not ω -categorical.

This divide makes ω -categoricity rather special! As Morley's theorem indicates, the toolkit for studying ω -categoricity is very different to that of uncountable categoricity.

6.1. Realising Types. This section introduces types, which are our main tool building and comparing non-isomorphic models of some first-order theory. All of the work in this section is useful in proving the Engeler, Ryll-Nardzewski, Svenonius Theorem (6.3.3) later on.

Definition 6.1.1. Let T be some L -theory, and $\bar{x}$ be a tuple of n variables for $n \in \mathbb{N}$. A *partial n -type* is a set of formulae $\Sigma(\bar{x})$ such that there exists $A \models T$ with $\bar{a}$ an n -tuple over A such that $A \models \Sigma(\bar{a})$. An n -type is a maximal partial type. Write $S_n(T)$ for the set of n -types in T .

By the strong version of the Downward-Löwenheim Skolem Theorem (discussed below 4.3.8), we can assume that A in the above definition is countable. Note that some authors write types for our partial types and write complete types for our types. This creates danger when comparing results from different textbooks!

Definition 6.1.2. For $A \models T$ and $\bar{a}$ a tuple over A , we define the *n -type generated by $\bar{a}$* :

$$\text{tp}_A(\bar{a}) = \{\phi(\bar{x}) \mid \phi(\bar{x}) \text{ is an } L\text{-formula and } A \models \phi(\bar{a})\}$$

The type generated by some tuple in A encodes all of the first-order information about that tuple in A .

Definition 6.1.3. Let T be an L -theory and $\Sigma(\bar{x}) \in S_n(T)$. Let $A \models T$ be given. We say A *realises* $\Sigma(\bar{x})$ if $A \models \Sigma(\bar{a})$ for $\bar{a}$ some n -tuple over A . If A does not realise $\Sigma(\bar{x})$ we say it *omits* $\bar{x}$.

We see an example of a type being omitted.

Example 6.1.4. Let T be the theory of Hall's universal group H . Recall Corollary 4.3.7, which gives us some group G with $G \equiv H$, with G containing some element of infinite order g . Then

$$\text{tp}_G(g) \supseteq \{x^n \neq 1 \mid n \in \mathbb{N}\},$$

so $\text{tp}_G(g)$ encodes that g is an element of infinite order, so it must be omitted by H .

We now see an example classifying the types of some theory. This can be very time-consuming, and often uses heavy technology such as quantifier elimination.

Example 6.1.5. ([40], Exercise 4.3.10) We show $\text{Th}(\mathbb{R})$ has only two 1-types, but countably many 2-types.

Throughout, write L for the natural language of abelian groups. Recalling Lemma 4.5.6, this theory has quantifier elimination, so every formula $\phi(x_1, \dots, x_n)$ is equivalent to some formula of the form:

$$\psi(x_1, \dots, x_n) : \bigvee_{i \in I} \bigwedge_{j \in J} \theta_{i,j}(x_1, \dots, x_n)$$

where $\theta_{i,j}(x_1, \dots, x_n)$ is an atomic formula or its negation. Thus, recalling the definitions of satisfaction for negations, conjunctions, and disjunctions (4.1.16), an n -type over $\text{Th}(\mathbb{R})$ is determined entirely by its atomic formulae. We know up to equivalence, the atomic formulae in L are of the form

$$\sum_{i=1}^n a_i x_i = 0,$$

for $a_1, \dots, a_n \in \mathbb{Z}$. i.e. if $\bar{a} \in A \models \text{Th}(\mathbb{R})$ and $\bar{b} \in B \models \text{Th}(\mathbb{R})$, we have

$\text{tp}_A(\bar{a}) = \text{tp}_B(\bar{b}) \iff \bar{a}$ and $\bar{b}$ satisfy the same atomic formulae in A and B respectively.

Now let $p(x)$ be some 1-type. Now as $p(x)$ is consistent and models of T are torsion-free groups, we see that either

$$p(x) \supseteq \{nx = 0 \mid n \in \mathbb{N}\} \text{ or } p(x) \supseteq \{nx \neq 0 \mid n \in \mathbb{N}\}.$$

By our discussion above these sets determine the type $p(x)$, so either

$$p(x) = \text{tp}_{\mathbb{R}}(0) \text{ or } p(x) = \text{tp}_{\mathbb{R}}(1).$$

We have shown $\text{Th}(\mathbb{R})$ has only two 1-types. Now let $p(x, y)$ be some 2-type. We need only consider the following atomic formulae:

- (1) $ax = 0$ for $a \in \mathbb{Z} \setminus \{0\}$
- (2) $by = 0$ for $b \in \mathbb{Z} \setminus \{0\}$
- (3) $ax + by = 0$ for $a, b \in \mathbb{Z} \setminus \{0\}$

Kinds (1) and (2) present us with three edge cases which we first address. If $p(x, y)$ contains a formula of the kind (1) and a formula of kind (2), then p must contain all formulae of kind (1) and all formulae of kind (2), so

$$p(x, y) = \text{tp}_{\mathbb{R}}(0, 0).$$

If $p(x, y)$ contains a formula of kind (1), but not a formula of kind (2), then

$$p(x, y) = \text{tp}_{\mathbb{R}}(0, 1).$$

If $p(x, y)$ contains a formula of kind (2), but not a formula of kind (1), then as above it must contain all formulae of kind (2), so

$$p(x, y) = \text{tp}_{\mathbb{R}}(1, 0).$$

Now for the remainder of our analysis, we may assume $p(x, y)$ contains no formulae of kind (1) or (2). If $p(x, y)$ contains no formulae of kind (3), then it contains no atomic formulae. One can check for all atomic formulae $\phi(x, y)$ of kind (1), (2), or (3) that $\mathbb{R} \not\models \phi(1, \sqrt{2})$, so

$$p(x, y) = \text{tp}_{\mathbb{R}}(1, \sqrt{2}).$$

Now if $p(x, y)$ contains some formula of type (3), we have

$$ax + by \in p(x, y)$$

for some $a, b \in \mathbb{Z}$. Now for $c = \gcd(a, b)$ we have, as models of $\text{Th}(\mathbb{R})$ are torsion-free groups:

$$c\left(\frac{a}{c}x + \frac{b}{c}y\right) = 0 \in p(x, y) \implies \frac{a}{c}x + \frac{b}{c}y = 0 \in p(x, y).$$

To save ink, we may then assume a and b are coprime. We now claim $rx + ty = 0 \in p(x, y)$ if and only if $r = ka, t = kb$ for some $k \in \mathbb{Z}$. The backwards direction is immediate. For the forwards direction, if $rx + ty = 0 \in p(x, y)$, then

$$\begin{aligned} (ar)x + (at)y &= 0 \in p(x, y) \text{ and } (ar)x + (br)y = 0 \in p(x, y) \\ \implies (at - br)y &= 0 \in p(x, y). \end{aligned}$$

Now if $at - br \in \mathbb{Z}$ were nonzero then $p(x, y)$ would contain a formula of kind (2), which would be a contradiction. Thus $at = br$. As a and b are coprime and $a \mid br$, some number theory says $a \mid r$, so $r = ka$ for some $k \in \mathbb{Z}$. This then implies $t = kb$, and our claim is proved.

Thus all atomic sentences in $p(x, y)$ are determined, and they are exactly those which are true in $\mathbb{R}$ under the assignment $x \mapsto 1$ and $y \mapsto \frac{-b}{a}$, so

$$p(x, y) = \text{tp}_{\mathbb{R}}(1, \frac{-b}{a}).$$

This gives countably many possibilities for this case, so $\text{Th}(\mathbb{R})$ has countably many 2-types, as required.

In fact, $S_n(\text{Th}(\mathbb{R}))$ is countably infinite for all $n \in \mathbb{N}_{\geq 2}$. With the below Theorem 6.1.14 in mind, this is a very nice property.

Lemma 6.1.6. *Let T be some L -theory and $A \models T$.*

- (1) *An n -type $\Sigma(\bar{x})$ is realised by $\bar{a}$ in A if and only if $\Sigma(\bar{x}) = \text{tp}_A(\bar{a})$.*
- (2) *Types are closed under finite conjunctions.*
- (3) *Every partial n -type is contained in an n -type.*

Proof. (1). The backward direction is immediate. For the forward direction, suppose $A \models \Sigma(\bar{a})$. Then $\Sigma(\bar{x}) \subseteq \text{tp}_A(\bar{a})$, but $\Sigma(\bar{x})$ is maximal, so $\Sigma(\bar{x}) = \text{tp}_A(\bar{a})$ as required.

(2). This is immediate from the maximality of types.

(3). Let $\Sigma(\bar{x})$ be a partial n -type. By definition, $\Sigma(\bar{x})$ is realised by some $\bar{a}$ a tuple in A . Then $\Sigma(\bar{x}) \subseteq \text{tp}_A(\bar{a})$. $\square$

Definition 6.1.7. We extend the definitions above. Let A be some L -structure with $B \subseteq \text{dom}(A)$ some finite subset, which we enumerate in the tuple $\bar{b}$. We say $\Sigma(\bar{x}, \bar{b})$ a set of $L(B)$ -formulae is a *partial n -type over B* if there is some elementary extension M of A , with $\bar{m}$ an n -tuple over M such that $M \models \Sigma(\bar{m}, \bar{b})$. An n -type over B is a maximal partial n -type over B . Write $S_n(B)$ for the set of n -types over B .

For $\bar{a}$ some n -tuple over A we define the n -type over B generated by $\bar{a}$:

$$\text{tp}_A(\bar{a}/B) = \{\phi(\bar{x}, \bar{b}) \mid \phi(\bar{x}, \bar{b}) \text{ is an } L(B)\text{-formula and } A \models \phi(\bar{a}, \bar{b})\}$$

We also define realising and omitting types as before.

Writing $T = \text{Th}(A)$, we see an n -type with respect to A over the finite set B in the language L is exactly the same thing as an n -type over T in the language $L(B)$. This lets us immediately recover any results we prove about types as defined before.

Lemma 6.1.8. *Let A be some L -structure with some finite subset B . Let $n \in \mathbb{N}$, $\bar{x} = (x_1, \dots, x_n)$, and let $\Sigma(\bar{x})$ be some consistent set of $L(B)$ -formulae. Then the following are equivalent:*

- (1) $\Sigma(\bar{x})$ is a partial n -type over B
- (2) Every finite subset of $\Sigma(\bar{x})$ is realised in A

Proof. (2 $\implies$ 1).

Suppose every finite subset of $\Sigma(\bar{x})$ is realised in A . Then consider some tuple of constants $\bar{c} = (c_1, \dots, c_n)$ disjoint from A and the following $L(\{\bar{c}\} \sqcup A)$ -theory:

$$T := \text{eldiag}(A) \sqcup \{\sigma(\bar{c}) \mid \sigma(\bar{x}) \in \Sigma(\bar{x})\}.$$

That the finite subsets of $\Sigma(\bar{x})$ are realised in A witnesses that A is a model for every finite subset of T , so T is finitely consistent. By the Compactness Theorem (4.3.2), there exists some model $M \models T$. The tuple $(c_1^M, \dots, c_n^M)$ then realises $\Sigma(\bar{x})$ in M , and $\text{eldiag}(A) \subseteq T$, so M is an elementary extension of A , as required.

(1 $\implies$ 2).

Suppose $\Sigma(\bar{x})$ is a partial n -type over B . Then unpacking Definition 6.1.7, we may find M some elementary extension of A and $\bar{m} \in M$ such that $M \models \Sigma(\bar{m})$. Now let $q(\bar{x})$ some finite subset of $\Sigma(\bar{x})$ be given. Then as q is finite, the conjunction of all formulae in q is a valid formula, and we can have

$$M \models (\exists \bar{x}) \bigwedge_{\phi \in q} \phi(\bar{x}).$$

But we know from the discussion below Definition 4.2.9 that, as M is an elementary extension of A , we have $A \equiv M$, giving

$$A \models (\exists \bar{x}) \bigwedge_{\phi \in q} \phi(\bar{x}).$$

This satisfaction gives some $\bar{a} \in A$ such that $A \models q(\bar{a})$, giving

$$A \models \bigwedge_{\phi \in q} \phi(\bar{a}) \implies A \models q(\bar{a}).$$

We have shown the arbitrary finite subset of $\Sigma(\bar{x})$ is realised in A , as required. □

Proposition 6.1.9. Let A be some L -structure. If $\bar{a} = (a_1, \dots, a_n), \bar{b} = (b_1, \dots, b_n) \in A$ with some $f \in \text{Aut}(A)$ such that $f(\bar{a}) = \bar{b}$, then $\text{tp}_A(\bar{a}) = \text{tp}_A(\bar{b})$.

Proof. This is an immediate consequence of Lemma 4.1.19. □

We show straight away the converse of this need not be true.

Example 6.1.10. Consider X some uncountable set and Y some countable set disjoint from X . Define the binary relation:

$$R \subseteq (X \sqcup Y)^2 \text{ where } R(a, b) \text{ holds if and only if } a, b \in X \text{ or } a, b \in Y.$$

Consider $X \sqcup Y$ as a $\{R\}$ -structure. One can show (via quantifier elimination), that any $x \in X$ and $y \in Y$ satisfy

$$\text{tp}(x) = \text{tp}(y).$$

However, if there was some automorphism $f \in \text{Aut}(X \sqcup Y)$ such that $f(x) \in Y$, then to preserve the relation R we would have $f(X) \subseteq Y$, violating the injectivity of f .

So elements of the same type need not be automorphically equivalent. We spend the rest of the section specialising to structures where a strong version of the converse to Proposition 6.1.9 holds.

Definition 6.1.11. We say a structure A is ω -saturated if every type over every finite subset of $\text{dom}(A)$ is realised in A .

Unfortunately we must build up some technology before we can meet examples of ω -saturated structures. The following lemma says ω -saturation does what we hoped it would do, and is one of the key parts of the Engeler, Ryll-Nardzewski, Svenonius Theorem (6.3.3) for ω -categorical structures.

Lemma 6.1.12. *Let A be some countable ω -saturated L -structure. Then for all $n \in \mathbb{N}$ and all n -tuples of distinct elements $\bar{a} = (a_1, \dots, a_n)$ and $\bar{b} = (b_1, \dots, b_n)$ over A , if $\text{tp}_A(\bar{a}) = \text{tp}_A(\bar{b})$, then there exists some $f \in \text{Aut}(A)$ such that $f(\bar{a}) = \bar{b}$.*

Proof. We employ a back-and-forth argument to inductively define some sequence of finite bijections

$$\{f_i : \bar{u}_i \rightarrow \bar{v}_i \mid i < \omega\},$$

such that

$$f := \bigcup_{i < \omega} f_i \in \text{Aut}(A),$$

where $f(\bar{a}) = \bar{b}$. To this end, extend the tuples $\bar{a}$ to $\{a_1, a_2, \dots\}$ and $\bar{b}$ to $\{b_1, b_2, \dots\}$, two enumerations of A .

Define $\bar{u}_0 = \bar{a}$ and $\bar{v}_0 = \bar{b}$, and let $f_0 : \bar{u}_0 \rightarrow \bar{v}_0$ be the function given by $f_0(\bar{a}) = \bar{b}$.

(Odd step) We use this step to ensure our function f will have each element of A in its domain. Suppose we have defined $f_{2i} : \bar{u}_{2i} \rightarrow \bar{v}_{2i}$ such that $a_1, \dots, a_i$ occur in the tuple $\bar{u}_{2i}$, and such that

$$(\dagger\dagger) \quad \text{tp}_A(\bar{u}_{2i}) = \text{tp}_A(\bar{v}_{2i}).$$

If a_{i+1} occurs in $\bar{u}_{2i}$, set $\bar{u}_{2i+1} = \bar{u}_{2i}$, $\bar{v}_{2i+1} = \bar{v}_{2i}$, and $f_{2i+1} = f_{2i}$.

Otherwise, if a_{i+1} does not occur in $\bar{u}_{2i}$, consider

$$\text{tp}_A(a_{i+1}/\bar{u}_{2i}),$$

the 1-type of a_i over the finite subset $\bar{u}_{2i}$. We claim that

$$p(x, \bar{v}_{2i}) := \{\phi(x, \bar{v}_{2i}) \mid \phi(x, \bar{u}_{2i}) \in \text{tp}_A(a_{i+1}/\bar{u}_{2i})\}$$

is a type over $\bar{v}_{2i}$. As the type $\text{tp}_A(a_{i+1}/\bar{u}_{2i})$ is maximal with respect to consistency, the set of formulae $p(x, \bar{v}_{2i})$ must also be maximal with respect to consistency. Along with Lemma 6.1.8, this tells us we will be done in proving the claim if we can show that every finite subset of $p(x, \bar{v}_{2i})$ is satisfied by some element in A . Let $q(x, \bar{v}_{2i})$ be some such finite subset. We use the shorthand:

$$(\exists x)q(x, \bar{u}_{2i}) : \bigwedge_{\phi(x, \bar{u}_{2i}) \in q(x, \bar{u}_{2i})} (\exists x)\phi(x, \bar{u}_{2i}).$$

Now

$$A \models q(a_{i+1}, \bar{u}_{2i}) \implies A \models (\exists x)q(x, \bar{u}_{2i}) \xrightarrow{(\dagger\dagger)} A \models (\exists x)q(x, \bar{v}_{2i}).$$

Thus there is some $a \in A$ such that

$$A \models q(a, \bar{v}_{2i}),$$

so the arbitrary finite subset $q(x, \bar{v}_{2i})$ is satisfied by some element in A . By our above discussion, we have that $p(x, \bar{v})$ is some type over $\bar{v}$, a finite subset of A . The structure

A is ω -saturated, so this type must be realised by some $d_{i+1} \in A$. Then we may define $\bar{u}_{2i+1} = (\bar{u}_{2i}, a_i)$ and $\bar{v}_{2i+1} = (\bar{v}_{2i}, d_{i+1})$, and define

$$f_{2i+1} : \bar{u}_{2i+1} \rightarrow \bar{v}_{2i+1} \text{ extending } f_{2i} \text{ such that } f_{2i+1}(a_{i+1}) = d_{i+1}.$$

(Even step) We use this step to ensure our function f will have each element of A in its image. This step is very similar to the odd step, but here we extend f_{2i+1} to $f_{2(i+1)}$, a bijection such that $b_{i+1} \in \text{im}(f_{2(i+1)})$, and such that the domain and image, when enumerated as a tuple, satisfy the same type over A . This then closes the induction loop.

Now we have inductively defined $\{f_i \mid i < \omega\}$, we may define f as discussed above. We will be done if we can show f is some automorphism. By our back-and-forth construction of the functions $\{f_i \mid i < \omega\}$, we know f is a bijection. Now we must show f satisfies the criteria on constants, relations, and functions to be an embedding from Definition 4.1.5. We will leave the case for functions to the reader.

(Constants) Let c be a constant in the language L . Then there is some $i < \omega$ such that c^A occurs in $\bar{u}_i$, which we enumerate as $u_{i,1}, \dots, u_{i,k}$. Also enumerate the tuple $\bar{v}_i$ as $v_{i,1}, \dots, v_{i,k}$ and recall $f(u_{i,l}) = v_{i,l}$ for each $1 \leq l \leq k$. Now $c^A = u_{i,j}$ for some $1 \leq j \leq k$. Giving names to the variables in the types,

$$p(x_1, \dots, x_k) = \text{tp}_A(\bar{u}_i) = \text{tp}_A(\bar{v}_i),$$

we have the following equivalences:

$$A \models u_{i,j} = c^A \iff x_j = c \in p(x, y) \iff A \models v_{i,j} = c^A,$$

where the last condition tell us $f(c^A) = c^A$, as required.

(Relations) Let R be some n -ary relation symbol in L and $p_1, \dots, p_n \in A$. We must show

$$A \models R^A(p_1, \dots, p_n) \iff R^A(f(p_1), \dots, f(p_n)).$$

We may find some $i < \omega$ such that $p_1, \dots, p_n \in \bar{u}_i$. Enumerating the tuples with the same notation as above, writing $p_s = u_{i,j_s}$, we see

$$\begin{aligned} A \models R^A(p_1, \dots, p_n) &\iff A \models R^A(u_{i,j_1}, \dots, u_{i,j_n}) \iff R(x_{j_1}, \dots, x_{j_n}) \in p(x_1, \dots, x_k) \\ &\iff A \models R^A(v_{i,j_1}, \dots, v_{i,j_n}) \iff A \models R^A(f(p_1), \dots, f(p_n)), \end{aligned}$$

as required. □

Note this property of ω -saturated structures is not homogeneity! Some elements of a structure can generate isomorphic substructures, but not satisfy the same type; in fact, recalling Example 5.1.10, this is exactly what stops $C_4 \oplus C_2$ from being homogeneous.

Hardcore model theorists love working in structures which have higher cardinality analogues of homogeneity and ω -saturation (they often call such structures Monster Models).

Definition 6.1.13. Throughout we will write $\mathcal{F}(A)$ for the set of finite subsets of A to save ink.

Theorem 6.1.14. *A consistent theory T has some countable ω -saturated model if and only if $S_n(T)$ is countable for all $n \in \mathbb{N}$.*

By Lemma 6.1.6 every realised type is generated by some element, so any ω -saturated structure has no more types than elements. This proves the forward direction. We will need to work harder for the backward direction.

Lemma 6.1.15. *Let T a consistent theory with $S_n(T)$ is countable for all $n \in \mathbb{N}$. If $A \models T$, then for all $B = \{b_1, \dots, b_k\} \in \mathcal{F}(A)$ the set of types $S_n(B)$ is countable for all $n \in \mathbb{N}$.*

Proof. Write $\bar{b} = (b_1, \dots, b_k)$. Let $\Sigma(\bar{x}, \bar{b}) \in S_n(B)$ be given. Then $\Sigma(\bar{x}, \bar{b})$ is realised by some tuple $\bar{m}$ in M an elementary extension of A . So

$$\Sigma(\bar{x}, \bar{b}) = \text{tp}_M(\bar{m}/B) = \{\phi(\bar{m}, \bar{b}) \mid M \models \phi(\bar{m}, \bar{b})\}.$$

Then define $\Sigma'(\bar{x}, \bar{y}) = \text{tp}_M(\bar{m}, \bar{b}) \in S_{n+k}(T)$. This discussion lets us define a function $S_n(B) \rightarrow S_{n+k}(T)$, and it is a quick check that this is injective: If $\Sigma_1(\bar{x}, \bar{b}) \neq \Sigma_2(\bar{x}, \bar{b})$, then we may find some $\phi(\bar{x}, \bar{b}) \in \Sigma_1 \setminus \Sigma_2$, so $\phi(\bar{x}, \bar{y}) \in \Sigma'_1 \setminus \Sigma'_2$, so $\Sigma'_1 \neq \Sigma'_2$, as required. We have shown $S_n(B)$ injects into a countable set, so $S_n(B)$ is countable. $\square$

Lemma 6.1.16. *Let A be some L -structure. There exists some elementary extension B of A , such that all types over finite subsets of A are realised in B . Furthermore, if A is countable and $S_n(T)$ is countable for all $n \in \mathbb{N}$, then B is countable.*

Proof. We extend the language L . First create a new language $L(A)$ by adding some new constant a for each $a \in A$, and recall the elementary diagram of A (see Definition 4.2.9). Now for all $B \in \mathcal{F}(A)$, for all $\Sigma \in S_n(B)$, we add new constant symbols $c_{\Sigma,1}, \dots, c_{\Sigma,n}$. Then define the new theory:

$$T = \text{eldiag}(A) \cup \left(\bigcup_{B \in \mathcal{F}(A)} \bigcup_{n \in \mathbb{N}} \bigcup_{\Sigma \in S_n(B)} \Sigma(c_{\Sigma,1}, \dots, c_{\Sigma,n}) \right)$$

Any model M of T will realise all types over finite subsets of A (An n -type Σ will have $M \models \Sigma(c_{\Sigma,1}^M, \dots, c_{\Sigma,n}^M)$). We will show T is consistent. By the Compactness Theorem (4.3.2) it is sufficient to show every finite subset of T is consistent. Let T' be some finite subset of T . Then T' is a union of finitely many subsets of types, along with finitely many formulae in $\text{eldiag}(A)$. By Lemma 6.1.8, we see that we can assign $c_{\Sigma,i}$ into A such that each of the finite subsets of types are realised in A , and clearly A realises any formula in $\text{eldiag}(A)$. Thus $A \models T'$. By compactness we then get some $M \models T$, where M is some elementary extension of A .

Now note that if $S_n(T)$ is countable for all $n \in \mathbb{N}$, then by Lemma 6.1.15, $S_n(B)$ is countable for all $B \in \mathcal{F}(A), n \in \mathbb{N}$. Thus if A is countable also, the language T built above is countable, so the Compactness Theorem in fact gives a countable model. $\square$

Our plan is to inductively repeat the above strategy. We need the following lemma to tell us that a union of a chain of elementary extensions is an elementary extension of each element of the chain.

Lemma 6.1.17. *If we have some chain of L -structures*

$$A_0 \leq A_1 \leq \dots$$

where A_{i+1} is an elementary extension for A_i , then $\bigcup_{i < \omega} A_i$ is an elementary extension of A_i for each $i < \omega$.

Proof. We leave to the reader a not very enlightening induction on the complexity of formulae. $\square$

We are now ready to combine our lemmas to prove our fundamental theorem on countable ω -structures.

Proof of Theorem 6.1.14. We iterate the process in Lemma 6.1.16. Suppose T is some consistent theory with countably many n -types for each n . By consistency, there exists an L -structure A_0 with $A_0 \models T$. Now given some countable $A_i \models T$, define A_{i+1} to be the countable structure from Lemma 6.1.16 containing A_i as a substructure which realises all types over finite subsets of A_i . We have built:

$$A_0 \leq A_1 \leq A_2 \leq \dots$$

A chain of elementary extensions. Define A to be the union of this chain. Certainly A is countable. Applying Lemma 6.1.17, we see that A is an elementary extension of A_i , for each $i < \omega$. We must now show A is ω -saturated.

Let B some finite subset of A be given, then B is contained in A_i for some $i < \omega$. Enumerate B as some tuple $\bar{b}$. Now let $p(\bar{x}, \bar{b}) \in S_n(B)$ be given. By construction, there exists $\bar{a} \in A_{i+1}$ which realises $p(\bar{x})$. By the Elementary Diagram Lemma (4.2.10), we have

$$A \models \phi(\bar{a}, \bar{b}) \iff A_{i+1} \models \phi(\bar{a}, \bar{b})$$

for all $\phi(\bar{x}, \bar{b}) \in p(\bar{x}, \bar{b})$ and the right hand side holds for all $\phi(\bar{x}, \bar{b}) \in p(\bar{x}, \bar{b})$. Thus

$$A \models p(\bar{a}, \bar{b}),$$

so A realises the arbitrary tuple in $S_n(B)$. We have shown A is an ω -saturated model of T . $\square$

Now with the help of Theorem 6.1.14, we can finally see an example of an ω -saturated structure.

Example 6.1.18. Consider $T = \text{Th}(\mathbb{Q})$, the theory of the additive group of the rationals (which by Theorem 4.5.5 is the theory of the additive group of the reals). We give a non-constructive proof that the countable ω -saturated model of T is $\mathbb{Q}^{(\omega)}$, using some linear algebra.

By Example 6.1.5 and its following discussion, we know $S_n(T)$ is countable or finite for all $n \in \mathbb{N}$. Along with Theorem 6.1.14, this tells us there is some countable ω -saturated model of T .

Let G be any countable model of T . We know G is a torsion-free divisible abelian group, so G can naturally be given a $\mathbb{Q}$ -vector space structure: For $\frac{a}{b} \in \mathbb{Q}$ and $x \in G$, there is some $y \in G$ such that $by = x$. Set $\frac{a}{b} \cdot x = ay$. By a standard linear algebra fact,

$$G \text{ has dimension } \kappa \text{ over } \mathbb{Q} \iff G \cong \bigoplus_{i \in \kappa} \mathbb{Q}.$$

The group G is countable by assumption, so κ is at most countable. Now we will be done if we can show $\bigoplus_{i=1}^n \mathbb{Q}$ is not ω -saturated for all $n \in \mathbb{N}$.

For all $n \in \mathbb{N}_{\geq 1}$ and for all subsets $I = \{\frac{p_{1,1}}{q_{1,1}}, \frac{p_{1,2}}{q_{1,2}}, \dots, \frac{p_{n-1,1}}{q_{n-1,1}}, \dots, \frac{p_{n-1,n}}{q_{n-1,n}}\} \subseteq \mathbb{Q}$ of size $n(n-1)$, where $q_{i,j} > 0$, and where $p_{i,j}$ and $q_{i,j}$ are coprime if $p_{i,j} \neq 0$, for all appropriate i, j , define the formula

$$\phi_{n,I}(x_1, \dots, x_n) : \neg((\exists y_1) \dots (\exists y_{n-1}) \bigwedge_{i=1}^n \sum_{j=1}^{n-1} (\prod_{j=1}^{n-1} q_{i,j}) p_{i,j} y_j = (\prod_{j=1}^{n-1} q_{i,j}) x_i).$$

If we were to allow ourselves shorthand for rational multiplication in our formulae, up to equivalence modulo T we could write the nicer formula

$$\phi_{n,I}(x_1, \dots, x_n) : \neg((\exists y_1) \dots (\exists y_{n-1}) \bigwedge_{i=1}^n \sum_{j=1}^{n-1} \frac{p_{i,j}}{q_{i,j}} y_j = x_i).$$

Then for all $n \in \mathbb{N}_{\geq 2}$ define a set collecting these formulae together:

$$\Phi_n(x_1, \dots, x_n) = \{\phi_{n,I}(x_1, \dots, x_n) \mid I = \{a_{1,1}, a_{1,2}, \dots, a_{n-1,n}\} \subset \mathbb{Q}\}.$$

So $\Phi_n(x_1, \dots, x_n)$ says that $x_1, \dots, x_n$ cannot all be found in the linear subspace generated by some smaller set $\{y_1, \dots, y_{n-1}\}$.

Writing $e_1, \dots, e_n$ for the standard basis vectors of $\mathbb{Q}^n$, we see $\mathbb{Q}^n \models \Phi_n(e_1, \dots, e_n)$ as a basis is linearly independent. Thus

$$\Phi_n(x_1, \dots, x_n) \subseteq \text{tp}_{\mathbb{Q}^n}(e_1, \dots, e_n).$$

But $\mathbb{Q}^{n-1}$ has some $(n-1)$ -dimensional basis, which witnesses that $\Phi_n(x_1, \dots, x_n)$ is not realised in $\mathbb{Q}^{n-1}$. In particular, $\text{tp}_{\mathbb{Q}^n}(e_1, \dots, e_n)$ is a type over the empty set which is not realised in $\mathbb{Q}^{n-1}$, so $\mathbb{Q}^{n-1}$ is not ω -saturated, and we are done.

The following lemma is another crucial component of the Engeler, Ryll-Nardzewski, Svenonius Theorem (6.3.3), which we are working our way towards proving.

Lemma 6.1.19. *Let A and B be countable ω -saturated L -structures with $A \equiv B$. Then $A \cong B$.*

Proof. This is a fairly straightforward back-and-forth argument, too similar to the proof of Lemma 6.1.12 to warrant full exposition. We leave it to the reader. $\square$

6.2. Omitting Types. Given some theory T and two models $A, B \models T$, if there is some type $p(\bar{x})$ realised in A but omitted in B then A and B cannot be isomorphic (recalling Lemma 4.1.19). Thus on our march towards characterising ω -categorical structures in Theorem 6.3.3, we look at when types may be omitted.

Definition 6.2.1. Let T be some L -theory and $\Sigma(\bar{x})$ an n -type over T . We say $\phi(\bar{x})$ *supports* $\Sigma(\bar{x})$ if for all $A \models T$ and for all n -tuples $\bar{a}$ over A , we have

$$A \models \phi(\bar{a}) \text{ implies } A \models \Sigma(\bar{a})$$

If no such $\phi(\bar{x})$ exists we say $\Sigma(x)$ is *unsupported*.

Example 6.2.2. In our analysis of the 2-types of $\text{Th}(\mathbb{R})$ in Example 6.1.5, we met some supported types. For example, $\text{tp}_{\mathbb{R}}(0, 0)$ is supported by the formula $(x = 0) \wedge (y = 0)$.

However, we claim the type $\text{tp}_{\mathbb{R}}(1, \sqrt{2})$ is unsupported. Suppose, looking for a contradiction, that this type is supported by some formula $\phi(x, y)$. Note from Theorem 4.5.5 that $\mathbb{Q} \models \text{Th}(\mathbb{R})$ and recall that types are finitely satisfiable by Lemma 6.1.8, so there is some $a, b \in \mathbb{Q}$ such that

$$\mathbb{Q} \models \phi(a, b).$$

As $\phi(x, y)$ supports $\text{tp}_{\mathbb{R}}(1, \sqrt{2})$ this implies that

$$\mathbb{Q} \models \sigma(a, b) \text{ for all } \sigma(x, y) \in \text{tp}_{\mathbb{R}}(1, \sqrt{2}).$$

Now write $a = \frac{p}{q}, b = \frac{r}{s}$ for $p, q, r, s \in \mathbb{Z}$. Then note

$$(q^2rs)x \neq (pqs^2)y \in \text{tp}_{\mathbb{R}}(1, \sqrt{2}) \implies \mathbb{Q} \models (q^2rs)a \neq (pqs^2)b.$$

But this is equivalent to the satisfaction

$$\mathbb{Q} \models pqr s \neq pqr s,$$

so we have arrived at the desired contradiction, and $\text{tp}_{\mathbb{R}}(1, \sqrt{2})$ is unsupported.

From our work applying Lemma 6.1.8 in the above example we can conclude a supported type in T cannot be omitted in any model of T . The next theorem says that unsupported types are fair game.

Theorem 6.2.3. (*Omitting Types Theorem*)

Let $\bar{x}$ be a tuple of n variables for some $n < \omega$. Let T be some countable consistent theory with an infinite model and $(\Sigma_i(\bar{x}))_{i \in I}$ a countable or finite collection of unsupported sets of L -formulae. Then there is some countable model $A \models T$ which omits $\Sigma_i(\bar{x})$ for all $i \in I$.

Tent and Zeigler [40] prove that a single unsupported 1-type is omitted in some model. We complete their Exercise 4.1.1 in generalising their proof to countable many n -types for any $n \in \mathbb{N}$.

By Lemma 4.2.2, every consistent theory T is contained in some maximal complete consistent theory T' , and any model for T' omitting a set of formulae $\Sigma(\bar{x})$ is also a model for T omitting $\Sigma(\bar{x})$. Thus we can assume T is a maximal complete consistent theory.

Proof. We first fix some bijection $f : \omega \rightarrow \omega \times I$, and write f_1 for the projection of this function onto the first coordinate, f_2 for the projection onto the second coordinate. These functions will manage all our indexing.

Choose C some new countable set of constants, and fix two enumerations: Firstly let

$$\{\psi_0(x), \psi_1(x), \dots\}$$

be an enumeration of all L -formulae in one variable. Also fix

$$\{\bar{c}_0, \bar{c}_1, \dots\},$$

a countable enumeration of the n -tuples of constants in C . We will enlarge T to some Henkin theory T^* such that the constants in C witness that each $\Sigma_i(\bar{x})$ is omitted. To that end we will define a chain of finite consistent $L(C)$ -theories:

$$T_0 \subseteq T_1 \subseteq \dots$$

such that $T^* = \bigcup_{i < \omega} T_i$. Set $T_0 = T$.

(Odd step) We use this step to ensure we are building a Henkin theory. Suppose we are given T_{2i} some finite consistent $L(C)$ -theory with a model $A \models T_{2i}$. As T_{2i} is finite, the set of constants used in some formula of T_{2i} is finite. We may choose some unused $d \in C$ and set

$$T_{2i+1} = T_{2i} \cup \{((\exists x)\psi_i(x)) \rightarrow \psi_i(d)\}.$$

We must now check that A is a model for T_{2i+1} under some assignment of the constants $\{c^A \mid c \in C\}$. We know by assumption there is some assignment giving $A \models T_{2i}$. Choose c^A to agree with this assignment for all $c \in C \setminus \{d\}$. Thus if we assign $d^A \in A$ such that

$$(\$) \quad A \models ((\exists x)\psi_i(x)) \rightarrow \psi_i(d^A)$$

we will be done.

If $A \models (\exists x)\psi_i(x)$, then there is some $a \in A$ witnessing this fact, so set $d^A = a$. If instead $A \not\models (\exists x)\psi_i(x)$, then arbitrarily assign $d^A \in A$. In both cases (\$) holds, so we are done.

(Even step) We use this step to ensure the n -tuples of Henkin constants do not realise any of the types we wish to omit. Write $s = f_1(i)$ and $t = f_2(i)$. We will ensure $\bar{c}_s$ does not realise $\Sigma_t(\bar{x})$. Suppose we are given $T_{2i+1} = T \cup \{\tau_1(\bar{c}_s, \bar{c}), \dots, \tau_k(\bar{c}_s, \bar{c})\}$ for some k , where $\bar{c}_s$ and $\bar{c}$ do not share any constants, and where $A \models T_{2i+1}$. Writing

$$\tau(\bar{c}_s, \bar{c}) = \bigwedge_{m=1}^k \tau_m(\bar{c}_s, \bar{c}),$$

note $A \models \{\tau_1(\bar{c}_s, \bar{c}), \dots, \tau_k(\bar{c}_s, \bar{c})\}$ if and only if $A \models \tau(\bar{c}_s, \bar{c})$. Thus

$$(\mathcal{L}) \quad T_{2i+1} \equiv T \cup \{\tau(\bar{c}_s, \bar{c})\}.$$

Now $\Sigma_t(\bar{x})$ is not supported, so in particular $(\exists \bar{y})\tau(\bar{x}, \bar{y})$ does not support $\Sigma_t(\bar{x})$. This is witnessed by some $\sigma_t(\bar{x}) \in \Sigma_t(\bar{x})$ where $((\exists \bar{y})\tau(\bar{x}, \bar{y})) \wedge (\neg \sigma_t(\bar{x}))$ is satisfied by some model of $B \models T$. Set

$$T_{2(i+1)} = T_{2i+1} \cup \{\neg \sigma_t(\bar{c}_s)\}.$$

We must show $A \models T_{2(i+1)}$. By definition, along with $(\mathcal{L})$, we see

$$T_{2(i+1)} \equiv T \cup \{\tau(\bar{c}_s, \bar{c}) \wedge \neg \sigma_t(\bar{c}_s)\}.$$

As discussed above, we have some $\bar{b} \in B$ such that

$$B \models ((\exists \bar{y})\tau(\bar{b}, \bar{y})) \wedge \neg \sigma_t(\bar{b}),$$

so we may find some $\bar{a} \in B$ such that

$$B \models \tau(\bar{b}, \bar{a}) \wedge \neg \sigma_t(\bar{b}).$$

Then the assignment $\bar{c}^B = \bar{a}$, $\bar{c}_s^B = \bar{b}$, along with an arbitrary assignment of all other constants in C witnesses that $B \models T_{2(i+1)}$, so in particular $T_{2(i+1)}$ is a consistent $L(C)$ -theory.

This concludes our construction of $\{T_i \mid i < \omega\}$, so we may define T^* as discussed above. Any finite subset of T^* is contained in T_n , a consistent set, for some $n < \omega$, so T^* is a finitely consistent Henkin theory. Also as $T \subseteq T^*$ and T is a maximal complete L -theory by assumption, so $\phi \in T^*$ or $\neg \phi \in T^*$ for all L -formulae ϕ . Thus T^* meets all the conditions necessary to apply Lemma 4.3.6, giving some countable model $M \models T$ of equivalence classes of the Henkin constants.

We now verify M omits each $\{\Sigma_j(\bar{x})\}_{j \in I}$. It is for this problem that we chose to use Henkin's proof of the Compactness (Theorem 4.3.2). Let $j \in I$ be given. For any $[c_1], \dots, [c_n] \in M$ and $\Sigma_j(x)$, by the even step of the construction there is some $\sigma_j(\bar{x}) \in \Sigma_j(\bar{x})$ such that $\neg \sigma_j(c_1, \dots, c_n) \in T^*$, so

$$M \models \neg \sigma_j([c_1], \dots, [c_n]) \text{ i.e. } M \not\models \sigma_j([c_1], \dots, [c_n]),$$

so the arbitrary n -tuple $([c_1], \dots, [c_n])$ does not realise $\Sigma_j(\bar{x})$. Thus M omits $\Sigma_j(\bar{x})$, as required. $\square$

The n -types of a theory form a Stone space, a compact Hausdorff totally disconnected topological space. We will steal some of the accessible lemmas this provides, but without the topological setup.

Definition 6.2.4. For some formula ϕ on n free variables $(x_1, \dots, x_n) = \bar{x}$, define the *type closure* of $\phi(\bar{x})$:

$$[\phi(\bar{x})] = \{p(\bar{x}) \in S_n(T) \mid \phi(\bar{x}) \in p(\bar{x})\}.$$

We can immediately list off some properties of the type closure.

Lemma 6.2.5. (1) $\bigcap_{i=1}^k [\phi_i(\bar{x})] = [\bigwedge_{i=1}^k \phi_i(\bar{x})]$

$$(2) \bigcup_{i=1}^k [\phi_i(\bar{x})] = [\bigvee_{i=1}^k \phi_i(\bar{x})]$$

$$(3) S_n(T) \setminus [\phi(\bar{x})] = [\neg \phi(\bar{x})]$$

$$(4) [\phi(\bar{x})] = \{p(\bar{x})\} \text{ if and only if } p(\bar{x}) \text{ is supported by } \phi(\bar{x})$$

Proof. (1), (2), and (3) are all of the same flavour and hopefully intuitive, so we only prove (4). The proof is just unpacking definitions.

For the backward direction, suppose $p(\bar{x})$ is supported by $\phi(\bar{x})$. By definition, for all $\sigma(\bar{x}) \in p(\bar{x})$:

$$T \vdash (\forall \bar{x})(\phi(\bar{x}) \rightarrow \sigma(\bar{x}))$$

Suppose q is some type containing $\phi(\bar{x})$. The type q is maximally consistent so q must contain $\sigma(\bar{x})$ for all $\sigma(\bar{x}) \in p(\bar{x})$. But now $p(\bar{x}) \subseteq q(\bar{x})$ and $p(\bar{x})$ is maximal, so $q(\bar{x}) = p(\bar{x})$. We have shown $[\phi(\bar{x})] = \{p(\bar{x})\}$.

Conversely, suppose $[\phi(\bar{x})] = \{p(\bar{x})\}$. Let $\sigma(\bar{x}) \in p(\bar{x})$ be given. Now if $A \models T$ with $\bar{a} \in A$ such that $A \models \phi(\bar{a})$, then $\phi(\bar{a}) \in \text{tp}_A(\bar{a})$. By our assumption we have

$$\text{tp}_A(\bar{a}) = p(\bar{x}).$$

This gives $A \models \sigma(\bar{a})$ for all $\sigma(\bar{x}) \in p(\bar{x})$. We have shown

$$\begin{aligned} A \models \phi(\bar{a}) \text{ implies } A \models \sigma(\bar{a}) \text{ for all } \bar{a} \in A \\ \implies A \models \phi(\bar{a}) \rightarrow \sigma(\bar{a}) \text{ for all } \bar{a} \in A \\ \implies A \models \forall(\bar{x})(\phi(\bar{x}) \rightarrow \sigma(\bar{x})) \end{aligned}$$

and A was some arbitrary model for T , so

$$T \vdash \forall(\bar{x})(\phi(\bar{x}) \rightarrow \sigma(\bar{x})).$$

The formula $\sigma(\bar{x}) \in p(\bar{x})$ was arbitrary, so $\phi(\bar{x})$ isolates $p(\bar{x})$ as required. $\square$

Lemma 6.2.6. *Let T be some complete consistent theory. If $S_n(T)$ is infinite for some $n \in \mathbb{N}$, then it contains some unsupported type.*

A topologically minded reader should think about the equivalence between the finite intersection property and topological compactness when reading the following proof.

Proof. Suppose, looking for a contradiction, that every type in $S_n(T)$ is supported. Then by Lemma 6.2.5 (4) each $p \in S_n(T)$ has some formula $\phi_p(\bar{x})$ such that $[\phi_p(\bar{x})] = \{p(\bar{x})\}$. Note:

$$(\star) \quad S_n(T) = \bigcup_{p \in S_n(T)} [\phi_p(\bar{x})].$$

Then for every finite subset $P \subseteq S_n(T)$, we have

$$\bigcup_{p \in P} [\phi_p(\bar{x})] \subsetneq S_n(T).$$

Taking complements:

$$\bigcap_{p \in P} (S_n(T) \setminus [\phi_p(\bar{x})]) \neq \emptyset.$$

Applying Lemma 6.2.5 (1) and (3):

$$[\bigwedge_{p \in P} \neg \phi_p(\bar{x})] = \bigcap_{p \in P} [\neg \phi_p(\bar{x})] = \bigcap_{p \in P} (S_n(T) \setminus [\phi_p(\bar{x})]) \neq \emptyset.$$

So some $q(\bar{x}) \in S_n(T)$ contains the formula $\bigwedge_{p \in P} \neg \phi_p(\bar{x})$. Thus some $A \models T$ has $\bar{a}$ some n -tuple over A with $A \models \bigwedge_{p \in P} \neg \phi_p(\bar{a})$.

Now $P \subseteq S_n(T)$ above was arbitrary, so every finite subset of $\Sigma(\bar{x}) = \{\neg \phi_p(\bar{x}) \mid p \in S_n(T)\}$ is realised. Applying Lemma 6.1.8, $\Sigma(\bar{x})$ is a partial type, so contained in some type $\tilde{\Sigma}(\bar{x})$. But then:

$$\tilde{\Sigma}(\bar{x}) \in (S_n \setminus \bigcup_{p \in S_n(T)} [\phi_p(\bar{x})]) \stackrel{(\star)}{=} \emptyset$$

A contradiction, giving us some unsupported type as required. $\square$

6.3. Engeler, Ryll-Nardzewski, Svenonius Theorem. Before proving the main theorem of the chapter, we must define a class of groups with very powerful actions.

Definition 6.3.1. Let G be a group acting on a set X . Note G has a pointwise action on X^n for all $n \in \mathbb{N}$. We say G is *oligomorphic* in its action on X if for all $n \in \mathbb{N}$, X^n has finitely many orbits under the action of G .

Oligomorphic groups admit interesting combinatorial questions which we do not consider here (see [6]). Our work here and in Chapter 7 will show that model theory is a great place to look for examples of oligomorphic groups.

Example 6.3.2. The infinite symmetric group $\text{Sym}(\omega)$ is oligomorphic in its action on ω . In fact, two n -tuples $(x_1, \dots, x_n)$ and $(y_1, \dots, y_n)$ are in the same orbit if and only if for all $i, j \in \{1, \dots, n\}$ the following holds:

$$x_i = x_j \iff y_i = y_j.$$

After all our work, we are now ready to prove the main theorem on ω -categoricity. Our proof is some mixture between the paths taken by Hodges [20] and Tent and Zeigler [40].

Theorem 6.3.3. *Let T be some countable complete L -theory which has infinite models. Then the following are equivalent:*

- (1) *For all $n \in \mathbb{N}$, there are only finitely many L -formulae in n free variables modulo T .*
- (2) *Every countable model of T is ω -saturated.*
- (3) *T is ω -categorical.*
- (4) *For all $n \in \mathbb{N}$, $S_n(T)$ is finite.*
- (5) *Every countable model of T has an oligomorphic automorphism group.*
- (6) *Some countable model of T has an oligomorphic automorphism group.*

This proof will require us to flex all of the muscles we have built throughout the chapter, proceeding as follows:

$$(1) \implies (2) \implies (3) \implies (4) \implies (1) \text{ and then } (4) \implies (5) \implies (6) \implies (4)$$

The first loop of implications is by far the hardest, but we do use the equivalence of (4) and (3) in the second loop of implications.

Proof. (1) $\implies$ (2).

Let $A \models T$ and let B be some finite subset of A . Enumerate B in a tuple $\bar{b}$. Now let $p(\bar{x})$ be some tuple over B . Then $p(\bar{x}) = q(\bar{x}, \bar{b})$, where $q(\bar{x}, \bar{y}) \in S_{n+|B|}(T)$. Now by assumption, there are finitely many formula with $n + |B|$ free variables modulo T . In particular, q contains a finite number of pairwise non-equivalent formulae $\phi_1(\bar{x}, \bar{y}), \dots, \phi_k(\bar{x}, \bar{y})$. We see

$$\left[\bigwedge_{i=1}^k \phi_i(\bar{x}, \bar{y}) \right] = \{q(\bar{x}, \bar{y})\},$$

so by Lemma 6.2.5, the formula $\bigwedge_{i=1}^k \phi_i(\bar{x}, \bar{y})$ supports the type $q(\bar{x}, \bar{y})$. But by Lemma 6.1.8, we know $\{\phi_1(\bar{x}, \bar{b}), \dots, \phi_k(\bar{x}, \bar{b})\}$ is realised in A , by some tuple $\bar{a}$. But then $A \models p(\bar{a})$. The type $p(\bar{x})$ and the finite subset $B \subseteq A$ were arbitrary, so A is ω -saturated.

$$(2) \implies (3).$$

This follows from Lemma 6.1.19.

$$(3) \implies (4).$$

We prove the contrapositive. Suppose that $S_n(T)$ is infinite for some $n \in \mathbb{N}$. By Lemma 6.2.6, we get some unsupported type $p(\bar{x}) \in S_n(T)$. By the Omitting Types Theorem (6.2.3), we may find some countable model $A \models T$ which omits the type $p(\bar{x})$. On the other hand, from the definition of types, we know there is some countable $B \models T$ with $\bar{b}$ a tuple over B such that $B \models p(\bar{b})$. Now $A \not\cong B$, as any isomorphism from B to A would map $\bar{b}$ to some tuple realising $p(\bar{x})$ in A . Thus T is not ω -categorical.

$$(4) \implies (1).$$

Let $n \in \mathbb{N}$ be given and let $p_1, \dots, p_{k_n}$ be an enumeration of the n -types in $S_n(T)$. We claim there are at most $k_n!$ non-equivalent formulae on n free variables modulo T . This comes from noting the following are equivalent:

- $\phi(\bar{x})$ and $\psi(\bar{x})$ are equivalent formulae modulo T .
- $\{i \in \{1, \dots, k_n\} \mid \phi(\bar{x}) \in p_i(\bar{x})\} = \{i \in \{1, \dots, k_n\} \mid \psi(\bar{x}) \in p_i(\bar{x})\}$

The second of these conditions then gives at most $k!$ non-equivalent formulae. We have succeeded in proving the first loop of implications.

(4) $\implies$ (5).

Suppose $S_n(T)$ is finite for all $n \in \mathbb{N}$ and let A be some countable model of T . Our goal is to show $\text{Aut}(A)$ is oligomorphic. As we have already shown the equivalence of (4) and (2), we know A is ω -saturated. Applying Lemma 6.1.12, we see that for all $n \in \mathbb{N}$, the orbits of A^n under $\text{Aut}(A)$ correspond to the n -types of T . This, along with our assumption, gives that $\text{Aut}(A)$ is oligomorphic.

(5) $\implies$ (6).

This is immediate.

(6) $\implies$ (4).

The idea here is to apply the completeness of the theory T .

Suppose T has some countable model A with an oligomorphic automorphism group. Let $n \in \mathbb{N}$ be given. By assumption, there are finitely many orbits of n -tuples in A . By Proposition 6.1.9, any two n -tuples over A in the same orbit under $\text{Aut}(A)$, satisfy the same n -tuple, so A satisfies finitely many n -types $p_1(\bar{x}), \dots, p_k(\bar{x})$. For each $1 \leq i \leq k$, we may choose some $\phi_i(\bar{x})$ such that

$$\phi_i(\bar{x}) \in p_j(\bar{x}) \iff i = j.$$

These formulae $\{\phi_i(\bar{x}) \mid 1 \leq i \leq k\}$ tell you exactly which type you are in, so for all formulae $\psi(\bar{x})$ on n variables and for each $1 \leq i \leq k$, we have

$$(\boxtimes) \quad A \models (\forall \bar{x})(\forall \bar{y}) ((\phi_i(\bar{x}) \wedge \phi_i(\bar{y})) \rightarrow (\psi(\bar{x}) \leftrightarrow \psi(\bar{y}))).$$

We also have the satisfaction

$$(b) \quad A \models (\forall \bar{x}) \bigvee_{i=1}^k \phi_i(\bar{x}),$$

as any n -tuple A satisfies one of the types $p_i(\bar{x})$ for some i . All of these satisfactions noted above are sentences. Recall that any two models of a complete theory are elementarily equivalent (see Lemma 4.2.4), so we must have that every sentence in $(\boxtimes)$, and the sentence in (b) are all in the theory T .

We now show these sentences guarantee the types $p_1, \dots, p_k$ are the only n -types in $S_n(T)$. Let $M \models T$ and some n -tuple $\bar{m} \in M$ be given. Now we have

$$M \models \bigvee_{i=1}^k \phi_i(\bar{m}),$$

so there is some i such that $M \models \phi_i(\bar{m})$. Now recall every finite subset of the type p_i is realised in M by Lemma 6.1.8. In particular for all $\psi(\bar{x}) \in p_i(\bar{x})$, the finite subset

$$\{\phi_i(\bar{x}), \psi(\bar{x})\}$$

is realised by some $\bar{q} \in M$. But now considering $(\boxtimes)$, we have

$$\begin{aligned} M \models (\phi_i(\bar{q}) \wedge \phi_i(\bar{m})) \rightarrow (\psi(\bar{q}) \leftrightarrow \psi(\bar{m})) \text{ and } M \models \phi_i(\bar{q}), \phi_i(\bar{m}), \psi(\bar{q}) \\ \implies M \models \psi(\bar{m}). \end{aligned}$$

Thus $\text{tp}_M(\bar{m}) = p_i(\bar{x})$. The model M and the tuple $\bar{m}$ and the n -tuple $\bar{m}$ were arbitrary, so we have shown:

$$S_n(T) = \{p_1(\bar{x}), \dots, p_k(\bar{x})\},$$

which is finite, as required. $\square$

We will see below one general application of Theorem 6.3.3, then specialise to groups for the remainder of the chapter, repeatedly applying the theorem. Chapter 7 is then another general discussion.

Definition 6.3.4. We say an L -structure A is *locally finite* if $\langle a_1, \dots, a_n \rangle$ is finite for all $a_1, \dots, a_n \in A$. We say A is *strongly locally finite* if there is some $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $f(n)$ is the upper bound for the size of substructure of A generated by n elements.

Corollary 6.3.5. *If T is a consistent complete ω -categorical theory with some countable model A , then A is strongly locally finite.*

Proof. Let $a_1, \dots, a_n \in A$ be given for some $n \in \mathbb{N}$. Let $b \in \langle a_1, \dots, a_n \rangle$ be given, then $\text{tp}(a_1, \dots, a_n, b)$ encodes how b is generated from $a_1, \dots, a_n$. Thus the function

$$\langle a_1, \dots, a_n \rangle \rightarrow S_{n+1}(T) \text{ where } b \mapsto \text{tp}(a_1, \dots, a_n, b)$$

is an injection. By Theorem 6.3.3, $S_{n+1}(T)$ is finite, and so we may well define the function

$$f : \mathbb{N} \rightarrow \mathbb{N} \text{ where } f(n) = |S_{n+1}(T)|.$$

As all countable models of T are isomorphic, the function f witnesses that A is strongly locally finite. $\square$

This agrees with all our work done on groups: Recall Corollary 4.3.7, which says any group of unbounded exponent is elementarily equivalent to some group with an element of infinite order. Groups of unbounded exponent have arbitrarily large cyclic subgroups, so are not strongly locally finite, so not ω -categorical.

6.4. ω -Categorical Groups.

Definition 6.4.1. If G is a countable group with an ω -categorical theory in the natural language of groups, we say G is an ω -categorical group.

In this section we give a thorough investigation of ω -categorical groups. We consider closure under group theoretic constructions (direct sums, subgroups, quotients, characteristic subgroups, wreath products) along the way recovering a specific result of Waskiewicz and Węglorz [43] and applying it to fill in a result of Rosenstein [35].

One should also view this section as a series of applications of Theorem 6.3.3, without which none of these proofs would work.

Lemma 6.4.2. ([20] Section 7.3, exercise 5, for groups) *Let G and H be ω -categorical groups, then $G \oplus H$ is also ω -categorical.*

Proof. We apply Theorem 6.3.3 throughout. It is sufficient to show $\text{Aut}(G \oplus H)$ is oligomorphic. By our assumption, $\text{Aut}(G)$ and $\text{Aut}(H)$ are both oligomorphic. Let $n \in \mathbb{N}$, then G has $k_{n,G}$ orbits of n -tuples in G under $\text{Aut}(G)$ and H has $k_{n,H}$ orbits of n -tuples in H under $\text{Aut}(H)$.

Let $((g_1, h_1), \dots, (g_n, h_n)), ((g'_1, h'_1), \dots, (g'_n, h'_n))$ be some n -tuples in $(G \oplus H)^n$. We claim they are in the same orbit under $\text{Aut}(G \oplus H)$ if both $(g_1, \dots, g_n)$ and $(g'_1, \dots, g'_n)$ are in the same orbit under $\text{Aut}(G)$ and $(h_1, \dots, h_n)$ and $(h'_1, \dots, h'_n)$ are in the same orbit under $\text{Aut}(H)$. Suppose the latter is true, then there is some $\phi \in \text{Aut}(G)$ such that

$$(g_1, \dots, g_n)\phi = (g'_1, \dots, g'_n)$$

and some $\sigma \in \text{Aut}(H)$ such that

$$(h_1, \dots, h_n)\sigma = (h'_1, \dots, h'_n).$$

Then the function

$$\phi\sigma : G \oplus H \rightarrow G \oplus H \text{ where } (g, h) \rightarrow (g\phi, h\sigma)$$

is an automorphism of $G \oplus H$ such that

$$\phi\sigma((g_1, h_1), \dots, (g_n, h_n))\phi\sigma = ((g'_1, h'_1), \dots, (g'_n, h'_n)).$$

We have shown our claim to be true. This gives that there are at most $k_{n,G}k_{n,H}$ orbits of n -tuples of $G \oplus H$ under $\text{Aut}(G \oplus H)$.

We have shown $\text{Aut}(G \oplus H)$ is oligomorphic in its action on $G \oplus H$, so $G \oplus H$ is ω -categorical. $\square$

Very similarly, one can show ω -categoricity is closed under taking direct sums with finite groups and under taking finite wreath products. It is unclear whether ω -categoricity is closed under an arbitrary semidirect product by a finite group. We have a complete classification for ω -categorical countable direct sums of finite groups:

Lemma 6.4.3. ([20], section 7.3, exercise 8)

Let $G = \bigoplus_{n \in \mathbb{N}} G_n$ be a direct sum of finite groups. Then G is ω -categorical if and only if both of the following hold:

- (1) Every group which occurs infinitely often in the direct sum is abelian
- (2) G has bounded exponent

Proof. ($\Rightarrow$) We saw in Corollary 4.3.7 that if (2) does not hold then G is not ω -categorical. Now suppose (1) does not hold. Certainly if $g, h \in G$ are in the same orbit under $\text{Aut}(G)$ then g and h have the same number of conjugates in G . Thus to show G is not ω -categorical we may show the sizes of conjugacy classes in G are unbounded. This argument is similar to one from Rosenstein [34].

Let $N \subseteq \mathbb{N}$ be the infinite index set where $G_n \cong K$ some non-abelian group for all $n \in N$. For all $n \in N$ we get some isomorphism $f_n : G_n \rightarrow K$. Then K is non-abelian, so we may find $g \in K$ such that $|g^K| > 1$. Let $g_n = gf_n$ for all $n \in N$. Now for all $n_1, \dots, n_k \in N$ and for all $x \in G$, from the definition of a direct sum, we may find $y_{n_1} \in G_{n_1}, \dots, y_{n_k} \in G_{n_k}, z \in \bigoplus_{i \in \mathbb{N} \setminus \{n_1, \dots, n_k\}} G_i$ such that $x = y_n z$. Thus

$$(g_{n_1} \dots g_{n_k})^x = (g_{n_1}^{y_1 \dots y_k z}) \dots (g_{n_k}^{y_1 \dots y_k z}) = g_{n_1}^{y_1} \dots g_{n_k}^{y_k}.$$

Varying $y_1, \dots, y_k$, we see $g_{n_1} \dots g_{n_k}$ has exactly $|g^K|^k$ conjugates in G . Thus

$$\{g_{n_1} \dots g_{n_k} \mid k \in \mathbb{N}, n_1, \dots, n_k \in N\}$$

is an infinite subset of G , where no two elements are in the same orbit under $\text{Aut}(G)$. Thus G is not ω -categorical.

($\Leftarrow$) Now supposing (1) and (2) both hold, using the fundamental theorem for finite abelian groups (a special case of Theorem 3.0.3) we may write

$$G = \left(\bigoplus_{i=1}^n H_i \right) \oplus \left(\bigoplus_{i=1}^k (K_i^{(\omega)}) \right).$$

Where K_i is a finite cyclic group of order q for all $1 \leq i \leq k$, and H_i is a finite group for all $1 \leq i \leq n$. Direct sums with finite groups preserve ω -categoricity, so by Lemma 6.4.2 we will be done if we can show $C_q^{(\omega)}$ is ω -categorical for all $q \in \mathbb{N}$. Let $q \in \mathbb{N}$ be given, and write $T = \text{Th}(C_q^{(\omega)})$. We note the following sentence is in T :

$$((\forall a)(\forall b)ab = ba) \wedge \left((\forall x)((\bigvee_{m|q} x^m = 1) \wedge (\exists y)((\bigvee_{1 \leq m < q} (y^m \neq 1)) \wedge (\bigvee_{1 \leq m \leq q} x = y^m))) \right).$$

Thus any countable model of $A \models T$ is a countable abelian group with exponent q , where any element of order less than q is a power of an element with order q . We know from Theorem 3.0.4 that any abelian group of bounded exponent is a direct sum of finite cyclic groups, so

$$A \cong \bigoplus_{i < \omega} C_{l_i},$$

for some $\{l_i \mid i < \omega\}$. Our other condition ensures $l_i = q$ for all $i < \omega$, so $A \cong C_q^{(\omega)}$, as required. $\square$

We will see an example on how ω -categoricity is well-behaved under taking characteristic subgroups, but we need the following quick lemma.

Lemma 6.4.4. *Let G be a group and $H \text{ char } G$, then for all $f \in \text{Aut}(G)$, the map:*

$$\hat{f} : G/H \rightarrow G/H \text{ where } (Hx)\hat{f} = H(xf)$$

is a well-defined automorphism of G/H .

Lemma 6.4.5. ([20], section 7.3, exercise 9)

ω -categoricity is closed under taking characteristic subgroups, and closed under taking quotients by infinite characteristic subgroups. In particular, for G a ω -categorical group, G has finitely many characteristic subgroups, and a finite series of characteristic subgroups $G = G_0 > \dots > G_n = 1$, where each factor is characteristically simple and ω -categorical.

Proof. Let G be some ω -categorical group and $H \text{ char } G$. Then every automorphism of G induces some automorphism of H . so for all $n \in \mathbb{N}$ there can be no more orbits of n -types in H under $\text{Aut}(H)$ than under $\text{Aut}(G)$. By assumption $\text{Aut}(G)$ is oligomorphic, so $\text{Aut}(H)$ is oligomorphic, so H is ω -categorical. We see ω -categoricity is closed under taking quotients by characteristic subgroups by a similar argument, using Lemma 6.4.4.

Now as characteristic subgroups are closed under automorphisms, each characteristic subgroup of G contains at least one orbit of 1-tuples. As $\text{Aut}(G)$ is oligomorphic, there can then be at most finitely many characteristic subgroups.

Now set $G_0 = G$. Given $G_i \text{ char } G$, if G_i is characteristically simple, then set $G_{i+1} = 1$, and otherwise note that G_i is ω -categorical by above, so has finitely many proper characteristic subgroups, so there must be some $H \text{ char } G$ maximal with respect to characteristic subgroups of G . Set $G_{i+1} = H \text{ char } G$. As G has finitely many characteristic subgroups, this process must terminate, giving some characteristic series $G = G_0 > \dots > G_n = 1$. Each term is ω -categorical, so G_i/G_{i+1} is ω -categorical by above. Using the correspondence theorem, we see G_i/G_{i+1} characteristically simple by the maximality of G_{i+1} among characteristic subgroups in G . $\square$

Olin [31] first showed ω -categoricity is not closed under taking subgroups, but claimed that ω -categoricity is closed under taking quotients. Building on Olin's work, Rosenstein [35] showed this is not true.

Theorem 6.4.6. *ω -categoricity is not closed under taking subgroups or quotients.*

We work towards a proof of the theorem. We follow Rosenstein [35], filling in some extra details to recover a result of Waskiewicz and Węglorz [43] and splashing on a different coat of paint. The following construction will seem pulled from a hat, but is in fact one of the central examples in ω -categorical groups.

Consider the Cantor set $\mathfrak{C}$, whose elements can be represented as infinite binary sequences $\bar{x}$. For some finite binary word w , we can define

$$[w] = \{w\bar{x} \mid \bar{x} \in \mathfrak{C}\}.$$

This is the *cone* at w , containing all the points under w in Figure 3. Now all the cones at finite binary words form the standard clopen basis for the standard topology on $\mathfrak{C}$.

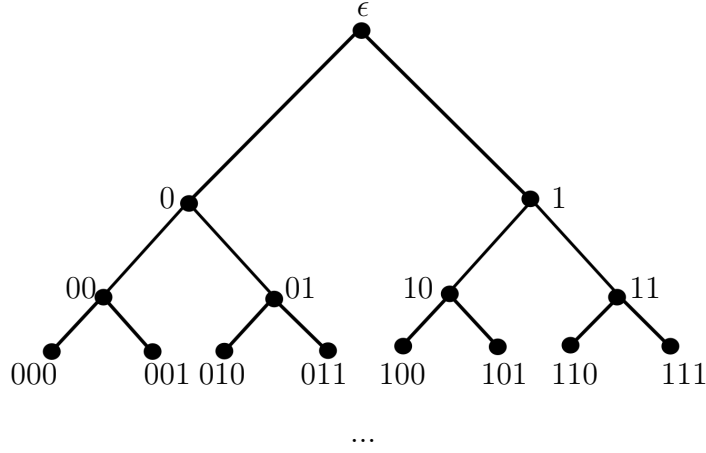


FIGURE 3. The Cantor set $\mathfrak{C}$

Definition 6.4.7. Let G be some ω -categorical group. Write $\mathfrak{C}$ for the middle third Cantor set. Define the group:

$$\text{Maps}(\mathfrak{C}, G) := \{f : \mathfrak{C} \rightarrow G \mid f \text{ is continuous with respect to the discrete topology on } G\}$$

where group multiplication is pointwise multiplication of the functions. We write the action of $\text{Maps}(\mathfrak{C}, G)$ on $\mathfrak{C}$ on the right.

For those so inclined, $\text{Maps}(\mathfrak{C}, G)$ is isomorphic to the Boolean power of the group G by the countable atomless Boolean algebra.

An easy topological argument shows that for $f \in \text{Maps}(\mathfrak{C}, G)$, we may find some finite set of words $w_1, \dots, w_n$, such that

$$\mathfrak{C} = \bigsqcup_{i=1}^n [w_i]$$

and such that f is constant on $[w_i]$ for each $1 \leq i \leq n$. Thus an element can be represented by some finite binary tree, with leaves labelled by elements of G .

If we have some $w_1, \dots, w_n$ as above, we can define some larger set of words with the same property:

$$(\bigcirc) \quad w_1, \dots, w_{i-1}, w_i 0, w_i 1, w_{i+1}, \dots, w_n.$$

As f was constant on $[w_i]$, it is constant on $[w_i 0]$ and $[w_i 1]$, taking the same value on both cones. This means there are infinitely many finite binary trees representing f . We see a visual example in Figure 4.

We now define two key automorphisms of $\text{Maps}(\mathfrak{C}, G)$:

($\mathfrak{C}$ -type) For $\phi : \mathfrak{C} \rightarrow \mathfrak{C}$ a homeomorphism, we define some corresponding automorphism of $\text{Maps}(\mathfrak{C}, G)$:

$$\Theta_\phi : f \mapsto \tilde{f} \text{ where } x\tilde{f} = x\phi^{-1}f \text{ for all } x \in \mathfrak{C}.$$

(G -type) For $\psi \in \text{Aut}(G)$ and w some finite binary word, we define some corresponding automorphism of $\text{Maps}(\mathfrak{C}, G)$:

$$\Phi_{w,\psi} : f \mapsto \tilde{f} \text{ where } x\tilde{f} = \begin{cases} x\psi & \text{if } x \in \hat{w} \\ xf & \text{otherwise} \end{cases}$$

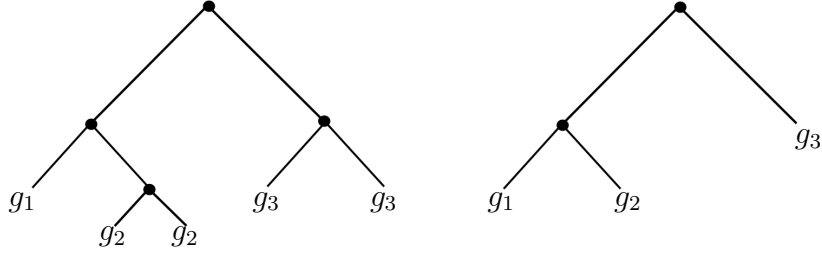


FIGURE 4. Two trees representing the same element in $\text{Maps}(\mathfrak{C}, G)$, where $g_1, g_2, g_3 \in G$

We now have everything needed to prove the following lemma. We are filling in Wilson's proof [44].

Lemma 6.4.8. *Let G be an ω -categorical group, then $\text{Maps}(\mathfrak{C}, G)$ is also ω -categorical.*

Proof. Let $n \in \mathbb{N}$ be given. Let $G_1, \dots, G_k$ be the orbits of G^n under $\text{Aut}(G)$. By Theorem 6.3.3, we need only show there are finitely many orbits of $(\text{Maps}(\mathfrak{C}, G))^n$ under the automorphism group of $\text{Maps}(\mathfrak{C}, G)$. Let $\bar{f} = (f_1, \dots, f_n)$ and $\bar{h} = (h_1, \dots, h_n)$ be some n -tuples over $\text{Maps}(\mathfrak{C}, G)$. We claim $\bar{f}$ and $\bar{h}$ are in the same orbit if

$$(\Delta) \quad \{i \in \{1, \dots, k\} \mid G_i \cap \text{im}(\bar{f})\} = \{i \in \{1, \dots, k\} \mid G_i \cap \text{im}(\bar{h})\}$$

If this were true, as there are at most $k!$ sets of the form in (Δ) , there are at most $k!$ orbits in $\text{Maps}(\mathfrak{C}, G)$.

Now suppose (Δ) holds. Our strategy is to line up the cones of $\bar{f}$ and $\bar{h}$ using the $\mathfrak{C}$ -type automorphisms, then line up the elements labelling the cones with the G -type automorphisms.

Write $t = |G|^n$. Choose some set of finite words $w_1, \dots, w_t$ such that

$$\mathfrak{C} = \bigsqcup_{i=1}^t [w_i].$$

We will find some $\mathfrak{C}$ -type automorphisms Θ_ϕ, Θ_ψ such that $(\bar{f})\Theta_\phi$ and $(\bar{h})\Theta_\psi$ are both constant on the cones $[w_1], \dots, [w_t]$.

Now for each $\bar{g} = (g_1, \dots, g_n) \in G^n$, define the set

$$F_{\bar{f}}(\bar{g}) = \{x \in \mathfrak{C} \mid f_i(x) = g_i \text{ for all } 1 \leq i \leq n\}.$$

Note that $F_{\bar{f}}(\bar{g})$ is a clopen set, so is some union of cones. Define the set

$$X_{\bar{f}} = \{\bar{g} \in G^n \mid F_{\bar{f}}(\bar{g}) \neq \emptyset\},$$

so we can write $X_{\bar{f}} = \{\bar{g}_1, \dots, \bar{g}_s\}$ for some $1 \leq s \leq t$.

If $s = t$, then define the function:

$$\phi : \mathfrak{C} \rightarrow \mathfrak{C} \text{ where } ([w_i])\phi = F_{\bar{f}}(\bar{g}_i).$$

If $s < t$ elements, then by splitting the cones as discussed in $(\bigcirc)$, split $F_{\bar{f}}(\bar{g}_1)$ into $t - s + 1$ clopen sets

$$F_{\bar{f},1}(\bar{g}_1), \dots, F_{\bar{f},t-s+1}(\bar{g}_1).$$

Now define the function:

$$\phi : \mathfrak{C} \rightarrow \mathfrak{C} \text{ where } ([w_i])\phi = \begin{cases} F_{\bar{f},i}(\bar{g}_1) & \text{if } 1 \leq i \leq t-s+1 \\ F_{\bar{f}}(\bar{g}_{i-(t-s+1)}) & \text{otherwise} \end{cases}$$

In either case the function ϕ maps clopen sets to clopen sets, so is a homeomorphism. Thus the $\mathfrak{C}$ -type automorphism of $\text{Maps}(\mathfrak{C}, G)$, write Θ_ϕ , sends $\bar{f}$ to some tuple of elements each constant on the cones $[w_1], \dots, [w_t]$. One can analogously define ψ and Θ_ψ which gives $(\bar{h})\Theta_\psi$ constant on the cones $[w_1], \dots, [w_t]$. We can also carefully note that

$$x\bar{f} = x\bar{h} \text{ if and only if } x(f\Theta_\phi) = x(\bar{h}\Theta_\psi).$$

So we have, by (Δ) , that

$$\{i \in \{1, \dots, k\} \mid G_i \cap \text{im}(\bar{f}\Theta_\phi)\} = \{i \in \{1, \dots, k\} \mid G_i \cap \text{im}(\bar{h}\Theta_\psi)\}.$$

We may extend the cones $\{[w_1], \dots, [w_t]\}$ to $\{[v_1], \dots, [v_l]\}$ such that

$$\{i \in \{1, \dots, l\} \mid G_j \cap [v_i](\bar{f}\Theta_\phi)\} = \{i \in \{1, \dots, l\} \mid G_j \cap [v_i](\bar{h}\Theta_\psi)\}$$

for each $1 \leq j \leq k$. From this we may define γ some permutation of the cones $[v_1], \dots, [v_l]$ such that

$$(x)(\bar{f}\Theta_\phi\Theta_\gamma) \in G_j \text{ if and only if } x(\bar{h}\Theta_\psi\Theta_\gamma) \in G_j$$

for all $1 \leq j \leq k$.

Then for each $1 \leq i \leq l$, we may find some $\sigma_i \in \text{Aut}(G)$ such that

$$([v_i](\bar{f}\Theta_\phi\Theta_\gamma))\sigma_i = [v_i](\bar{h}\Theta_\psi\Theta_\gamma).$$

Then taking the product of G -type automorphisms

$$\Phi = \prod_{i=1}^l \Phi_{v_i, \sigma_i},$$

we finally get

$$\bar{f}\Theta_\phi\Theta_\gamma\Phi = \bar{h}\Theta_\psi\Theta_\gamma \implies \bar{f}\Theta_\phi\Theta_\gamma\Phi\Theta_\gamma\Theta_\psi = \bar{h}.$$

So $\bar{f}$ and $\bar{h}$ are in the same orbit under the automorphism group of $\text{Maps}(\mathfrak{C}, G)$, as required. $\square$

Proof of Theorem 6.4.6. Let G be any non-abelian finite simple group, then by Lemma 6.4.8, $\text{Maps}(\mathfrak{C}, G)$ is ω -categorical. Define the following subgroup of $\text{Maps}(\mathfrak{C}, G)$:

$$K = \{f \in \text{Maps}(\mathfrak{C}, G) \mid f \text{ is constant on } [1^n 0] \text{ for all } n < \omega\}$$

where we use the notation $1^0 = \epsilon$, the empty word. We see a visual example of an element of K in Figure 5.

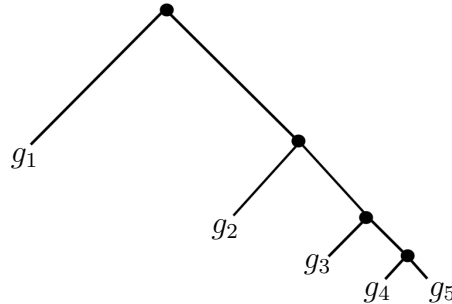


FIGURE 5. A typical element of K , where $g_1, g_2, g_3, g_4, g_5 \in G$.

We claim K is not ω -categorical. This comes from the fact it that K is nearly isomorphic to $G^{(\omega)}$, which we know is not ω -categorical by Theorem 6.4.3. Choose $g \in G$ with $k > 1$ conjugates. Then for each $n \in \mathbb{N}$, define

$$f_n \in \text{Maps}(\mathfrak{C}, G) \text{ where } f(x) = \begin{cases} g & \text{if } x \in [1^i 0] \text{ for } 1 \leq i \leq n \\ 1 & \text{otherwise} \end{cases}$$

i.e. with a tree-structure as in Figure 5, working left to right f_n labels the first n leaves with g and labels all the other leaves with 1. As in the proof of Theorem 6.4.3, we see f_n has k^n conjugates for all $n \in \mathbb{N}$. The number of conjugates is preserved under automorphism, so f_i and f_j cannot be in the same orbit under $\text{Aut}(K)$ for $i \neq j$. We have shown $\text{Aut}(K)$ has infinitely many orbits on K , so the group K cannot be ω -categorical by Theorem 6.3.3.

As K is a subgroup of $\text{Maps}(\mathfrak{C}, G)$, we have shown ω -categoricity is not closed under taking subgroups. To see ω -categoricity is not closed under taking quotients, we show there is some surjective homomorphism

$$\phi : \text{Maps}(\mathfrak{C}, G) \rightarrow K.$$

Then by the first isomorphism theorem

$$G / \ker(\phi) \cong \text{im}(\phi) = K$$

and we will be done. We define ϕ as follows:

$$(x)(f\phi) = (1^n \bar{0})f \text{ for } x \in [1^n 0],$$

where $\bar{0}$ is the infinite zero word. The function ϕ picks out images of f on the left branch of each of the cones in Figure 5.

The check that ϕ is a homomorphism is quick. Let $f_1, f_2 \in \text{Maps}(\mathfrak{C}, G)$ be given. Let $x \in \mathfrak{C}$ be given, then we may find $n < \omega$ such that $x \in [1^n 0]$. We see

$$x((f_1 \cdot f_2)\phi) = (1^n \bar{0})(f_1 \cdot f_2) = (1^n \bar{0})f_1 \cdot ((1^n \bar{0})f_2) = (x(f_1\phi)) \cdot (x(f_2\phi)) = x((f_1\phi) \cdot (f_2\phi)).$$

The element $x \in \mathfrak{C}$ was arbitrary, so

$$(f_1\phi) \cdot (f_2\phi) = (f_1 \cdot f_2)\phi,$$

confirming ϕ is a homomorphism.

Note that the restriction $\phi|_K$ is the identity on K , so the map ϕ is certainly surjective. By above, G has some quotient which is not ω -categorical. $\square$

We discussed above that $\text{Maps}(\mathfrak{C}, G)$ is a central construction for ω -categorical groups. The classification of characteristically simple ω -categorical groups, first stated by Wilson [44], then proved by Apps [1] explains why this is so.

Theorem 6.4.9. *Every characteristically simple ω -categorical group is isomorphic to one of the following:*

- (1) *An elementary abelian p -group for some prime p*
- (2) *$\text{Maps}(\mathfrak{C}, G)$ for some non-abelian finite simple group G .*
- (3) *A p -group G such that $G = G'$ (i.e. G is perfect) for some prime p .*

We dare not venture towards this proof, which requires the classification of finite simple groups. We have met characteristically simple ω -categorical groups of type (1) and (2) on our journey so far; what about type (3)? It is unknown whether there even exists a locally finite characteristically simple perfect p -group of finite exponent for p some prime.

This leads to the following conjecture of Wilson.

Conjecture 6.4.10. *There are no type (3) characteristically simple ω -categorical groups.*

If we try to step out further, say to look at nilpotent ω -categorical groups, things become much more complicated. For example, Saracino and Wood [36] construct uncountably many pairwise non-isomorphic nilpotent ω -categorical groups. Archer and Macpherson [2] have a model theoretic result (whose exact details would need too many extra definitions) which warns that there may be no possible nice classification for soluble ω -categorical groups.

7. CONNECTIONS

Here we borrow an idea from Johnstone [22], who recalls a story from E. J. Robinson about South India:

'Four men, who had been blind from birth, wanted to know what an elephant was like; so they asked an elephant-driver for information. He led them to an elephant, and invited them to examine it; so one man felt the elephant's leg, another its trunk, another its tail and the fourth its ear. Then they attempted to describe the elephant to one another. The first man said "The elephant is like a tree". "No," said the second, "the elephant is like a snake". "Nonsense!" said the third, "the elephant is like a broom". "You are all wrong," said the fourth, "the elephant is like a fan". And so they went on arguing amongst themselves, while the elephant stood watching them quietly.'

In chapter 5, we investigated something which felt like a broom. In Chapter 6, we investigated something which felt like a snake. In this chapter, we see that we have been investigating an elephant this whole time.

7.1. Homogeneity and ω -categoricity.

Lemma 7.1.1. *If A is some strongly locally finite L -structure for L a finite language, then $A \equiv B$ implies $\text{age}(A) = \text{age}(B)$. In particular, B is also strongly locally finite.*

Proof. Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be the function witnessing that A is strongly locally finite. Let $n \in \mathbb{N}$ be given, and write $\bar{x} = (x_1, \dots, x_n)$. Write

$$\text{age}_n(A) = \{D \in \text{age}(A) \mid D \text{ is generated by some set of size } n\}.$$

Now as L is a finite language, up to isomorphism there are finitely many L -structures of order at most $f(n)$. Thus the following set is finite:

$$I := \{(D, \bar{d}) \mid D \in \text{age}_n(A) \text{ and } \bar{d} \text{ is a generating set for } D \text{ of size } n\}$$

Let $(D, \bar{d}) \in I$ be given and write $D = \{m_1, \dots, m_k\}$. Recall from Lemma 4.2.7, D is completely characterised by some sentence

$$\lambda_D : (\exists y_1) \dots (\exists y_k) \psi_D(y_1, \dots, y_k)$$

We know $D = \langle \bar{d} \rangle$, so for each $1 \leq i \leq k$, we can write $m_i = t_i(\bar{d})$. Then we encode the generation of D by $\bar{d}$ in the formula:

$$\psi_{(D, \bar{d})}(\bar{x}) : \psi_D(t_1(\bar{x}), \dots, t_k(\bar{x}))$$

So some L -structure B with elements $\bar{b}$ has $B \models \psi_{(D, \bar{d})}(\bar{b})$ if and only if $\langle \bar{b} \rangle_B \cong D$.

Now as $A \equiv B$ we have

$$A \models (\forall \bar{x}) \bigvee_{(D, \bar{d}) \in I} \psi_{(D, \bar{d})}(\bar{x}), \text{ which implies } B \models (\forall \bar{x}) \bigvee_{(D, \bar{d}) \in I} \psi_{(D, \bar{d})}(\bar{x}),$$

and from our discussion, the above sentence holding in B forces $\text{age}_n(B) \subseteq \text{age}_n(A)$. We now show the other inclusion. For all $(D, \bar{d}) \in I$, note that $A \models (\exists \bar{x}) \psi_{(D, \bar{d})}(\bar{x})$, so $B \models (\exists \bar{x}) \psi_{(D, \bar{d})}(\bar{x})$. Thus there exists $\bar{b} \in B$ such that $B \models \psi_{(D, \bar{d})}(\bar{b})$. By above, this means $\langle \bar{b} \rangle_B \cong D$, so $D \in \text{age}_n(B)$. The tuple $(D, \bar{d}) \in I$ was arbitrary, so we have $\text{age}(A)_n \subseteq \text{age}_n(B)$.

We have shown $\text{age}_n(A) = \text{age}_n(B)$ for all $n \in \mathbb{N}$, so certainly $\text{age}(A) = \text{age}(B)$. □

This implication does not hold if we drop the strongly locally finite assumption: Recall

$$\mathbb{Z} \equiv \mathbb{Z} \oplus \mathbb{Q}$$

is an elementary equivalence in the natural language of groups. The left has age $\{1, \mathbb{Z}\}$, the right has age $\{1, \mathbb{Z}, \mathbb{Z} \oplus \mathbb{Z}\}$.

Remark 7.1.2. We can apply a similar trick to the one used in the proof of Lemma 4.2.7 to strengthen Lemma 7.1.1 to apply to arbitrary languages.

We extract from the proof of the above lemma a useful fact.

Remark 7.1.3. Let D be some L -structure finitely generated by $\bar{d}$. There is some formula $\psi_{(D, \bar{d})}(\bar{x})$, which is in fact quantifier-free, such that any L -structure A with $\bar{a} \in A$ has $A \models \psi_{(D, \bar{d})}(\bar{a})$ if and only if $\langle \bar{a} \rangle_A \cong D$. This encodes one way of embedding D into A . These formulae are key tools in proving the following theorem.

Theorem 7.1.4. (*Connections Theorem*) *Let L be some finite signature and A a countable L -structure where $\text{age}(A) := \mathbf{K}$ is a Fraïssé class with at most countably many automorphism types. The following are equivalent:*

- (1) *A is homogeneous and strongly locally finite*
- (2) *$\text{Th}(A)$ is ω -categorical and has quantifier elimination.*

Proof. (1) $\implies$ (2).

Suppose A is homogeneous and strongly locally finite, with this fact witnessed by some $f : \mathbb{N} \rightarrow \mathbb{N}$.

ω -Categoricity

We know from Lemma 5.1.16 that homogeneity is equivalent to weak homogeneity. We will build some theory T whose models are the weakly homogeneous structures with age $\mathbf{K}$.

Let $n \in \mathbb{N}$ be given. Write $\bar{x} = (x_1, \dots, x_n)$. Let us inherit the notation from the proof of Lemma 7.1.1. Define a theory whose models have age $\mathbf{K}$:

$$T_{1,n} = \{(\exists \bar{x})\psi_{(D, \bar{d})}(\bar{x}) \mid (D, \bar{d}) \in I\} \cup \{(\forall \bar{x}) \bigvee_{(D, \bar{d}) \in I} \psi_{(D, \bar{d})}(\bar{x})\}$$

Now we add formulae which force our theory to force a one-point weak homogeneity:

$$T_{2,n} = \{(\forall \bar{x})(\psi_{\langle \bar{a} \rangle, \bar{a}}(\bar{x}) \rightarrow (\exists y)\psi_{\langle \bar{a}b \rangle, \bar{a}b}(\bar{x}, y)) \mid b \in A \text{ and } \bar{a} \text{ is an } n\text{-tuple over } A\}.$$

Now consider that

$$\bigcup_{n \in \mathbb{N}} (T_{1,n} \cup T_{2,n}) \subseteq \text{Th}(A).$$

As $\text{Th}(A)$ contains every $T_{1,n}$, we know from Lemma 7.1.1 that any model of T has age $\mathbf{K}$. We must now show that any model of $\text{Th}(A)$ is weakly homogeneous. As A is a homogeneous countable L -structure, by Fraïssé's Theorem (5.2.1) A is isomorphic to any homogeneous L -structure with age $\mathbf{K}$. This will give that A is isomorphic to every model of $\text{Th}(A)$, and hence $\text{Th}(A)$ is ω -categorical.

Towards proving the weak homogeneity of any model of $\text{Th}(A)$, let $M \models \text{Th}(A)$ be given. Let C and D be finitely generated substructures of M , and suppose $f : C \hookrightarrow M$ is an embedding. Now choose some generating set $c_1, \dots, c_n$ for C and enlarge this to a generating set $c_1, \dots, c_n, d_1, \dots, d_k$ for D . Define $C = D_0$, $D_i = \langle c_1, \dots, c_n, d_1, \dots, d_i \rangle$ for each $1 \leq i \leq k$. Set $f_0 = f$. We show inductively for all $0 \leq i \leq k-1$ that there exists some $f_{i+1} : D_{i+1} \hookrightarrow M$ extending f_i , and hence extending $f_0 = f$.

Now suppose we have some embedding $f_i : D_i \rightarrow M$ extending f . We build some $f_{i+1} : D_{i+1} \rightarrow M$ extending f_i .

Write $\bar{d}_i = (c_1, \dots, c_n, d_1, \dots, d_i)$. As $M \models T_{2,n}$ for all $n \in \mathbb{N}$ we have

$$M \models (\forall \bar{x})(\psi_{(D_i, \bar{d}_i)}(\bar{x}) \rightarrow (\exists y)\psi_{(D_{i+1}, \bar{d}_i d_{i+1})}(\bar{x}, y))$$

and we know f_i is an embedding, giving

$$M \models \psi_{(D_i, \bar{d}_i)}(f_i(\bar{d}_i)).$$

The above two satisfactions in M together imply

$$M \models (\exists y)\psi_{(D_{i+1}, \bar{d}_i d_{i+1})}(f_i(\bar{d}_i), y)$$

so there exists some $m \in M$ such that

$$(\diamond) \quad M \models \psi_{(D_{i+1}, \bar{d}_i d_{i+1})}(f_i(\bar{d}_i), m).$$

Then define $f_{i+1}(d) = f_i(d)$ for all $d \in \bar{d}_i$, and define $f_{i+1}(d_{i+1}) = m$. By $(\diamond)$, we know this defines some isomorphism $f_{i+1} : D_{i+1} \rightarrow \langle f_i(\bar{d}_i), m \rangle_M$, which defines an embedding from D_{i+1} into M which extends f_i , and hence extending $f_0 = f$, this closes the induction loop.

Thus $f_k : D = D_k \hookrightarrow M$ is some embedding extending f , witnessing the weak homogeneity of M and proving the claim.

Quantifier Elimination

The theory $\text{Th}(A)$ is complete, so certainly every sentence is equivalent to $\perp$ or $\top$ modulo $\text{Th}(A)$. Now consider some formula $\phi(\bar{x})$ over the n -tuple of variables $\bar{x}$ for $n \in \mathbb{N}$. Define

$$X = \{\bar{a} \in A \mid \phi(\bar{a})\}$$

then as $\text{Th}(A)$ is ω -categorical, there are finitely many orbits of n -tuples, so we may reduce the set X to some finite subset X' which contains at most one representative from each orbit. We claim:

$$\phi(\bar{x}) \equiv_{\text{Th}(A)} \bigvee_{\bar{a} \in X'} \psi_{(\langle \bar{a} \rangle_A, \bar{a})}(\bar{x}).$$

The right hand side is quantifier free, so we will be done if the claim is true. To see the claim is true, let $\bar{m}$ be some n -tuple over M . We must show

$$M \models \phi(\bar{m}) \iff M \models \bigvee_{\bar{a} \in X'} \psi_{(\langle \bar{a} \rangle_A, \bar{a})}(\bar{m}).$$

The theory $\text{Th}(A)$ is ω -categorical, so A is ω -saturated (see Theorem 6.3.3), so we may find $\bar{b} \in A$ such that $\text{tp}_M(\bar{m}) = \text{tp}_A(\bar{b})$. Thus, we are done if we can show

$$(\otimes) \quad A \models \phi(\bar{b}) \iff A \models \bigvee_{\bar{a} \in X'} \psi_{(\langle \bar{a} \rangle_A, \bar{a})}(\bar{b}).$$

This is now more clear. If $A \models \phi(\bar{b})$ then $\bar{b} \in X$, so there is some $\bar{d}$ in the orbit of $\bar{b}$ under $\text{Aut}(A)$ such that $\bar{d} \in X'$. Then $A \models \psi_{(\langle \bar{d} \rangle_A, \bar{d})}(\bar{b})$, and we have proven the forward direction of the implication $(\otimes)$. The backward direction follows from this argument in reverse.

$$(2) \implies (1).$$

Suppose $\text{Th}(A)$ is ω -categorical and has quantifier elimination. We must show A is strongly locally finite and homogeneous.

Strongly Locally Finite

This was proven in Corollary 6.3.5.

Homogeneity

Let $\bar{a} = (a_1, \dots, a_n)$ and $\bar{b} = (b_1, \dots, b_n)$ be tuples in A such that

$$(\natural) \quad a_i \mapsto b_i \text{ gives an isomorphism } \langle \bar{a} \rangle \rightarrow \langle \bar{b} \rangle.$$

We claim $\text{tp}_A(\bar{a}) = \text{tp}_A(\bar{b})$. We know from Theorem 6.3.3 that A is ω -saturated. Thus if the claim is true, by Lemma 6.1.12, the tuples $\bar{a}$ and $\bar{b}$ are in the same orbit under $\text{Aut}(A)$, i.e. our isomorphism in (‡) can be extended to an automorphism of A .

Towards proving the claim, let $\phi(\bar{x})$ some L -formula be given. As $\text{Th}(A)$ has quantifier elimination, $\phi(\bar{x})$ is equivalent modulo $\text{Th}(A)$ to some quantifier-free formula $\sigma(\bar{x})$. We see

$$A \models \sigma(\bar{a}) \iff \langle \bar{a} \rangle \models \sigma(\bar{a}) \xLeftrightarrow{(h)} \langle \bar{b} \rangle \models \sigma(\bar{b}) \iff A \models \sigma(\bar{b}),$$

where the unlabelled equivalences follow from $\sigma(\bar{x})$ being quantifier free. This gives the following:

$$A \models \phi(\bar{a}) \iff A \models \sigma(\bar{a}) \iff A \models \sigma(\bar{b}) \iff A \models \phi(\bar{b}).$$

We have shown

$$\phi(\bar{x}) \in \text{tp}_A(\bar{a}) \iff \phi(\bar{x}) \in \text{tp}_A(\bar{b}),$$

as required to prove the claim. As $\langle \bar{a} \rangle$ and $\langle \bar{b} \rangle$ were two arbitrary finitely generated substructures of A , we conclude that A is homogeneous. $\square$

The above theorem gives us plenty of good news. Most importantly, we get a logical characterisation of some non-logical properties, which tells us that model theory is worth doing!

The theorem also tells us that Fraïssé's Theorem (5.2.1) is a bountiful tool for building ω -categorical relational structures, and therefore for building oligomorphic groups. These structures will also have quantifier elimination. We saw in the proof of Theorem 4.5.5 that explicitly proving quantifier elimination can take up lots of paper and time.

We will end the main body of the project by outlining a nice investigation where some of our techniques have been used.

We saw that in strongly locally finite structures, the first-order theory controls the age. Fraïssé's Theorem (5.2.1) goes the other way, saying that if we put some restrictions on the age, it controls the isomorphism class (and hence the first-order theory). Can we classify those structures where the age behaves rigidly? To this end, the following notion was introduced by Cameron in a 1984 seminar talk (see the introduction of [3]).

Definition 7.1.5. A countable strongly locally finite L -structure A is *absolutely ubiquitous* if any countable L -structure B with $\text{age}(A) = \text{age}(B)$ in fact has $A \cong B$.

The natural example is a Fraïssé limit in some relational language. Hodkinson and Macpherson [21] characterise those relational structures which are absolutely ubiquitous. For groups, Macpherson [26] proved the following.

Theorem 7.1.6. *Let G be a countable absolutely ubiquitous group, then G is ω -categorical and has some characteristic subgroup of finite index which is a direct product of elementary abelian groups of infinite rank.*

We have the tools to see that any absolutely ubiquitous group G is ω -categorical. Lemma 7.1.1 tells us that any group H with $G \equiv H$ must have $\text{age}(G) = \text{age}(H)$, which implies $G \cong H$ by the absolute ubiquity of G .

The other part of the theorem is much harder, and relies on Wilson's classification of characteristically simple ω -categorical groups (Theorem 6.4.9), and a result similar to our Lemma 6.4.5.

8. CONCLUSION

By now, the reader is familiar with the main techniques of model theory, including quantifier elimination, omitting types, and back-and-forth constructions. We have seen many of the defining results of classical model theory, including Fraïssé's Theorem (5.2.1), the Engeler, Ryll-Nardzewski, Svenonius Theorem (6.3.3), and the Connections Theorem (7.1.4).

We applied these techniques and theorems to homogeneous groups (with a focus on Hall's universal group) and ω -categorical groups. In the case of ω -categorical groups, we were near the edge of the theory. With more heavy model-theoretic restrictions (often related to the classification theory of Shelah we discussed in the introduction) there has been more progress [12], but Wilson's conjectures are still unresolved.

The crown jewel of the project was the Connections Theorem, which said a structure is homogeneous and strongly locally finite if and only if it is ω -categorical and has quantifier elimination. The first two conditions require no model theory at all and could be quickly explained to any pure mathematician, while the second two conditions are model-theoretic and required a lot of this project to properly understand. This should motivate the reader to see model theory not as an island separate from the rest of mathematics, but as something hiding beneath all of the mathematics we do every day.

An especially interesting research programme which could be a next step for the author (or the interested reader) is the generalisation of some of the results in this project to semigroups.

The class of finite semigroups does not have (AP), but Dolinka and Gray [11] build semigroup and inverse semigroup analogues of Hall's group H , and show they are maximally homogeneous (where maximal refers to those the subsets of the age whose partial automorphisms admit automorphisms). The author would love to see whether there is also an analogue of verbal completeness in these cases.

Gould and Quinn-Gregson [18] have recently considered ω -categorical semigroups, making heavy use of the Engeler, Ryll-Nardzewski, Svenonius Theorem (6.3.3), picking up many results similar to those we had for groups in Chapter 6. With these foundations, it could be worthwhile to see if Macpherson's investigations into absolutely ubiquitous groups [26] can be translated into the semigroup case.

The author hopes the reader has enjoyed this project as much as he has.

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APPENDIX A. HOMOGENEITY AND ORDER CONJUGACY IN FINITE GROUPS

Recall Hall's group H is a very rich group which is order conjugate and homogeneous. In this appendix we see how order conjugacy and homogeneity are very restrictive when translated to finite groups.

This was attached as an appendix as our investigation will include no model theory.

Theorem A.0.1. *A finite group G is homogeneous if and only if $G \cong U \oplus V$ where U is abelian with all Sylow subgroups homocyclic, and either V is trivial or is one of the following:*

- (1) $W \rtimes C_{2^n}$ where W is abelian of odd order with all Sylow subgroups homocyclic, and C_{2^n} acts by inversion on W .
- (2) Q_8 and $Q(64)$, a Sylow 2-subgroup of $PSU(3, 16)$.
- (3) $A_4, Q_8 \rtimes C_3, C_3^2 \rtimes Q_8$, or $Q(64) \rtimes C_3$.
- (4) $L_2(5), L_2(7), SL_2(5)$, or $SL_2(7)$

A proof of this theorem is long and would require some group theory technology (see [9] for the proof). Other than a brief appeal to the classification of finite simple groups, we can present the proof of the following:

Theorem A.0.2. *Let G be a group which is order conjugate. Then $G \in \{1, C_2, S_3\}$.*

The result falls out as a corollary in [13] but this requires extra technology, so we will instead follow the path marked by [15].

Remark A.0.3. For some group G with a normal subgroup N and $x \in G$, in this appendix we will write $\bar{x}$ for Nx , the image of x under the quotient map. We have no logical formulae with tuples of variables to worry about here, so there should be no confusion.

Lemma A.0.4. *Let G be an order conjugate group with $x, y \in G$ some p -elements for p some prime dividing $|G|$. Let $N \trianglelefteq G$, and suppose $\bar{x}$ and $\bar{y}$ have equal nonzero order in G/N . Then $|x| = |y|$.*

Proof. Suppose, looking for a contradiction, that the claim is false. Then, without loss of generality, we may write $|x| = p^{n+k}$, $|y| = p^n$ for $k > 0$. Also note here $|\bar{x}| = |\bar{y}| = p^m$ for some $m \in \mathbb{Z}$. Then x^{p^k} and y are of the same order, so are conjugate by our assumption on G . This conjugacy is passed down to the quotient group, giving

$$(*) \quad \bar{y}^g = \overline{x^{p^k}} = \bar{x}^{p^k},$$

for some $g \in G$. Now if $k \geq m$ we have $\bar{x}^{p^k} = 1_{G/N}$, which has lesser order than $\bar{y}$. Otherwise if $k < m$, then p^k properly divides p^m , and so

$$|\bar{x}^{p^k}| = p^{m-k} < p^m = |\bar{y}|.$$

Thus the left and right sides of the equation in $(*)$ have different orders, which is a contradiction. $\square$

This lemma has immediate application in the following more important result.

Lemma A.0.5. *Order conjugacy is closed under taking quotients.*

Proof. Let G be an order conjugate group and $N \trianglelefteq G$. Suppose $\bar{x}, \bar{y} \in G/N$ are elements of the same order $p_1^{\alpha_1} \dots p_n^{\alpha_n}$. First assume x has order $p_1^{\beta_1} \dots p_n^{\beta_n}$ and y has order $p_1^{\gamma_1} \dots p_n^{\gamma_n}$, and note immediately $\alpha_i < \beta_i$ and $\alpha_i < \gamma_i$ for $1 \leq i \leq n$.

Set $c = p_2^{\beta_2} \dots p_n^{\beta_n}$, $d = p_2^{\gamma_2} \dots p_n^{\gamma_n}$. Then x^c and y^d are both p_1 -elements of G , and $|\bar{x}^c| = |\bar{y}^d| = p^{\alpha_1}$. Applying the above lemma, x^c and y^d have the same order, so $\beta_1 = \gamma_1$. By

inductively applying this argument, we can get $\beta_i = \gamma_i$ for all $1 \leq i \leq n$. Thus $|x| = |y|$, and G is order conjugate so $x^g = y$ for some $g \in G$, so $\bar{x}^g = \bar{y}$.

If our assumption about the order of x does not hold, then x has order $p_1^{\beta_1} \dots p_n^{\beta_n} k$ for some $k \in \mathbb{Z}$ with $p_i \nmid k$ for all $1 \leq i \leq n$. Then we may apply Bezout's Lemma to write $tp_1^{\beta_1} \dots p_n^{\beta_n} + sk = 1$, and $\bar{x}^{sk} = \bar{x}$, and $|x^{sk}| = p_1^{\beta_1} \dots p_n^{\beta_n}$, so we can run through the same argument as above with x^{sk} in place of x . Our assumption about the order of y is similarly justified. $\square$

Lemma A.0.6. *There are no non-abelian finite simple order conjugate groups.*

Proof. By the classification of finite simple groups, we need only check the alternating groups, the groups of Lie type, and the sporadic simple groups. Unfortunately, we can only discuss alternating groups here, but Fitzpatrick cites a survey of Syskin [39] as containing the information to show each sporadic simple group is not order conjugate.

First consider the alternating groups. In A_5 , there are two conjugacy classes of 5-cycles, so A_5 is not order conjugate. For $n > 5$, we have the elements $(123)(456)$ and (123) , which have different cycle structures, so are not conjugate in A_n , so A_n is not order conjugate. $\square$

Before we get to our classification result, we will fly through a few more technical results on order conjugate groups.

Lemma A.0.7. *Let G be an order conjugate group.*

- (1) *If $N \trianglelefteq G$ and $x \in N$, then for all $y \in G$ with $|y| \mid |x|$, we have $y \in N$.*
- (2) *For all $x \in G$, we have $N_G(\langle x \rangle)/C_G(x) \cong \text{Aut}(\langle x \rangle)$.*
- (3) *G does not have a subgroup N such that $|G : N| = 2$ and $|N|$ is even.*
- (4) *If G has a nontrivial subgroup N such that $|G : N| = 2$ and $|N|$ is odd, then $G \cong S_3$.*

Proof. (1). Let $y \in G$ with $|y| \mid |x|$ be given. Then $z := x^{\frac{|x|}{|y|}} \in N$, and has order $|y|$. G is order conjugate, so there is some $g \in G$ such that $y = z^g \in N$.

(2). This is similar to part of our proof of Lemma 5.1.12, and follows from the fact that an automorphism of $\langle x \rangle$ is determined by sending x to some other generator, which can always be done by conjugation as G is order conjugate.

(3). This proof is left to the reader.

(4). As G has even order but N does not, we may find $x \in G \setminus N$ with $|x| = 2$. We will show that x acts on N by inversion.

Let $z \in N \setminus \{1\}$ be given, and suppose, looking for a contradiction, that $z^x = z$. Then $C_G(z)$ has even order by Lagrange's theorem, so $\text{Aut}\langle z \rangle \cong N_G\langle z \rangle/C_G(z)$ must have odd order. It is a standard fact that $|\text{Aut}(C_n)| = \varphi(n)$ where φ is Euler's totient function, which is even for $n \geq 2$, so we have arrived at a contradiction, so x cannot centralize z . The element z was arbitrary, so x is a derangement on $N \setminus \{1\}$.

Now define $\varphi : N \rightarrow N$ by $y\varphi = y^{-1}y^x$. Showing φ is injective, we suppose $z^{-1}z^x = y^{-1}y^x$. Then

$$yz^{-1} = y^x(z^x)^{-1} = (yz^{-1})^x,$$

but x is a derangement on $N \setminus \{1\}$, so this forces $yz^{-1} = 1$. Thus $y = z$ and φ is an injection from a finite set to itself, so is a bijection. Thus each element of N may be written $y^{-1}y^x$ for some $y \in N$.

Now observe:

$$(y^{-1}y^x)^x = (y^{-1})^x y^{x^2} = (y^x)^{-1}y = (y^{-1}y^x)^{-1}.$$

As each element of N may be written as $y^{-1}y^x$ for some $y \in N$, this means x acts by inversion. Then for all $y, z \in N$:

$$yz = (y^{-1})^{-1}(z^{-1})^{-1} = (y^{-1})^x(z^{-1})^x = (y^{-1}z^{-1})^x = (y^{-1}z^{-1})^{-1} = zy.$$

So N is an abelian group of odd order, so $N \cong \prod_{i=1}^k C_{n_i}$ where $n_i > 1$ is odd for all $1 \leq i \leq n$. We know the action of $\langle x \rangle$ on G , so we conclude G is isomorphic to the outer semidirect product $P = \prod_{i=1}^k C_{n_i} \rtimes_{\phi} C_2$, where ϕ induces the inversion action. Now let $1 \leq i \leq n$ be given and consider n_i . Then P has a subgroup

$$K := (1 \oplus \dots 1 \oplus \oplus C_{n_i} \oplus 1 \dots \oplus 1) \rtimes C_2 \cong D_{2n_i}.$$

It is The group K is normal, and must be order conjugate itself. Writing α for a rotation of order n_i in D_{2n_i} , we know that $\alpha^{D_{2n_i}} = \{\alpha, \alpha^{-1}\}$, but as n_i is odd, $|\alpha^2| = |\alpha|$. K is order conjugate, so we must have $\alpha^2 = \alpha^{-1}$, forcing $n_i = 3$. Considering that the different coordinates cannot interact with each other, $k = 1$ is then forced, so $G \cong C_3 \rtimes_{\phi} C_2$ where ϕ induces inversion, so $G \cong S_3$. $\square$

Proof of Theorem A.0.2. First suppose G is soluble. By a result from MT5864, we may find a composition series with composition factors cyclic of prime order:

$$G = G_0 > G_1 > G_2 > \dots > G_k = 1.$$

If $k = 0$, then $G = 1$, so we may suppose $k > 0$. Now G is order conjugate, so by Lemma A.0.5 G/G_1 is order conjugate. By above, G/G_1 must also be cyclic. This forces $G/G_1 \cong C_2$, and so G_1 is a subgroup of index 2, and therefore has odd order by Lemma A.0.7. In particular this means the 2 is the highest power of 2 which divides $|G|$. If $G_1 = 1$, then $G \cong C_2$. Otherwise G is a nontrivial subgroup of index 2, in which case $G \cong S_3$.

We have shown that if G is soluble, then G is isomorphic to one of $1, C_2, S_3$.

Now suppose, looking for a contradiction, that G is not soluble. Take G to be minimal with this property. By Lemma A.0.6, G cannot be non-abelian simple, so has some maximal nontrivial proper normal subgroup N . The quotient group G/N is order conjugate by Lemma A.0.5, and has a lesser order than G . By minimality of G , the group G/N must be soluble, so G/N is isomorphic to C_2 or S_3 . By maximality of N along with the correspondence theorem, $G/N \cong C_2$, so $G \cong S_3$, a soluble group, by Lemma A.0.7. We have arrived at the desired contradiction. $\square$

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